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Yeast expressing a heterologous phospholipase for ethanol production

Abstract

Described herein are recombinant fermenting organisms having a heterologous polynucleotide encoding a phospholipase. Also described are processes for producing a fermentation product, such as ethanol, from starch or cellulosic-containing material with the recombinant fermenting organisms.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/283,760 which is a 35 U.S.C. 371 national application of PCT/US2019/054996, filed Oct. 7, 2019, which claims priority or the benefit from U.S. Provisional Application Ser. No. 62/742,805, filed Oct. 8, 2018. The contents of these applications are fully incorporated herein by reference.

REFERENCE TO A SEQUENCE LISTING

(1) This application contains a Sequence Listing in computer readable form, that was submitted as an xml file named SQ_ST26 (created on Oct. 3, 2023, containing 665 kb), which is incorporated herein by reference.

BACKGROUND

(2) Production of ethanol from starch and cellulosic containing materials is well-known in the art.

(3) The most commonly industrially used commercial process for starch-containing material, often referred to as a “conventional process”, includes liquefying gelatinized starch at high temperature (about 85° C.) using typically a bacterial alpha-amylase, followed by simultaneous saccharification and fermentation (SSF) carried out anaerobically in the presence of typically a glucoamylase and a *Saccharomyces cerevisiae* yeast.

(4) Yeasts which are used for production of ethanol for use as fuel, such as in the corn ethanol industry, require several characteristics to ensure cost effective production of the ethanol. These characteristics include ethanol tolerance, low by-product yield, rapid fermentation, and the ability to limit the amount of residual sugars remaining in the ferment. Such characteristics have a marked effect on the viability of the industrial process.

(5) Yeast of the genus *Saccharomyces* exhibits many of the characteristics required for production of ethanol. In particular, strains of *Saccharomyces cerevisiae* are widely used for the production of ethanol in the fuel ethanol industry. Strains of *Saccharomyces cerevisiae* that are widely used in the fuel ethanol industry have the ability to produce high yields of ethanol under fermentation conditions found in, for example, the fermentation of corn mash. An example of such a strain is the yeast used in commercially available ethanol yeast product called ETHANOL RED®.

(6) *Saccharomyces cerevisiae* yeast have been genetically engineered to express alpha-amylase and/or glucoamylase to improve yield and decrease the amount of exogenously added enzymes necessary during SSF (e.g., WO2018/098381, WO2017/087330, WO2017/037614, WO2011/128712, WO2011/153516, US2018/0155744). Yeast have also been engineered to express trehalase in an attempt to increase fermentation yield by breaking down residual trehalose (e.g., WO2017/077504).

(7) WO2008/135547 concerns reducing foam in processes for production of a fermentation product by contacting the fermentation media comprising a fermenting organism with a lipolytic enzyme selected from the group consisting of phospholipase, lyso-phospholipase and lipase, and a metal salt.

(8) WO2014/147219 concerns a phospholipase A from *Talaromyces leycettanus*.

(9) WO2015/140275 discloses a phospholipase C from *Bacillus thuringiensis*.

(10) Despite significant improvement of ethanol production processes over the past decade there is still a desire and need for providing improved processes of ethanol fermentation from starch and cellulosic containing material in an economically and commercially relevant scale.

(11) For example, foam generation during ethanol fermentation is a major problem, especially in ethanol production processes where starch-containing material is liquefied with an alpha-amylase and a protease before saccharification and fermentation. Additionally, the use of nitrogen supplements (e.g., urea) is an added expense during fermentation. Therefore, there is a desire to, inter alia, reduce foam and/or reduce supplemental nitrogen requirements in ethanol fermentation.

SUMMARY

(12) Described herein are, inter alia, methods for producing a fermentation product, such as

ethanol, from starch or cellulosic-containing material, and yeast suitable for use in such processes. The Applicant has surprisingly found that yeasts expressing a phospholipase provide beneficial properties during fermentation, such as reduced foaming, improved oil extraction yield, and improved ethanol yield.

(13) A first aspect relates to methods of producing a fermentation product from a starch-containing or cellulosic-containing material comprising: (a) saccharifying the starch-containing or cellulosic-containing material; and (b) fermenting the saccharified material of step (a) with a fermenting organism; wherein the fermenting organism comprises a heterologous polynucleotide encoding a phospholipase. In some embodiments, the phospholipase is a Phospholipase A or a Phospholipase C.

(14) In some embodiments of the methods, fermentation and saccharification are performed simultaneously in a simultaneous saccharification and fermentation (SSF). In other embodiments, fermentation and saccharification are performed sequentially (SHF).

(15) In some embodiments of the methods, the method comprises recovering the fermentation product from the fermentation (e.g., by distillation).

(16) In some embodiments of the methods, the fermentation product is ethanol.

(17) In some embodiments of the methods, fermentation is performed under reduced nitrogen conditions (e.g., less than 1000 ppm urea or ammonium hydroxide, such as less than 750 ppm, less than 500 ppm, less than 400 ppm, less than 300 ppm, less than 250 ppm, less than 200 ppm, less than 150 ppm, less than 100 ppm, less than 75 ppm, less than 50 ppm, less than 25 ppm, or less than 10 ppm).

(18) In some embodiments of the methods, the method results in higher yield of fermentation product (e.g., ethanol) and/or reduced foam accumulation when compared to the same process using an identical cell without the heterologous polynucleotide encoding the phospholipase under the same conditions (e.g., at about or after 54 hours fermentation, such as the conditions described in Examples 3 or 4). In some embodiments, the method results in at least 0.25% (e.g., 0.5%, 0.75%, 1.0%, 1.25%, 1.5%, 1.75%, 2%, 3% or 5%) higher yield of fermentation product.

(19) In some embodiments of the methods, the phospholipase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342. In some embodiments of the methods, the phospholipase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242. In some embodiments of the methods, the heterologous polynucleotide encodes a phospholipase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). In some embodiments of the methods, the heterologous polynucleotide encodes a phospholipase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242).

(20) In some embodiments of the methods, saccharification of step occurs on a starch-containing material, and wherein the starch-containing material is either gelatinized or ungelatinized starch.

(21) In some embodiments of the methods, the method comprises liquefying the starch-containing material by contacting the material with an alpha-amylase prior to saccharification.

(22) In some embodiments of the methods, liquefying the starch-containing material and/or saccharifying the starch-containing material is conducted in presence of exogenously added protease.

(23) In some embodiments of the methods, the fermenting organism comprises a heterologous

polynucleotide encoding a glucoamylase, such as a glucoamylase having a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of a *Pycnopus* glycoamylase (e.g., a *Pycnopus sanguineus* glucoamylase of SEQ ID NO: 229), a *Gloeophyllum* glucoamylase (e.g. a *Gloeophyllum sepiarium* of SEQ ID NO: 8), or a glucoamylase of any one of SEQ ID NOs: 102-113 (e.g., a *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103 or 104, or a *Trichoderma reesei* glucoamylase of SEQ ID NO: 230).

(24) In some embodiments of the methods, the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase, such as an alpha-amylase having a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231. In some embodiments of the methods, the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231. In some embodiments of the methods, the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: SEQ ID NOs: 76-101, 121-174 and 231.

(25) In some embodiments of the methods, the fermenting organism comprises a heterologous polynucleotide encoding a trehalase, such as a trehalase having a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 175-226. In some embodiments of the methods, the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 175-226. In some embodiments of the methods, the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: SEQ ID NOs: 175-226.

(26) In some embodiments of the methods, the fermenting organism comprises a heterologous polynucleotide encoding a protease, such as a protease having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69).

(27) In some embodiments of the methods, saccharification of step occurs on a cellulosic-containing material, and wherein the cellulosic-containing material is pretreated (e.g. a dilute acid pretreatment).

(28) In some embodiments of the methods, saccharification occurs on a cellulosic-containing material, and wherein the enzyme composition comprises one or more enzymes selected from a cellulase (e.g., endoglucanase, a cellobiohydrolase, or a beta-glucosidase), an AA9 polypeptide, a hemicellulase (e.g., a xylanase, an acetylxylan esterase, a feruloyl esterase, an arabinofuranosidase, a xylosidase, or a glucuronidase), a CIP, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin.

(29) In some embodiments of the methods, the fermenting organism is a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell. In some embodiments, the fermenting organism is a *Saccharomyces cerevisiae* cell.

(30) Another aspect relates to a recombinant yeast cell comprising a heterologous polynucleotide

encoding a phospholipase. In some embodiments, the phospholipase is a Phospholipase A or a Phospholipase C.

(31) In some embodiments of the yeast cell, the cell is capable of higher yield of fermentation product and/or reduced foam accumulation when compared to fermentation using the same process and an identical cell without the heterologous polynucleotide encoding the phospholipase under the same conditions (e.g., at about or after 54 hours fermentation, such as the conditions described in Examples 3 or 4). In some embodiments, the cell is capable of at least 0.25% (e.g., 0.5%, 0.75%, 1.0%, 1.25%, 1.5%, 1.75%, 2%, 3% or 5%) higher yield of fermentation product.

(32) In some embodiments, the recombinant yeast cell is a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell. In some embodiments, the recombinant yeast cell is a *Saccharomyces cerevisiae* cell.

(33) In some embodiments of the yeast cell, the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase, such as a glucoamylase having a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of a *Pycnoporus* glycoamylase (e.g., a *Pycnoporus sanguineus* glucoamylase of SEQ ID NO: 229), a *Gloeophyllum* glucoamylase (e.g. a *Gloeophyllum sepiarium* of SEQ ID NO: 8), or a glucoamylase of any one of SEQ ID NOs: 102-113 (e.g., a *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103 or 104, or a *Trichoderma reesei* glucoamylase of SEQ ID NO: 230).

(34) In some embodiments of the yeast cell, the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase, wherein the alpha-amylase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231. In some embodiments of the methods, the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231. In some embodiments of the methods, the heterologous polynucleotide encodes an alpha-amylase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: SEQ ID NOs: 76-101, 121-174 and 231.

(35) In some embodiments of the yeast cell, the fermenting organism comprises a heterologous polynucleotide encoding a trehalase, wherein the trehalase has mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 175-226. In some embodiments of the methods, the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 175-226. In some embodiments of the methods, the heterologous polynucleotide encodes a trehalase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: SEQ ID NOs: 175-226.

(36) In some embodiments of the yeast cell, the fermenting organism comprises a heterologous polynucleotide encoding a protease, such as a protease having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69).

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 shows % ethanol improvement for phospholipase-expressing yeast strains and control strain MHCT-484 described in Example 3.
- (2) FIG. 2 shows normalized mean ethanol improvement after 54 hours of fermentation of AMP mash at 0 and 150 ppm urea as described in Example 4.
- (3) FIG. 3 shows normalized mean ethanol improvement after 54 hours of fermentation of the non-AMP mash at 0 and 300 ppm urea as described in Example 4.
- (4) FIG. 4 shows improved defoaming capability during fermentation of a phospholipase-expressing yeast strain HP21-F04 (right) compared to control strain yMHCT48 (left) as described in Example 9.

DEFINITIONS

- (5) Unless defined otherwise or clearly indicated by context, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art.
- (6) Allelic variant: The term “allelic variant” means any of two or more alternative forms of a gene occupying the same chromosomal locus. Allelic variation arises naturally through mutation, and may result in polymorphism within populations. Gene mutations can be silent (no change in the encoded polypeptide) or may encode polypeptides having altered amino acid sequences. An allelic variant of a polypeptide is a polypeptide encoded by an allelic variant of a gene.
- (7) Alpha-amylase: The term “alpha amylase” means an 1,4-alpha-D-glucan glucanohydrolase, EC. 3.2.1.1, which catalyze hydrolysis of starch and other linear and branched 1,4-glucosidic oligo- and polysaccharides. For purposes of the present invention, alpha amylase activity can be determined using an alpha amylase assay described in the examples section below.
- (8) Auxiliary Activity 9: The term “Auxiliary Activity 9” or “AA9” means a polypeptide classified as a lytic polysaccharide monooxygenase (Quinlan et al., 2011, *Proc. Natl. Acad. Sci. USA* 208: 15079-15084; Phillips et al., 2011, *ACS Chem. Biol.* 6: 1399-1406; Lin et al., 2012, *Structure* 20: 1051-1061). AA9 polypeptides were formerly classified into the glycoside hydrolase Family 61 (GH61) according to Henrissat, 1991, *Biochem. J.* 280: 309-316, and Henrissat and Bairoch, 1996, *Biochem. J.* 316: 695-696.
- (9) AA9 polypeptides enhance the hydrolysis of a cellulosic-containing material by an enzyme having cellulolytic activity. Cellulolytic enhancing activity can be determined by measuring the increase in reducing sugars or the increase of the total of cellobiose and glucose from the hydrolysis of a cellulosic-containing material by cellulolytic enzyme under the following conditions: 1-50 mg of total protein/g of cellulose in pretreated corn stover (PCS), wherein total protein is comprised of 50-99.5% w/w cellulolytic enzyme protein and 0.5-50% w/w protein of an AA9 polypeptide for 1-7 days at a suitable temperature, such as 40 C-80° C., e.g., 50° C., 55° C., 60° C., 65° C., or 70° C., and a suitable pH, such as 4-9, e.g., 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, or 8.5, compared to a control hydrolysis with equal total protein loading without cellulolytic enhancing activity (1-50 mg of cellulolytic protein/g of cellulose in PCS).
- (10) AA9 polypeptide enhancing activity can be determined using a mixture of CELLUCLAST® 1.5 L (Novozymes A/S, Bagsvaerd, Denmark) and beta-glucosidase as the source of the cellulolytic activity, wherein the beta-glucosidase is present at a weight of at least 2-5% protein of the cellulase protein loading. In one embodiment, the beta-glucosidase is an *Aspergillus oryzae* beta-glucosidase (e.g., recombinantly produced in *Aspergillus oryzae* according to WO 02/095014). In another embodiment, the beta-glucosidase is an *Aspergillus fumigatus* beta-glucosidase (e.g., recombinantly produced in *Aspergillus oryzae* as described in WO 02/095014).
- (11) AA9 polypeptide enhancing activity can also be determined by incubating an AA9 polypeptide with 0.5% phosphoric acid swollen cellulose (PASC), 100 mM sodium acetate pH 5, 1 mM

MnSO₄·0.1% gallic acid, 0.025 mg/mL of *Aspergillus fumigatus* beta-glucosidase, and 0.01% TRITON® X-100 (4-(1,1,3,3-tetramethylbutyl)phenyl-polyethylene glycol) for 24-96 hours at 40° C. followed by determination of the glucose released from the PASC.

(12) AA9 polypeptide enhancing activity can also be determined according to WO 2013/028928 for high temperature compositions.

(13) AA9 polypeptides enhance the hydrolysis of a cellulosic-containing material catalyzed by enzyme having cellulolytic activity by reducing the amount of cellulolytic enzyme required to reach the same degree of hydrolysis preferably at least 1.01-fold, e.g., at least 1.05-fold, at least 1.10-fold, at least 1.25-fold, at least 1.5-fold, at least 2-fold, at least 3-fold, at least 4-fold, at least 5-fold, at least 10-fold, or at least 20-fold.

(14) Beta-glucosidase: The term “beta-glucosidase” means a beta-D-glucoside glucohydrolase (E.C. 3.2.1.21) that catalyzes the hydrolysis of terminal non-reducing beta-D-glucose residues with the release of beta-D-glucose. Beta-glucosidase activity can be determined using p-nitrophenyl-beta-D-glucopyranoside as substrate according to the procedure of Venturi et al., 2002, *J. Basic Microbiol.* 42: 55-66. One unit of beta-glucosidase is defined as 1.0 μmole of p-nitrophenolate anion produced per minute at 25° C., pH 4.8 from 1 mM p-nitrophenyl-beta-D-glucopyranoside as substrate in 50 mM sodium citrate containing 0.01% TWEEN® 20.

(15) Beta-xylosidase: The term “beta-xylosidase” means a beta-D-xyloside xylohydrolase (E.C. 3.2.1.37) that catalyzes the exo-hydrolysis of short beta (1.fwdarw.4)-xylooligosaccharides to remove successive D-xylose residues from non-reducing termini. Beta-xylosidase activity can be determined using 1 mM p-nitrophenyl-beta-D-xyloside as substrate in 100 mM sodium citrate containing 0.01% TWEEN® 20 at pH 5, 40° C. One unit of beta-xylosidase is defined as 1.0 μmole of p-nitrophenolate anion produced per minute at 40° C., pH 5 from 1 mM p-nitrophenyl-beta-D-xyloside in 100 mM sodium citrate containing 0.01% TWEEN® 20.

(16) Catalase: The term “catalase” means a hydrogen-peroxide:hydrogen-peroxide oxidoreductase (EC 1.11.1.6) that catalyzes the conversion of 2 H₂O₂ to O₂+2 H₂O. For purposes of the present invention, catalase activity is determined according to U.S. Pat. No. 5,646,025. One unit of catalase activity equals the amount of enzyme that catalyzes the oxidation of 1 μmole of hydrogen peroxide under the assay conditions.

(17) Catalytic domain: The term “catalytic domain” means the region of an enzyme containing the catalytic machinery of the enzyme.

(18) Cellobiohydrolase: The term “cellobiohydrolase” means a 1,4-beta-D-glucan cellobiohydrolase (E.C. 3.2.1.91 and E.C. 3.2.1.176) that catalyzes the hydrolysis of 1,4-beta-D-glucosidic linkages in cellulose, celooligosaccharides, or any beta-1,4-linked glucose containing polymer, releasing cellobiose from the reducing end (cellobiohydrolase I) or non-reducing end (cellobiohydrolase II) of the chain (Teeri, 1997, *Trends in Biotechnology* 15: 160-167; Teen et al., 1998, *Biochem. Soc. Trans.* 26:173-178). Cellobiohydrolase activity can be determined according to the procedures described by Lever et al., 1972, *Anal. Biochem.* 47: 273-279; van Tilbeurgh et al., 1982, *FEBS Letters* 149: 152-156; van Tilbeurgh and Claeysens, 1985, *FEBS Letters* 187: 283-288; and Tomme et al., 1988, *Eur. J. Biochem.* 170: 575-581.

(19) Cellulolytic enzyme or cellulase: The term “cellulolytic enzyme” or “cellulase” means one or more (e.g., several) enzymes that hydrolyze a cellulosic-containing material. Such enzymes include endoglucanase(s), cellobiohydrolase(s), beta-glucosidase(s), or combinations thereof. The two basic approaches for measuring cellulolytic enzyme activity include: (1) measuring the total cellulolytic enzyme activity, and (2) measuring the individual cellulolytic enzyme activities (endoglucanases, cellobiohydrolases, and beta-glucosidases) as reviewed in Zhang et al., 2006, *Biotechnology Advances* 24: 452-481. Total cellulolytic enzyme activity can be measured using insoluble substrates, including Whatman No 1 filter paper, microcrystalline cellulose, bacterial cellulose, algal cellulose, cotton, pretreated lignocellulose, etc. The most common total cellulolytic activity assay is the filter paper assay using Whatman No 1 filter paper as the substrate. The assay

was established by the International Union of Pure and Applied Chemistry (IUPAC) (Ghose, 1987, *Pure Appl. Chem.* 59: 257-68).

(20) Cellulolytic enzyme activity can be determined by measuring the increase in production/release of sugars during hydrolysis of a cellulosic-containing material by cellulolytic enzyme(s) under the following conditions: 1-50 mg of cellulolytic enzyme protein/g of cellulose in pretreated corn stover (PCS) (or other pretreated cellulosic-containing material) for 3-7 days at a suitable temperature such as 40° C.-80° C., e.g., 50° C., 55° C., 60° C., 65° C., or 70° C., and a suitable pH such as 4-9, e.g., 5.0, 5.5, 6.0, 6.5, or 7.0, compared to a control hydrolysis without addition of cellulolytic enzyme protein. Typical conditions are 1 mL reactions, washed or unwashed PCS, 5% insoluble solids (dry weight), 50 mM sodium acetate pH 5, 1 mM MnSO₄, 50° C., 55° C., or 60° C., 72 hours, sugar analysis by AMINEX® HPX-87H column chromatography (Bio-Rad Laboratories, Inc., Hercules, CA, USA).

(21) Coding sequence: The term “coding sequence” or “coding region” means a polynucleotide sequence, which specifies the amino acid sequence of a polypeptide. The boundaries of the coding sequence are generally determined by an open reading frame, which usually begins with the ATG start codon or alternative start codons such as GTG and TTG and ends with a stop codon such as TAA, TAG, and TGA. The coding sequence may be a sequence of genomic DNA, cDNA, a synthetic polynucleotide, and/or a recombinant polynucleotide.

(22) Control sequence: The term “control sequence” means a nucleic acid sequence necessary for polypeptide expression. Control sequences may be native or foreign to the polynucleotide encoding the polypeptide, and native or foreign to each other. Such control sequences include, but are not limited to, a leader sequence, polyadenylation sequence, propeptide sequence, promoter sequence, signal peptide sequence, and transcription terminator sequence. The control sequences may be provided with linkers for the purpose of introducing specific restriction sites facilitating ligation of the control sequences with the coding region of the polynucleotide encoding a polypeptide.

(23) Disruption: The term “disruption” means that a coding region and/or control sequence of a referenced gene is partially or entirely modified (such as by deletion, insertion, and/or substitution of one or more nucleotides) resulting in the absence (inactivation) or decrease in expression, and/or the absence or decrease of enzyme activity of the encoded polypeptide. The effects of disruption can be measured using techniques known in the art such as detecting the absence or decrease of enzyme activity using from cell-free extract measurements referenced herein; or by the absence or decrease of corresponding mRNA (e.g., at least 25% decrease, at least 50% decrease, at least 60% decrease, at least 70% decrease, at least 80% decrease, or at least 90% decrease); the absence or decrease in the amount of corresponding polypeptide having enzyme activity (e.g., at least 25% decrease, at least 50% decrease, at least 60% decrease, at least 70% decrease, at least 80% decrease, or at least 90% decrease); or the absence or decrease of the specific activity of the corresponding polypeptide having enzyme activity (e.g., at least 25% decrease, at least 50% decrease, at least 60% decrease, at least 70% decrease, at least 80% decrease, or at least 90% decrease). Disruptions of a particular gene of interest can be generated by methods known in the art, e.g., by directed homologous recombination (see *Methods in Yeast Genetics* (1997 edition), Adams, Gottschling, Kaiser, and Stems, Cold Spring Harbor Press (1998)).

(24) Endogenous gene: The term “endogenous gene” means a gene that is native to the referenced host cell. “Endogenous gene expression” means expression of an endogenous gene.

(25) Endoglucanase: The term “endoglucanase” means a 4-(1,3;1,4)-beta-D-glucan 4-glucanohydrolase (E.C. 3.2.1.4) that catalyzes endohydrolysis of 1,4-beta-D-glycosidic linkages in cellulose, cellulose derivatives (such as carboxymethyl cellulose and hydroxyethyl cellulose), lichenin, beta-1,4 bonds in mixed beta-1,3-1,4 glucans such as cereal beta-D-glucans or xyloglucans, and other plant material containing cellulosic components. Endoglucanase activity can be determined by measuring reduction in substrate viscosity or increase in reducing ends determined by a reducing sugar assay (Zhang et al., 2006, *Biotechnology Advances* 24: 452-481).

Endoglucanase activity can also be determined using carboxymethyl cellulose (CMC) as substrate according to the procedure of Ghose, 1987, *Pure and Appl. Chem.* 59: 257-268, at pH 5, 40° C.

(26) Expression: The term “expression” includes any step involved in the production of the polypeptide including, but not limited to, transcription, post-transcriptional modification, translation, post-translational modification, and secretion. Expression can be measured—for example, to detect increased expression—by techniques known in the art, such as measuring levels of mRNA and/or translated polypeptide.

(27) Expression vector: The term “expression vector” means a linear or circular DNA molecule that comprises a polynucleotide encoding a polypeptide and is operably linked to control sequences that provide for its expression.

(28) Fermentable medium: The term “fermentable medium” or “fermentation medium” refers to a medium comprising one or more (e.g., two, several) sugars, such as glucose, fructose, sucrose, cellobiose, xylose, xylulose, arabinose, mannose, galactose, and/or soluble oligosaccharides, wherein the medium is capable, in part, of being converted (fermented) by a host cell into a desired product, such as ethanol. In some instances, the fermentation medium is derived from a natural source, such as sugar cane, starch, or cellulose, and may be the result of pretreating the source by enzymatic hydrolysis (saccharification). The term fermentation medium is understood herein to refer to a medium before the fermenting organism is added, such as, a medium resulting from a saccharification process, as well as a medium used in a simultaneous saccharification and fermentation process (SSF).

(29) Glucoamylase: The term “glucoamylase” (1,4- α -D-glucan glucohydrolase, EC 3.2.1.3) is defined as an enzyme that catalyzes the release of D-glucose from the non-reducing ends of starch or related oligo- and polysaccharide molecules. For purposes of the present invention, glucoamylase activity may be determined according to the procedures known in the art, such as those described in the Examples of U.S. Provisional Patent Application No. 62/703,103, filed Jul. 25, 2018.

(30) Hemicellulolytic enzyme or hemicellulase: The term “hemicellulolytic enzyme” or “hemicellulase” means one or more (e.g., several) enzymes that hydrolyze a hemicellulosic material. See, for example, Shallom and Shoham, 2003, *Current Opinion In Microbiology* 6(3): 219-228). Hemicellulases are key components in the degradation of plant biomass. Examples of hemicellulases include, but are not limited to, an acetylmannan esterase, an acetylxylan esterase, an arabinanase, an arabinofuranosidase, a coumaric acid esterase, a feruloyl esterase, a galactosidase, a glucuronidase, a glucuronoyl esterase, a mannanase, a mannosidase, a xylanase, and a xylosidase. The substrates for these enzymes, hemicelluloses, are a heterogeneous group of branched and linear polysaccharides that are bound via hydrogen bonds to the cellulose microfibrils in the plant cell wall, crosslinking them into a robust network. Hemicelluloses are also covalently attached to lignin, forming together with cellulose a highly complex structure. The variable structure and organization of hemicelluloses require the concerted action of many enzymes for its complete degradation. The catalytic modules of hemicellulases are either glycoside hydrolases (GHs) that hydrolyze glycosidic bonds, or carbohydrate esterases (CEs), which hydrolyze ester linkages of acetate or ferulic acid side groups. These catalytic modules, based on homology of their primary sequence, can be assigned into GH and CE families. Some families, with an overall similar fold, can be further grouped into clans, marked alphabetically (e.g., GH-A). A most informative and updated classification of these and other carbohydrate active enzymes is available in the Carbohydrate-Active Enzymes (CAZy) database. Hemicellulolytic enzyme activities can be measured according to Ghose and Bisaria, 1987, *Pure & Appl. Chem.* 59: 1739-1752, at a suitable temperature such as 40° C.-80° C., e.g., 50° C., 55° C., 60° C., 65° C., or 70° C., and a suitable pH such as 4-9, e.g., 5.0, 5.5, 6.0, 6.5, or 7.0.

(31) Heterologous polynucleotide: The term “heterologous polynucleotide” is defined herein as a polynucleotide that is not native to the host cell; a native polynucleotide in which structural

modifications have been made to the coding region; a native polynucleotide whose expression is quantitatively altered as a result of a manipulation of the DNA by recombinant DNA techniques, e.g., a different (foreign) promoter; or a native polynucleotide in a host cell having one or more extra copies of the polynucleotide to quantitatively alter expression. A “heterologous gene” is a gene comprising a heterologous polynucleotide.

(32) High stringency conditions: The term “high stringency conditions” means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/mL sheared and denatured salmon sperm DNA, and 50% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 65° C.

(33) Host cell: The term “host cell” means any cell type that is susceptible to transformation, transfection, transduction, and the like with a nucleic acid construct or expression vector comprising a polynucleotide described herein (e.g., a polynucleotide encoding an alpha-amylase and/or trehalase). The term “host cell” encompasses any progeny of a parent cell that is not identical to the parent cell due to mutations that occur during replication. The term “recombinant cell” is defined herein as a non-naturally occurring host cell comprising one or more (e.g., two, several) heterologous polynucleotides.

(34) Low stringency conditions: The term “low stringency conditions” means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/mL sheared and denatured salmon sperm DNA, and 25% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 50° C.

(35) Mature polypeptide: The term “mature polypeptide” is defined herein as a polypeptide having biological activity that is in its final form following translation and any post-translational modifications, such as N-terminal processing, C-terminal truncation, glycosylation, phosphorylation, etc. The mature polypeptide sequence lacks a signal sequence, which may be determined using techniques known in the art (See, e.g., Zhang and Henzel, 2004, *Protein Science* 13: 2819-2824).

(36) Medium stringency conditions: The term “medium stringency conditions” means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/mL sheared and denatured salmon sperm DNA, and 35% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 55° C.

(37) Medium-high stringency conditions: The term “medium-high stringency conditions” means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5×SSPE, 0.3% SDS, 200 micrograms/mL sheared and denatured salmon sperm DNA, and 35% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2×SSC, 0.2% SDS at 60° C.

(38) Nucleic acid construct: The term “nucleic acid construct” means a polynucleotide comprises one or more (e.g., two, several) control sequences. The polynucleotide may be single-stranded or double-stranded, and may be isolated from a naturally occurring gene, modified to contain segments of nucleic acids in a manner that would not otherwise exist in nature, or synthetic.

(39) Operably linked: The term “operably linked” means a configuration in which a control sequence is placed at an appropriate position relative to the coding sequence of a polynucleotide such that the control sequence directs expression of the coding sequence.

(40) Phospholipase: The term “phospholipase” means an enzyme that catalyzes the conversion of phospholipids into fatty acids and other lipophilic substances, such as phospholipase A (EC numbers 3.1.1.4, 3.1.1.5 and 3.1.1.32) or phospholipase C (EC numbers 3.1.4.3 and 3.1.4.11). For purposes of the present invention, phospholipase activity may be determined using activity assays known in the art, or according to the procedures described in the Examples herein (Example 2).

(41) Pretreated corn stover: The term “Pretreated Corn Stover” or “PCS” means a cellulosic-containing material derived from corn stover by treatment with heat and dilute sulfuric acid, alkaline pretreatment, neutral pretreatment, or any pretreatment known in the art.

(42) Protease: The term “protease” is defined herein as an enzyme that hydrolyses peptide bonds. It includes any enzyme belonging to the EC 3.4 enzyme group (including each of the thirteen subclasses thereof). The EC number refers to Enzyme Nomenclature 1992 from NC-IUBMB, Academic Press, San Diego, California, including supplements 1-5 published in *Eur. J. Biochem.* 223: 1-5 (1994); *Eur. J. Biochem.* 232: 1-6 (1995); *Eur. J. Biochem.* 237: 1-5 (1996); *Eur. J. Biochem.* 250: 1-6 (1997); and *Eur. J. Biochem.* 264: 610-650 (1999); respectively. The term “subtilases” refer to a sub-group of serine protease according to Siezen et al., 1991, *Protein Engng.* 4: 719-737 and Siezen et al., 1997, *Protein Science* 6: 501-523. Serine proteases or serine peptidases is a subgroup of proteases characterised by having a serine in the active site, which forms a covalent adduct with the substrate. Further the subtilases (and the serine proteases) are characterised by having two active site amino acid residues apart from the serine, namely a histidine and an aspartic acid residue. The subtilases may be divided into 6 sub-divisions, i.e. the Subtilisin family, the Thermitase family, the Proteinase K family, the Lantibiotic peptidase family, the Kexin family and the Pyrolysin family. The term “protease activity” means a proteolytic activity (EC 3.4). Protease activity may be determined using methods described in the art (e.g., US 2015/0125925) or using commercially available assay kits (e.g., Sigma-Aldrich).

(43) Pullulanase: The term “pullulanase” means a starch debranching enzyme having pullulan 6-glucano-hydrolase activity (EC 3.2.1.41) that catalyzes the hydrolysis the α -1,6-glycosidic bonds in pullulan, releasing maltotriose with reducing carbohydrate ends. For purposes of the present invention, pullulanase activity can be determined according to a PHADEBAS assay or the sweet potato starch assay described in WO2016/087237.

(44) Sequence Identity: The relatedness between two amino acid sequences or between two nucleotide sequences is described by the parameter “sequence identity”.

(45) For purposes described herein, the degree of sequence identity between two amino acid sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, *J. Mol. Biol.* 1970, 48, 443-453) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., *Trends Genet* 2000, 16, 276-277), preferably version 3.0.0 or later. The optional parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EBLOSUM62 (EMBOSS version of BLOSUM62) substitution matrix. The output of Needle labeled “longest identity” (obtained using the -nobrief option) is used as the percent identity and is calculated as follows:

$$(\text{Identical Residues} \times 100) / (\text{Length of the Referenced Sequence} - \text{Total Number of Gaps in Alignment})$$

(46) For purposes described herein, the degree of sequence identity between two deoxyribonucleotide sequences is determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, *supra*) as implemented in the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, *supra*), preferably version 3.0.0 or later. The optional parameters used are gap open penalty of 10, gap extension penalty of 0.5, and the EDNAFULL (EMBOSS version of NCBI NUC4.4) substitution matrix. The output of Needle labeled “longest identity” (obtained using the -nobrief option) is used as the percent identity and is calculated as follows:

$$(\text{Identical Deoxyribonucleotides} \times 100) / (\text{Length of Referenced Sequence} - \text{Total Number of Gaps in Alignment})$$

(47) Signal peptide: The term “signal peptide” is defined herein as a peptide linked (fused) in frame to the amino terminus of a polypeptide having biological activity and directs the polypeptide into the cell's secretory pathway. Signal sequences may be determined using techniques known in the art (See, e.g., Zhang and Henzel, 2004, *Protein Science* 13: 2819-2824). The polypeptides

described herein may comprise any suitable signal peptide known in the art, or any signal peptide described herein (e.g., the *S. cerevisiae* MF α 1 signal peptide of SEQ ID NO: 7, the *S. cerevisiae* EXG1 signal peptide of SEQ ID NO: 227, or the *S. cerevisiae* AG2 signal peptide of SEQ ID NO: 234, or a signal peptide having at least 80%, 85%, 90%, 95%, 96%, 97%, 98% or 99% sequence identity thereof).

(48) Trehalase: The term “trehalase” means an enzyme which degrades trehalose into its unit monosaccharides (i.e., glucose). Trehalases are classified in EC 3.2.1.28 (alpha,alpha-trehalase) and EC. 3.2.1.93 (alpha,alpha-phosphotrehalase). The EC classes are based on recommendations of the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (IUBMB). Description of EC classes can be found on the internet, e.g., on “<http://www.expasy.org/enzyme/>”. Trehalases are enzymes that catalyze the following reactions: EC 3.2.1.28: Alpha,alpha-trehalose+H.sub.2O $\rightleftharpoons$ 2 D-glucose; EC 3.2.1.93: Alpha,alpha-trehalose 6-phosphate+H.sub.2O $\rightleftharpoons$ D-glucose+D-glucose 6-phosphate.

(49) For purposes of the present invention, trehalase activity may be determined according to the trehalase activity assay described herein in the experimental section.

(50) Very high stringency conditions: The term “very high stringency conditions” means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5 $\times$ SSPE, 0.3% SDS, 200 micrograms/mL sheared and denatured salmon sperm DNA, and 50% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2 $\times$ SSC, 0.2% SDS at 70° C.

(51) Very low stringency conditions: The term “very low stringency conditions” means for probes of at least 100 nucleotides in length, prehybridization and hybridization at 42° C. in 5 $\times$ SSPE, 0.3% SDS, 200 micrograms/mL sheared and denatured salmon sperm DNA, and 25% formamide, following standard Southern blotting procedures for 12 to 24 hours. The carrier material is finally washed three times each for 15 minutes using 0.2 $\times$ SSC, 0.2% SDS at 45° C.

(52) Xylanase: The term “xylanase” means a 1,4-beta-D-xylan-xylohydrolase (E.C. 3.2.1.8) that catalyzes the endohydrolysis of 1,4-beta-D-xylosidic linkages in xylans. Xylanase activity can be determined with 0.2% AZCL-arabinoxylan as substrate in 0.01% TRITON® X-100 and 200 mM sodium phosphate pH 6 at 37° C. One unit of xylanase activity is defined as 1.0 μ mole of azurine produced per minute at 37° C., pH 6 from 0.2% AZCL-arabinoxylan as substrate in 200 mM sodium phosphate pH 6.

(53) Xylose Isomerase: The term “Xylose Isomerase” or “XI” means an enzyme which can catalyze D-xylose into D-xylulose in vivo, and convert D-glucose into D-fructose in vitro. Xylose isomerase is also known as “glucose isomerase” and is classified as E.C. 5.3.1.5. As the structure of the enzyme is very stable, the xylose isomerase is a good model for studying the relationships between protein structure and functions (Karimaki et al., Protein Eng Des Sel, 12004, 17 (12):861-869). Xylose Isomerase activity may be determined using techniques known in the art (e.g., a coupled enzyme assay using D-sorbitol dehydrogenase, as described by Verhoeven et. al., 2017, *Sci Rep* 7, 46155).

(54) Reference to “about” a value or parameter herein includes embodiments that are directed to that value or parameter per se. For example, description referring to “about X” includes the embodiment “X”. When used in combination with measured values, “about” includes a range that encompasses at least the uncertainty associated with the method of measuring the particular value, and can include a range of plus or minus two standard deviations around the stated value.

(55) Likewise, reference to a gene or polypeptide that is “derived from” another gene or polypeptide X, includes the gene or polypeptide X.

(56) As used herein and in the appended claims, the singular forms “a,” “or,” and “the” include plural referents unless the context clearly dictates otherwise.

(57) It is understood that the embodiments described herein include “consisting” and/or “consisting essentially of” embodiments. As used herein, except where the context requires otherwise due to

express language or necessary implication, the word “comprise” or variations such as “comprises” or “comprising” is used in an inclusive sense, i.e. to specify the presence of the stated features but not to preclude the presence or addition of further features in various embodiments.

DETAILED DESCRIPTION

(58) Described herein, inter alia, are methods for producing a fermentation product, such as ethanol, from starch or cellulosic containing material.

(59) During industrial scale fermentation, yeast encounter various physiological challenges including variable concentrations of sugars, high concentrations of yeast metabolites such as ethanol, glycerol, organic acids, osmotic stress, as well as potential competition from contaminating microbes such as wild yeasts and bacteria. As a consequence, many yeasts are not suitable for use in industrial fermentation. The most widely used commercially available industrial strain of *Saccharomyces* (i.e. for industrial scale fermentation) is the *Saccharomyces cerevisiae* strain used, for example, in the product ETHANOL RED®. This strain is well suited to industrial ethanol production; however, it remains unclear how modifications to the yeast will impact performance. In particular, the functional expression of heterologous enzymes by an industrially-relevant *Saccharomyces cerevisiae* yeast is uncertain (See, for example U.S. Pat. No. 9,206,444 where the applicant was unable to functionally express numerous enzymes/enzyme classes).

(60) The Applicant has surprisingly found that yeast expressing a phospholipase provide beneficial properties that may be useful for ethanol fermentation, such as reduced need for supplemental nitrogen.

(61) In one aspect is a method of producing a fermentation product from a starch-containing or cellulosic-containing material comprising: (a) saccharifying the starch-containing or cellulosic-containing material; and (b) fermenting the saccharified material of step (a) with a fermenting organism; wherein the fermenting organism comprises a heterologous polynucleotide encoding a phospholipase.

(62) Steps of saccharifying and fermenting are carried out either sequentially or simultaneously (SSF). In one embodiment, steps of saccharifying and fermenting are carried out simultaneously (SSF). In another embodiment, steps of saccharifying and fermenting are carried out sequentially.

(63) Fermenting Organism

(64) The fermenting organism described herein may be derived from any host cell known to the skilled artisan capable of producing a fermentation product, such as ethanol. As used herein, a “derivative” of strain is derived from a referenced strain, such as through mutagenesis, recombinant DNA technology, mating, cell fusion, or cytoduction between yeast strains. Those skilled in the art will understand that the genetic alterations, including metabolic modifications exemplified herein, may be described with reference to a suitable host organism and their corresponding metabolic reactions or a suitable source organism for desired genetic material such as genes for a desired metabolic pathway. However, given the complete genome sequencing of a wide variety of organisms and the high level of skill in the area of genomics, those skilled in the art can apply the teachings and guidance provided herein to other organisms. For example, the metabolic alterations exemplified herein can readily be applied to other species by incorporating the same or analogous encoding nucleic acid from species other than the referenced species.

(65) The host cells for preparing the recombinant cells described herein can be from any suitable host, such as a yeast strain, including, but not limited to, a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell. In particular, *Saccharomyces* host cells are contemplated, such as *Saccharomyces cerevisiae*, *bayanus* or *carlsbergensis* cells. Preferably, the yeast cell is a *Saccharomyces cerevisiae* cell. Suitable cells can, for example, be derived from commercially available strains and polyploid or aneuploid industrial strains, including but not limited to those from Superstart™, THERMOSACC®, C5 FUEL™, XyloFerm®, etc. (Lallemand); RED STAR and ETHANOL RED® (Fermentis/Lesaffre); FALI (AB Mauri); Baker's Best Yeast,

Baker's Compressed Yeast, etc. (Fleishmann's Yeast); BIOFERM AFT, XP, CF, and XR (North American Bioproducts Corp.); Turbo Yeast (Gert Strand AB); and FERMIOL® (DSM Specialties). Other useful yeast strains are available from biological depositories such as the American Type Culture Collection (ATCC) or the Deutsche Sammlung von Mikroorganismen und Zellkulturen GmbH (DSMZ), such as, e.g., BY4741 (e.g., ATCC 201388); Y108-1 (ATCC PTA. 10567) and NRRL YB-1952 (ARS Culture Collection). Still other *S. cerevisiae* strains suitable as host cells DBY746, [Alpha][Eta]22, S150-2B, GPY55-15Ba, CEN.PK, USM21, TMB3500, TMB3400, VTT-A-63015, VTT-A-85068, VTT-c-79093 and their derivatives as well as *Saccharomyces* sp. 1400, 424A (LNH-ST), 259A (LNH-ST) and derivatives thereof. In one embodiment, the recombinant cell is a derivative of a strain *Saccharomyces cerevisiae* CIBTS1260 (deposited under Accession No. NRRL Y-50973 at the Agricultural Research Service Culture Collection (NRRL), Illinois 61604 U.S.A.).

(66) The fermenting organism may be *Saccharomyces* strain, e.g., *Saccharomyces cerevisiae* strain produced using the method described and concerned in U.S. Pat. No. 8,257,959-BB.

(67) The strain may also be a derivative of *Saccharomyces cerevisiae* strain NMI V14/004037 (See, WO2015/143324 and WO2015/143317 each incorporated herein by reference), strain nos. V15/004035, V15/004036, and V15/004037 (See, WO 2016/153924 incorporated herein by reference), strain nos. V15/001459, V15/001460, V15/001461 (See, WO2016/138437 incorporated herein by reference), strain no. NRRL Y67342 (See, WO2017/063159 incorporated herein by reference), or any strain described in WO2017/087330 (incorporated herein by reference).

(68) The fermenting organisms according to the invention have been generated in order to, e.g., improve fermentation yield and to improve process economy by cutting enzyme costs since part or all of the necessary enzymes needed to improve method performance are produced by the fermenting organism.

(69) The fermenting organisms described herein may utilize expression vectors comprising the coding sequence of one or more (e.g., two, several) heterologous genes linked to one or more control sequences that direct expression in a suitable cell under conditions compatible with the control sequence(s). Such expression vectors may be used in any of the cells and methods described herein. The polynucleotides described herein may be manipulated in a variety of ways to provide for expression of a desired polypeptide. Manipulation of the polynucleotide prior to its insertion into a vector may be desirable or necessary depending on the expression vector. The techniques for modifying polynucleotides utilizing recombinant DNA methods are well known in the art.

(70) A construct or vector (or multiple constructs or vectors) comprising the one or more (e.g., two, several) heterologous genes may be introduced into a cell so that the construct or vector is maintained as a chromosomal integrant or as a self-replicating extra-chromosomal vector as described earlier.

(71) The various nucleotide and control sequences may be joined together to produce a recombinant expression vector that may include one or more (e.g., two, several) convenient restriction sites to allow for insertion or substitution of the polynucleotide at such sites. Alternatively, the polynucleotide(s) may be expressed by inserting the polynucleotide(s) or a nucleic acid construct comprising the sequence into an appropriate vector for expression. In creating the expression vector, the coding sequence is located in the vector so that the coding sequence is operably linked with the appropriate control sequences for expression.

(72) The recombinant expression vector may be any vector (e.g., a plasmid or virus) that can be conveniently subjected to recombinant DNA procedures and can bring about expression of the polynucleotide. The choice of the vector will typically depend on the compatibility of the vector with the host cell into which the vector is to be introduced. The vector may be a linear or closed circular plasmid.

(73) The vector may be an autonomously replicating vector, i.e., a vector that exists as an

extrachromosomal entity, the replication of which is independent of chromosomal replication, e.g., a plasmid, an extrachromosomal element, a minichromosome, or an artificial chromosome. The vector may contain any means for assuring self-replication. Alternatively, the vector may be one that, when introduced into the host cell, is integrated into the genome and replicated together with the chromosome(s) into which it has been integrated. Furthermore, a single vector or plasmid or two or more vectors or plasmids that together contain the total DNA to be introduced into the genome of the cell, or a transposon, may be used.

(74) The expression vector may contain any suitable promoter sequence that is recognized by a cell for expression of a gene described herein. The promoter sequence contains transcriptional control sequences that mediate the expression of the polypeptide. The promoter may be any polynucleotide that shows transcriptional activity in the cell of choice including mutant, truncated, and hybrid promoters, and may be obtained from genes encoding extracellular or intracellular polypeptides either homologous or heterologous to the cell.

(75) Each heterologous polynucleotide described herein may be operably linked to a promoter that is foreign to the polynucleotide. For example, in one embodiment, the heterologous polynucleotide encoding the hexose transporter is operably linked to a promoter foreign to the polynucleotide. The promoters may be identical to or share a high degree of sequence identity (e.g., at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, or at least about 99%) with a selected native promoter.

(76) Examples of suitable promoters for directing the transcription of the nucleic acid constructs in a yeast cells, include, but are not limited to, the promoters obtained from the genes for enolase, (e.g., *S. cerevisiae* enolase or *I. orientalis* enolase (ENO1)), galactokinase (e.g., *S. cerevisiae* galactokinase or *I. orientalis* galactokinase (GAL1)), alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (e.g., *S. cerevisiae* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase or *I. orientalis* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (ADH1, ADH2/GAP)), triose phosphate isomerase (e.g., *S. cerevisiae* triose phosphate isomerase or *I. orientalis* triose phosphate isomerase (TPI)), metallothionein (e.g., *S. cerevisiae* metallothionein or *I. orientalis* metallothionein (CUP1)), 3-phosphoglycerate kinase (e.g., *S. cerevisiae* 3-phosphoglycerate kinase or *I. orientalis* 3-phosphoglycerate kinase (PGK)), PDC1, xylose reductase (XR), xylitol dehydrogenase (XDH), L-(+)-lactate-cytochrome c oxidoreductase (CYB2), translation elongation factor-1 (TEF1), translation elongation factor-2 (TEF2), glyceraldehyde-3-phosphate dehydrogenase (GAPDH), and orotidine 5'-phosphate decarboxylase (URA3) genes. Other suitable promoters may be obtained from *S. cerevisiae* TDH3, HXT7, PGK1, RPL18B and CCW12 genes. Additional useful promoters for yeast host cells are described by Romanos et al., 1992, *Yeast* 8: 423-488.

(77) The control sequence may also be a suitable transcription terminator sequence, which is recognized by a host cell to terminate transcription. The terminator sequence is operably linked to the 3'-terminus of the polynucleotide encoding the polypeptide. Any terminator that is functional in the yeast cell of choice may be used. The terminator may be identical to or share a high degree of sequence identity (e.g., at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, or at least about 99%) with the selected native terminator.

(78) Suitable terminators for yeast host cells may be obtained from the genes for enolase (e.g., *S. cerevisiae* or *I. orientalis* enolase cytochrome C (e.g., *S. cerevisiae* or *I. orientalis* cytochrome (CYC1)), glyceraldehyde-3-phosphate dehydrogenase (e.g., *S. cerevisiae* or *I. orientalis* glyceraldehyde-3-phosphate dehydrogenase (gpd)), PDC1, XR, XDH, transaldolase (TAL), transketolase (TKL), ribose 5-phosphate ketol-isomerase (RKI), CYB2, and the galactose family of genes (especially the GAL10 terminator). Other suitable terminators may be obtained from *S. cerevisiae* ENO2 or TEF1 genes. Additional useful terminators for yeast host cells are described by Romanos et al., 1992, *supra*.

(79) The control sequence may also be an mRNA stabilizer region downstream of a promoter and upstream of the coding sequence of a gene which increases expression of the gene.

(80) Examples of suitable mRNA stabilizer regions are obtained from a *Bacillus thuringiensis* cryIIIA gene (WO 94/25612) and a *Bacillus subtilis* SP82 gene (Hue et al., 1995, *Journal of Bacteriology* 177: 3465-3471).

(81) The control sequence may also be a suitable leader sequence, when transcribed is a nontranslated region of an mRNA that is important for translation by the host cell. The leader sequence is operably linked to the 5'-terminus of the polynucleotide encoding the polypeptide. Any leader sequence that is functional in the yeast cell of choice may be used.

(82) Suitable leaders for yeast host cells are obtained from the genes for enolase (e.g., *S. cerevisiae* or *I. orientalis* enolase (ENO-1)), 3-phosphoglycerate kinase (e.g., *S. cerevisiae* or *I. orientalis* 3-phosphoglycerate kinase), alpha-factor (e.g., *S. cerevisiae* or *I. orientalis* alpha-factor), and alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (e.g., *S. cerevisiae* or *I. orientalis* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (ADH2/GAP)).

(83) The control sequence may also be a polyadenylation sequence; a sequence operably linked to the 3'-terminus of the polynucleotide and, when transcribed, is recognized by the host cell as a signal to add polyadenosine residues to transcribed mRNA. Any polyadenylation sequence that is functional in the host cell of choice may be used. Useful polyadenylation sequences for yeast cells are described by Guo and Sherman, 1995, *Mol. Cellular Biol.* 15: 5983-5990.

(84) It may also be desirable to add regulatory sequences that allow the regulation of the expression of the polypeptide relative to the growth of the host cell. Examples of regulatory systems are those that cause the expression of the gene to be turned on or off in response to a chemical or physical stimulus, including the presence of a regulatory compound. Regulatory systems in prokaryotic systems include the lac, tac, and trp operator systems. In yeast, the ADH2 system or GAL1 system may be used.

(85) The vectors may contain one or more (e.g., two, several) selectable markers that permit easy selection of transformed, transfected, transduced, or the like cells. A selectable marker is a gene the product of which provides for biocide or viral resistance, resistance to heavy metals, prototrophy to auxotrophs, and the like. Suitable markers for yeast host cells include, but are not limited to, ADE2, HIS3, LEU2, LYS2, MET3, TRP1, and URA3.

(86) The vectors may contain one or more (e.g., two, several) elements that permit integration of the vector into the host cell's genome or autonomous replication of the vector in the cell independent of the genome.

(87) For integration into the host cell genome, the vector may rely on the polynucleotide's sequence encoding the polypeptide or any other element of the vector for integration into the genome by homologous or non-homologous recombination. Alternatively, the vector may contain additional polynucleotides for directing integration by homologous recombination into the genome of the host cell at a precise location(s) in the chromosome(s). To increase the likelihood of integration at a precise location, the integrational elements should contain a sufficient number of nucleic acids, such as 100 to 10,000 base pairs, 400 to 10,000 base pairs, and 800 to 10,000 base pairs, which have a high degree of sequence identity to the corresponding target sequence to enhance the probability of homologous recombination. The integrational elements may be any sequence that is homologous with the target sequence in the genome of the host cell. Furthermore, the integrational elements may be non-encoding or encoding polynucleotides. On the other hand, the vector may be integrated into the genome of the host cell by non-homologous recombination. Potential integration loci include those described in the art (e.g., See US2012/0135481).

(88) For autonomous replication, the vector may further comprise an origin of replication enabling the vector to replicate autonomously in the yeast cell. The origin of replication may be any plasmid replicator mediating autonomous replication that functions in a cell. The term "origin of replication" or "plasmid replicator" means a polynucleotide that enables a plasmid or vector to

replicate in vivo. Examples of origins of replication for use in a yeast host cell are the 2 micron origin of replication, ARS1, ARS4, the combination of ARS1 and CEN3, and the combination of ARS4 and CEN6.

(89) More than one copy of a polynucleotide described herein may be inserted into a host cell to increase production of a polypeptide. An increase in the copy number of the polynucleotide can be obtained by integrating at least one additional copy of the sequence into the yeast cell genome or by including an amplifiable selectable marker gene with the polynucleotide where cells containing amplified copies of the selectable marker gene, and thereby additional copies of the polynucleotide, can be selected for by cultivating the cells in the presence of the appropriate selectable agent.

(90) The procedures used to ligate the elements described above to construct the recombinant expression vectors described herein are well known to one skilled in the art (see, e.g., Sambrook et al., 1989, *Molecular Cloning, A Laboratory Manual*, 2d edition, Cold Spring Harbor, New York).

(91) Additional procedures and techniques known in the art for the preparation of recombinant cells for ethanol fermentation, are described in, e.g., WO 2016/045569, the content of which is hereby incorporated by reference.

(92) The fermenting organism may be in the form of a composition comprising a fermenting organism (e.g., a yeast strain described herein) and a naturally occurring and/or a none naturally occurring component.

(93) The fermenting organism described herein may be in any viable form, including crumbled, dry, including active dry and instant, compressed, cream (liquid) form etc. In one embodiment, the fermenting organism (e.g., a *Saccharomyces cerevisiae* yeast strain) is dry yeast, such as active dry yeast or instant yeast. In one embodiment, the fermenting organism (e.g., a *Saccharomyces cerevisiae* yeast strain) is crumbled yeast. In one embodiment, the fermenting organism (e.g., a *Saccharomyces cerevisiae* yeast strain) is compressed yeast. In one embodiment, the fermenting organism (e.g., a *Saccharomyces cerevisiae* yeast strain) is cream yeast.

(94) In one embodiment is a composition comprising a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain), and one or more of the component selected from the group consisting of: surfactants, emulsifiers, gums, swelling agent, and antioxidants and other processing aids.

(95) The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable surfactants. In one embodiment, the surfactant(s) is/are an anionic surfactant, cationic surfactant, and/or nonionic surfactant.

(96) The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable emulsifier. In one embodiment, the emulsifier is a fatty-acid ester of sorbitan. In one embodiment, the emulsifier is selected from the group of sorbitan monostearate (SMS), citric acid esters of monodiglycerides, polyglycerolester, fatty acid esters of propylene glycol.

(97) In one embodiment, the composition comprises a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain), and Olindronal SMS, Olindronal SK, or Olindronal SPL including composition concerned in European Patent No. 1,724,336 (hereby incorporated by reference). These products are commercially available from Bussetti, Austria, for active dry yeast.

(98) The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable gum. In one embodiment, the gum is selected from the group of carob, guar, tragacanth, arabic, xanthan and acacia gum, in particular for cream, compressed and dry yeast.

(99) The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable swelling agent. In one embodiment, the swelling agent is methyl cellulose or carboxymethyl cellulose.

(100) The compositions described herein may comprise a fermenting organism described herein (e.g., a *Saccharomyces cerevisiae* yeast strain) and any suitable anti-oxidant. In one embodiment,

the antioxidant is butylated hydroxyanisol (BHA) and/or butylated hydroxytoluene (BHT), or ascorbic acid (vitamin C), particular for active dry yeast.

(101) Gene Disruptions

(102) The fermenting organisms described herein may also comprise one or more (e.g., two, several) gene disruptions, e.g., to divert sugar metabolism from undesired products to ethanol. In some aspects, the recombinant host cells produce a greater amount of ethanol compared to the cell without the one or more disruptions when cultivated under identical conditions. In some aspects, one or more of the disrupted endogenous genes is inactivated.

(103) In certain embodiments, the fermenting organism provided herein comprises a disruption of one or more endogenous genes encoding enzymes involved in producing alternate fermentative products such as glycerol or other byproducts such as acetate or diols. For example, the cells provided herein may comprise a disruption of one or more of glycerol 3-phosphate dehydrogenase (GPD, catalyzes reaction of dihydroxyacetone phosphate to glycerol 3-phosphate), glycerol 3-phosphatase (GPP, catalyzes conversion of glycerol-3 phosphate to glycerol), glycerol kinase (catalyzes conversion of glycerol 3-phosphate to glycerol), dihydroxyacetone kinase (catalyzes conversion of dihydroxyacetone phosphate to dihydroxyacetone), glycerol dehydrogenase (catalyzes conversion of dihydroxyacetone to glycerol), and aldehyde dehydrogenase (ALD, e.g., converts acetaldehyde to acetate).

(104) Modeling analysis can be used to design gene disruptions that additionally optimize utilization of the pathway. One exemplary computational method for identifying and designing metabolic alterations favoring biosynthesis of a desired product is the OptKnock computational framework, Burgard et al., 2003, *Biotechnol. Bioeng.* 84: 647-657.

(105) The fermenting organisms comprising a gene disruption may be constructed using methods well known in the art, including those methods described herein. A portion of the gene can be disrupted such as the coding region or a control sequence required for expression of the coding region. Such a control sequence of the gene may be a promoter sequence or a functional part thereof, i.e., a part that is sufficient for affecting expression of the gene. For example, a promoter sequence may be inactivated resulting in no expression or a weaker promoter may be substituted for the native promoter sequence to reduce expression of the coding sequence. Other control sequences for possible modification include, but are not limited to, a leader, propeptide sequence, signal sequence, transcription terminator, and transcriptional activator.

(106) The fermenting organisms comprising a gene disruption may be constructed by gene deletion techniques to eliminate or reduce expression of the gene. Gene deletion techniques enable the partial or complete removal of the gene thereby eliminating their expression. In such methods, deletion of the gene is accomplished by homologous recombination using a plasmid that has been constructed to contiguously contain the 5' and 3' regions flanking the gene.

(107) The fermenting organisms comprising a gene disruption may also be constructed by introducing, substituting, and/or removing one or more (e.g., two, several) nucleotides in the gene or a control sequence thereof required for the transcription or translation thereof. For example, nucleotides may be inserted or removed for the introduction of a stop codon, the removal of the start codon, or a frame-shift of the open reading frame. Such a modification may be accomplished by site-directed mutagenesis or PCR generated mutagenesis in accordance with methods known in the art. See, for example, Botstein and Shortle, 1985, *Science* 229: 4719; Lo et al., 1985, *Proc. Natl. Acad. Sci. U.S.A.* 81: 2285; Higuchi et al., 1988, *Nucleic Acids Res* 16: 7351; Shimada, 1996, *Meth. Mol. Biol.* 57: 157; Ho et al., 1989, *Gene* 77: 61; Horton et al., 1989, *Gene* 77: 61; and Sarkar and Sommer, 1990, *BioTechniques* 8: 404.

(108) The fermenting organisms comprising a gene disruption may also be constructed by inserting into the gene a disruptive nucleic acid construct comprising a nucleic acid fragment homologous to the gene that will create a duplication of the region of homology and incorporate construct DNA between the duplicated regions. Such a gene disruption can eliminate gene expression if the

inserted construct separates the promoter of the gene from the coding region or interrupts the coding sequence such that a non-functional gene product results. A disrupting construct may be simply a selectable marker gene accompanied by 5' and 3' regions homologous to the gene. The selectable marker enables identification of transformants containing the disrupted gene.

(109) The fermenting organisms comprising a gene disruption may also be constructed by the process of gene conversion (see, for example, Iglesias and Trautner, 1983, *Molecular General Genetics* 189: 73-76). For example, in the gene conversion method, a nucleotide sequence corresponding to the gene is mutagenized in vitro to produce a defective nucleotide sequence, which is then transformed into the recombinant strain to produce a defective gene. By homologous recombination, the defective nucleotide sequence replaces the endogenous gene. It may be desirable that the defective nucleotide sequence also comprises a marker for selection of transformants containing the defective gene.

(110) The fermenting organisms comprising a gene disruption may be further constructed by random or specific mutagenesis using methods well known in the art, including, but not limited to, chemical mutagenesis (see, for example, Hopwood, *The Isolation of Mutants in Methods in Microbiology* (J. R. Norris and D. W. Ribbons, eds.) pp. 363-433, Academic Press, New York, 1970). Modification of the gene may be performed by subjecting the parent strain to mutagenesis and screening for mutant strains in which expression of the gene has been reduced or inactivated. The mutagenesis, which may be specific or random, may be performed, for example, by use of a suitable physical or chemical mutagenizing agent, use of a suitable oligonucleotide, or subjecting the DNA sequence to PCR generated mutagenesis. Furthermore, the mutagenesis may be performed by use of any combination of these mutagenizing methods.

(111) Examples of a physical or chemical mutagenizing agent suitable for the present purpose include ultraviolet (UV) irradiation, hydroxylamine, N-methyl-N'-nitro-N-nitrosoguanidine (MNNG), N-methyl-N'-nitrosoguanidine (NTG) O-methyl hydroxylamine, nitrous acid, ethyl methane sulphonate (EMS), sodium bisulphite, formic acid, and nucleotide analogues. When such agents are used, the mutagenesis is typically performed by incubating the parent strain to be mutagenized in the presence of the mutagenizing agent of choice under suitable conditions, and selecting for mutants exhibiting reduced or no expression of the gene.

(112) A nucleotide sequence homologous or complementary to a gene described herein may be used from other microbial sources to disrupt the corresponding gene in a recombinant strain of choice.

(113) In one aspect, the modification of a gene in the recombinant cell is unmarked with a selectable marker. Removal of the selectable marker gene may be accomplished by culturing the mutants on a counter-selection medium. Where the selectable marker gene contains repeats flanking its 5' and 3' ends, the repeats will facilitate the looping out of the selectable marker gene by homologous recombination when the mutant strain is submitted to counter-selection. The selectable marker gene may also be removed by homologous recombination by introducing into the mutant strain a nucleic acid fragment comprising 5' and 3' regions of the defective gene, but lacking the selectable marker gene, followed by selecting on the counter-selection medium. By homologous recombination, the defective gene containing the selectable marker gene is replaced with the nucleic acid fragment lacking the selectable marker gene. Other methods known in the art may also be used.

(114) Methods Using a Starch-Containing Material

(115) In some aspects, the methods described herein produce a fermentation product from a starch-containing material. Starch-containing material is well-known in the art, containing two types of homopolysaccharides (amylose and amylopectin) and is linked by alpha-(1-4)-D-glycosidic bonds. Any suitable starch-containing starting material may be used. The starting material is generally selected based on the desired fermentation product, such as ethanol. Examples of starch-containing starting materials include cereal, tubers or grains. Specifically, the starch-containing material may

be corn, wheat, barley, rye, milo, sago, cassava, tapioca, sorghum, oat, rice, peas, beans, or sweet potatoes, or mixtures thereof. Contemplated are also waxy and non-waxy types of corn and barley. (116) In one embodiment, the starch-containing starting material is corn. In one embodiment, the starch-containing starting material is wheat. In one embodiment, the starch-containing starting material is barley. In one embodiment, the starch-containing starting material is rye. In one embodiment, the starch-containing starting material is milo. In one embodiment, the starch-containing starting material is sago. In one embodiment, the starch-containing starting material is cassava. In one embodiment, the starch-containing starting material is tapioca. In one embodiment, the starch-containing starting material is sorghum. In one embodiment, the starch-containing starting material is rice. In one embodiment, the starch-containing starting material is peas. In one embodiment, the starch-containing starting material is beans. In one embodiment, the starch-containing starting material is sweet potatoes. In one embodiment, the starch-containing starting material is oats.

(117) The methods using a starch-containing material may include a conventional process (e.g., including a liquefaction step described in more detail below) or a raw starch hydrolysis process. In some embodiments using a starch-containing material, saccharification of the starch-containing material is at a temperature above the initial gelatinization temperature. In some embodiments using a starch-containing material, saccharification of the starch-containing material is at a temperature below the initial gelatinization temperature.

(118) Liquefaction

(119) In aspects using a starch-containing material, the methods may further comprise a liquefaction step carried out by subjecting the starch-containing material at a temperature above the initial gelatinization temperature to an alpha-amylase and optionally a protease and/or a glucoamylase. Other enzymes such as a pullulanase and phytase may also be present and/or added in liquefaction. In some embodiments, the liquefaction step is carried out prior to steps a) and b) of the described methods.

(120) Liquefaction step may be carried out for 0.5-5 hours, such as 1-3 hours, such as typically about 2 hours.

(121) The term “initial gelatinization temperature” means the lowest temperature at which gelatinization of the starch-containing material commences. In general, starch heated in water begins to gelatinize between about 50° C. and 75° C.; the exact temperature of gelatinization depends on the specific starch and can readily be determined by the skilled artisan. Thus, the initial gelatinization temperature may vary according to the plant species, to the particular variety of the plant species as well as with the growth conditions. The initial gelatinization temperature of a given starch-containing material may be determined as the temperature at which birefringence is lost in 5% of the starch granules using the method described by Gorinstein and Lii, 1992, *Starch/Stärke* 44(12): 461-466.

(122) Liquefaction is typically carried out at a temperature in the range from 70-100° C. In one embodiment, the temperature in liquefaction is between 75-95° C., such as between 75-90° C., between 80-90° C., or between 82-88° C., such as about 85° C.

(123) A jet-cooking step may be carried out prior to liquefaction in step, for example, at a temperature between 110-145° C., 120-140° C., 125-135° C., or about 130° C. for about 1-15 minutes, for about 3-10 minutes, or about 5 minutes.

(124) The pH during liquefaction may be between 4 and 7, such as pH 4.5-6.5, pH 5.0-6.5, pH 5.0-6.0, pH 5.2-6.2, or about 5.2, about 5.4, about 5.6, or about 5.8.

(125) In one embodiment, the process further comprises, prior to liquefaction, the steps of: i) reducing the particle size of the starch-containing material, preferably by dry milling; ii) forming a slurry comprising the starch-containing material and water.

(126) The starch-containing starting material, such as whole grains, may be reduced in particle size, e.g., by milling, in order to open up the structure, to increase surface area, and allowing for further

processing. Generally, there are two types of processes: wet and dry milling. In dry milling whole kernels are milled and used. Wet milling gives a good separation of germ and meal (starch granules and protein). Wet milling is often applied at locations where the starch hydrolysate is used in production of, e.g., syrups. Both dry milling and wet milling are well known in the art of starch processing. In one embodiment the starch-containing material is subjected to dry milling. In one embodiment, the particle size is reduced to between 0.05 to 3.0 mm, e.g., 0.1-0.5 mm, or so that at least 30%, at least 50%, at least 70%, or at least 90% of the starch-containing material fit through a sieve with a 0.05 to 3.0 mm screen, e.g., 0.1-0.5 mm screen. In another embodiment, at least 50%, e.g., at least 70%, at least 80%, or at least 90% of the starch-containing material fit through a sieve with #6 screen.

(127) The aqueous slurry may contain from 10-55 w/w-% dry solids (DS), e.g., 25-45 w/w-% dry solids (DS), or 30-40 w/w-% dry solids (DS) of starch-containing material.

(128) The alpha-amylase, optionally a protease, and optionally a glucoamylase may initially be added to the aqueous slurry to initiate liquefaction (thinning). In one embodiment, only a portion of the enzymes (e.g., about $\frac{1}{3}$) is added to the aqueous slurry, while the rest of the enzymes (e.g., about $\frac{2}{3}$) are added during liquefaction step.

(129) A non-exhaustive list of alpha-amylases used in liquefaction can be found below in the "Alpha-Amylases" section. Examples of suitable proteases used in liquefaction include any protease described supra in the "Proteases" section. Examples of suitable glucoamylases used in liquefaction include any glucoamylase found in the "Glucoamylases" section.

(130) Saccharification and Fermentation of Starch-Containing Material

(131) In aspects using a starch-containing material, a glucoamylase may be present and/or added in saccharification step a) and/or fermentation step b) or simultaneous saccharification and fermentation (SSF). The glucoamylase of the saccharification step a) and/or fermentation step b) or simultaneous saccharification and fermentation (SSF) is typically different from the glucoamylase optionally added to any liquefaction step described supra. In one embodiment, the glucoamylase is present and/or added together with a fungal alpha-amylase.

(132) In some aspects, the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase, for example, as described in WO2017/087330, the content of which is hereby incorporated by reference.

(133) Examples of glucoamylases can be found in the "Glucoamylases" section below.

(134) When doing sequential saccharification and fermentation, saccharification step a) may be carried out under conditions well-known in the art. For instance, saccharification step a) may last up to from about 24 to about 72 hours. In one embodiment, pre-saccharification is done. Pre-saccharification is typically done for 40-90 minutes at a temperature between 30-65° C., typically about 60° C. Pre-saccharification is, in one embodiment, followed by saccharification during fermentation in simultaneous saccharification and fermentation (SSF). Saccharification is typically carried out at temperatures from 20-75° C., preferably from 40-70° C., typically about 60° C., and typically at a pH between 4 and 5, such as about pH 4.5.

(135) Fermentation is carried out in a fermentation medium, as known in the art and, e.g., as described herein. The fermentation medium includes the fermentation substrate, that is, the carbohydrate source that is metabolized by the fermenting organism. With the processes described herein, the fermentation medium may comprise nutrients and growth stimulator(s) for the fermenting organism(s). Nutrient and growth stimulators are widely used in the art of fermentation and include nitrogen sources, such as ammonia; urea, vitamins and minerals, or combinations thereof.

(136) Generally, fermenting organisms such as yeast, including *Saccharomyces cerevisiae* yeast, require an adequate source of nitrogen for propagation and fermentation. Many sources of supplemental nitrogen, if necessary, can be used and such sources of nitrogen are well known in the art. The nitrogen source may be organic, such as urea, DDGs, wet cake or corn mash, or inorganic,

such as ammonia or ammonium hydroxide. In one embodiment, the nitrogen source is urea.

(137) Fermentation can be carried out under low nitrogen conditions, e.g., when using a protease-expressing yeast. In some embodiments, the fermentation step is conducted with less than 1000 ppm supplemental nitrogen (e.g., urea or ammonium hydroxide), such as less than 750 ppm, less than 500 ppm, less than 400 ppm, less than 300 ppm, less than 250 ppm, less than 200 ppm, less than 150 ppm, less than 100 ppm, less than 75 ppm, less than 50 ppm, less than 25 ppm, or less than 10 ppm, supplemental nitrogen. In some embodiments, the fermentation step is conducted with no supplemental nitrogen.

(138) Simultaneous saccharification and fermentation (“SSF”) is widely used in industrial scale fermentation product production processes, especially ethanol production processes. When doing SSF the saccharification step a) and the fermentation step b) are carried out simultaneously. There is no holding stage for the saccharification, meaning that a fermenting organism, such as yeast, and enzyme(s), may be added together. However, it is also contemplated to add the fermenting organism and enzyme(s) separately. SSF is typically carried out at a temperature from 25° C. to 40° C., such as from 28° C. to 35° C., such as from 30° C. to 34° C., or about 32° C. In one embodiment, fermentation is ongoing for 6 to 120 hours, in particular 24 to 96 hours. In one embodiment, the pH is between 4-5.

(139) In one embodiment, a cellulolytic enzyme composition is present and/or added in saccharification, fermentation or simultaneous saccharification and fermentation (SSF). Examples of such cellulolytic enzyme compositions can be found in the “Cellulolytic Enzymes and Compositions” section below. The cellulolytic enzyme composition may be present and/or added together with a glucoamylase, such as one disclosed in the “Glucoamylases” section below.

(140) Phospholipases

(141) The expressed phospholipase may be any phospholipase that is suitable for the host cells and/or the methods described herein, such as a naturally occurring phospholipase (e.g., a native phospholipase from another species or an endogenous phospholipase expressed from a modified expression vector) or a variant thereof that retains phospholipase activity.

(142) In some embodiments, the fermenting organism comprises a heterologous polynucleotide encoding a phospholipase, for example, as described in WO2018/075430, the content of which is hereby incorporated by reference. In some embodiments, the phospholipase is classified as a phospholipase A. In other embodiments, the phospholipase is classified as a phospholipase C. Any phospholipase described or referenced herein is contemplated for expression in the fermenting organism.

(143) In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a phospholipase has an increased level of phospholipase activity compared to the host cells without the heterologous polynucleotide encoding the phospholipase, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of phospholipase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the phospholipase, when cultivated under the same conditions.

(144) Exemplary phospholipases that can be used with the host cells and/or the methods described herein include bacterial, yeast, or filamentous fungal phospholipases, e.g., derived from any of the microorganisms described or referenced herein.

(145) Additional phospholipases that may be expressed with the fermenting organisms and used with the methods described herein are described in the examples, and include, but are not limited to phospholipases shown in Table 1 (or derivatives thereof).

(146) TABLE-US-00001 TABLE 1 Donor Organism SEQ ID NO: (catalytic domain) (mature polypeptide) *Thermomyces lanuginosus* 235 *Talaromyces leycettanus* 236 *Penicillium emersonii* 237 *Bacillus thuringiensis* 238 *Pseudomonas* sp. 239 *Kionochaeta* sp. 240 *Mariannaea pinicola*

241 *Fictibacillus macauensis* 242 *Aspergillus wentii* 252 *Penicillium cylindrosporum* 253
Penicillium meridianum 254 *Penicillium bialowiezense* 255 *Penicillium sclerotiorum* 256
Rasamsonia byssochlamydoides 257 *Rasamsonia eburnea* 258 *Penicillium brefeldianum* 259
Penicillium adametzii 260 *Rasamsonia brevistipitata* 261 *Penicillium scabrosum* 262 *Penicillium*
manginii 263 *Penicillium emersonii* 264 *Rasamsonia argillacea* 265 *Penicillium parviverrucosum*
266 *Penicillium flavescens* 267 *Penicillium hispanicum* 268 *Penicillium simplicissimum* 269
Penicillium vasconiae 270 *Talaromyces columbinus* 271 *Talaromyces variabilis* 272 *Talaromyces*
rugulosus 273 *Hamigera terricola* 274 *Penicillium piscarium* 275 *Talaromyces bacillisporus* 276
Galactomyces candidus 277 *Penicillium megasporum* 278 *Penicillium jensenii* 279 *Aspergillus*
stramenius 280 *Bacillus pseudomycoides* 281 *Bacillus mycoides* 282 *Bacillus thuringiensis* 283
Listeria innocua 284 *Aspergillus egyptiacus* 285 *Aspergillus tamarii* 286 *Aspergillus niger* 287
Bacillus luciferensis 288 *Bacillus mycoides* 289 *Bacillus mycoides* 290 *Bacillus* sp. 291 *Bacillus*
drentensis 292 *Aspergillus turcosus* 293 *Talaromyces subinflatus* 294 *Aspergillus tubingensis* 295
Bacillus acidicer 296 *Lysinibacillus xylanilyticus* 297 *Bacillus toyonensis* 298 *Bacillus*
wiedmannii 299 *Listeria seeligeri* 300 *Penicillium swiecickii* 301 *Talaromyces boninensis* 302
Hamigera striata 303 *Bacillus* sp. 304 *Bacillus thuringiensis* 305 *Bacillus mycoides* 306
Fictibacillus macauensis 307 *Listeria seeligeri* 308 *Penicillium donkii* 309 *Hamigera paravellanea*
310 *Talaromyces leycettanus* 311 *Paenibacillus* sp. 312 *Bacillus toyonensis* 313 *Bacillus*
thuringiensis 314 *Bacillus thuringiensis* 315 *Talaromyces rugulosus* 316 *Penicillium* sp. 317
Hamigera avellanea 318 *Penicillium spikei* 319 *Paenibacillus alginolyticus* 320 *Bacillus mycoides*
321 *Bacillus bingmayongensis* 322 *Bacillus mycoides* 323 *Brevibacillus* sp. 324 *Penicillium*
vasconiae 325 *Talaromyces diversus* 326 *Aspergillus wentii* 327 *Bacillus acidicer* 328 *Bacillus*
luti 329 *Bacillus pseudomycoides* 330 *Bacillus mycoides* 331 *Penicillium cinnamopurpureum* 332
Talaromyces verruculosus 333 *Talaromyces cellulolyticus* 334 *Penicillium megasporum* 335
Bacillus toyonensis 336 *Bacillus* sp. 337 *Bacillus manliponensis* 338 *Penicillium simplicissimum*
339 *Penicillium arenicola* 340 *Aspergillus aculeatus* 341 *Bacillus acidicer* 342

(147) Additional phospholipases contemplated for use with the present invention can be found in WO2018/075430 (the content of which is incorporated herein).

(148) Additional polynucleotides encoding suitable phospholipases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org).

(149) The phospholipase may be a bacterial phospholipase. For example, the phospholipase may be derived from a Gram-positive bacterium such as a *Bacillus*, *Clostridium*, *Enterococcus*, *Geobacillus*, *Lactobacillus*, *Lactococcus*, *Oceanobacillus*, *Staphylococcus*, *Streptococcus*, or *Streptomyces*, or a Gram-negative bacterium such as a *Campylobacter*, *E. coli*, *Flavobacterium*, *Fusobacterium*, *Helicobacter*, *Ilyobacter*, *Neisseria*, *Pseudomonas*, *Salmonella*, or *Ureaplasma*.

(150) In one embodiment, the phospholipase is derived from *Bacillus alkalophilus*, *Bacillus amyloliquefaciens*, *Bacillus brevis*, *Bacillus circulans*, *Bacillus clausii*, *Bacillus coagulans*, *Bacillus firmus*, *Bacillus lautus*, *Bacillus lentus*, *Bacillus licheniformis*, *Bacillus megaterium*, *Bacillus pumilus*, *Bacillus stearothermophilus*, *Bacillus subtilis*, or *Bacillus thuringiensis*.

(151) In another embodiment, the phospholipase is derived from *Streptococcus equisimilis*, *Streptococcus pyogenes*, *Streptococcus uberis*, or *Streptococcus equi* subsp. *Zooepidemicus*.

(152) In another embodiment, the phospholipase is derived from *Streptomyces achromogenes*, *Streptomyces avermitilis*, *Streptomyces coelicolor*, *Streptomyces griseus*, or *Streptomyces lividans*.

(153) The phospholipase may be a fungal phospholipase. For example, the phospholipase may be derived from a yeast such as a *Candida*, *Kluyveromyces*, *Pichia*, *Saccharomyces*, *Schizosaccharomyces*, *Yarrowia* or *Issatchenkia*; or derived from a filamentous fungus such as an *Acremonium*, *Agaricus*, *Alternaria*, *Aspergillus*, *Aureobasidium*, *Botryosphaeria*, *Ceriporiopsis*, *Chaetomidium*, *Chrysosporium*, *Claviceps*, *Cochliobolus*, *Coprinopsis*, *Coptotermes*, *Corynascus*, *Cryphonectria*, *Cryptococcus*, *Diplodia*, *Exidia*, *Filibasidium*, *Fusarium*, *Gibberella*,

Holomastigotoides, *Humicola*, *Irpex*, *Lentinula*, *Leptosphaeria*, *Magnaporthe*, *Melanocarpus*, *Meripilus*, *Mucor*, *Myceliophthora*, *Neocallimastix*, *Neurospora*, *Paecilomyces*, *Penicillium*, *Phanerochaete*, *Piromyces*, *Poitrasia*, *Pseudoplectania*, *Pseudotriconympha*, *Rhizomucor*, *Schizophyllum*, *Scytalidium*, *Talaromyces*, *Thermoascus*, *Thielavia*, *Tolypocladium*, *Trichoderma*, *Trichophaea*, *Verticillium*, *Volvariella*, or *Xylaria*.

(154) In another embodiment, the phospholipase is derived from *Saccharomyces carlsbergensis*, *Saccharomyces cerevisiae*, *Saccharomyces diastaticus*, *Saccharomyces douglasii*, *Saccharomyces kluyveri*, *Saccharomyces norbensis*, or *Saccharomyces oviformis*.

(155) In another embodiment, the phospholipase is derived from *Acremonium cellulolyticus*, *Aspergillus aculeatus*, *Aspergillus awamori*, *Aspergillus foetidus*, *Aspergillus fumigatus*, *Aspergillus japonicus*, *Aspergillus nidulans*, *Aspergillus niger*, *Aspergillus oryzae*, *Chrysosporium inops*, *Chrysosporium keratinophilum*, *Chrysosporium lucknowense*, *Chrysosporium merdarium*, *Chrysosporium pannicola*, *Chrysosporium queenslandicum*, *Chrysosporium tropicum*, *Chrysosporium zonatum*, *Fusarium bactridioides*, *Fusarium cerealis*, *Fusarium crookwellense*, *Fusarium culmorum*, *Fusarium graminearum*, *Fusarium graminum*, *Fusarium heterosporum*, *Fusarium negundi*, *Fusarium oxysporum*, *Fusarium reticulatum*, *Fusarium roseum*, *Fusarium sambucinum*, *Fusarium sarcochromum*, *Fusarium sporotrichioides*, *Fusarium sulphureum*, *Fusarium torulosum*, *Fusarium trichothecioides*, *Fusarium venenatum*, *Humicola grisea*, *Humicola insolens*, *Humicola lanuginosa*, *Irpex lacteus*, *Mucor miehei*, *Myceliophthora thermophila*, *Neurospora crassa*, *Penicillium funiculosum*, *Penicillium purpurogenum*, *Phanerochaete chrysosporium*, *Thielavia achromatica*, *Thielavia albomyces*, *Thielavia albopilosa*, *Thielavia australeinsis*, *Thielavia fimeti*, *Thielavia microspora*, *Thielavia ovispora*, *Thielavia peruviana*, *Thielavia setosa*, *Thielavia spededonium*, *Thielavia subthermophila*, *Thielavia terrestris*, *Trichoderma harzianum*, *Trichoderma koningii*, *Trichoderma longibrachiatum*, *Trichoderma reesei*, or *Trichoderma viride*.

(156) It will be understood that for the aforementioned species, the invention encompasses both the perfect and imperfect states, and other taxonomic equivalents, e.g., anamorphs, regardless of the species name by which they are known. Those skilled in the art will readily recognize the identity of appropriate equivalents.

(157) Strains of these species are readily accessible to the public in a number of culture collections, such as the American Type Culture Collection (ATCC), Deutsche Sammlung von Mikroorganismen und Zellkulturen GmbH (DSMZ), Centraalbureau Voor Schimmelcultures (CBS), and Agricultural Research Service Patent Culture Collection, Northern Regional Research Center (NRRL).

(158) The phospholipase coding sequences described or referenced herein, or a subsequence thereof, as well as the phospholipases described or referenced herein, or a fragment thereof, may be used to design nucleic acid probes to identify and clone DNA encoding a phospholipase from strains of different genera or species according to methods well known in the art. In particular, such probes can be used for hybridization with the genomic DNA or cDNA of a cell of interest, following standard Southern blotting procedures, in order to identify and isolate the corresponding gene therein. Such probes can be considerably shorter than the entire sequence, but should be at least 15, e.g., at least 25, at least 35, or at least 70 nucleotides in length. Preferably, the nucleic acid probe is at least 100 nucleotides in length, e.g., at least 200 nucleotides, at least 300 nucleotides, at least 400 nucleotides, at least 500 nucleotides, at least 600 nucleotides, at least 700 nucleotides, at least 800 nucleotides, or at least 900 nucleotides in length. Both DNA and RNA probes can be used. The probes are typically labeled for detecting the corresponding gene (for example, with .sup.32P, .sup.3H, .sup.35S, biotin, or avidin).

(159) A genomic DNA or cDNA library prepared from such other strains may be screened for DNA that hybridizes with the probes described above and encodes a parent. Genomic or other DNA from such other strains may be separated by agarose or polyacrylamide gel electrophoresis, or other separation techniques. DNA from the libraries or the separated DNA may be transferred to and immobilized on nitrocellulose or other suitable carrier material. In order to identify a clone or DNA

that hybridizes with a coding sequence, or a subsequence thereof, the carrier material is used in a Southern blot.

(160) In one embodiment, the nucleic acid probe is a polynucleotide, or subsequence thereof, that encodes the phospholipase of any one of SEQ ID NOs: 235-242 and 252-342 (such as the coding sequence of SEQ ID NOs: 244-251 and 343-433, respectively), or a fragment thereof.

(161) In one embodiment, the nucleic acid probe is a polynucleotide, or subsequence thereof, that encodes the phospholipase of any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242 (such as the coding sequence of SEQ ID NO: 244, 245, 246, 247, 248, 249, 250 or 251, respectively), or a fragment thereof.

(162) For purposes of the probes described above, hybridization indicates that the polynucleotide hybridizes to a labeled nucleic acid probe, or the full-length complementary strand thereof, or a subsequence of the foregoing; under very low to very high stringency conditions. Molecules to which the nucleic acid probe hybridizes under these conditions can be detected using, for example, X-ray film. Stringency and washing conditions are defined as described supra.

(163) In one embodiment, the phospholipase is encoded by a polynucleotide that hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence for any one of the phospholipases described or referenced herein (e.g., the coding sequence that encodes any one of SEQ ID NOs: 235-242 and 252-342; such as the corresponding coding sequence of SEQ ID NO: 244-251 or 343-433, respectively, or the coding sequence that encodes any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242; such as the corresponding coding sequence of SEQ ID NO: 244, 245, 246, 247, 248, 249, 250 or 251, respectively). (Sambrook et al., 1989, *Molecular Cloning, A Laboratory Manual*, 2d edition, Cold Spring Harbor, New York).

(164) The phospholipase may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, silage, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, silage, etc.) using the above-mentioned probes. Techniques for isolating microorganisms and DNA directly from natural habitats are well known in the art. The polynucleotide encoding a phospholipase may then be derived by similarly screening a genomic or cDNA library of another microorganism or mixed DNA sample.

(165) Once a polynucleotide encoding a phospholipase has been detected with a suitable probe as described herein, the sequence may be isolated or cloned by utilizing techniques that are known to those of ordinary skill in the art (see, e.g., Sambrook et al., 1989, *Molecular Cloning, A Laboratory Manual*, 2d edition, Cold Spring Harbor, New York). Techniques used to isolate or clone polynucleotides encoding alpha-amylases include isolation from genomic DNA, preparation from cDNA, or a combination thereof. The cloning of the polynucleotides from such genomic DNA can be effected, e.g., by using the well-known polymerase chain reaction (PCR) or antibody screening of expression libraries to detect cloned DNA fragments with shares structural features. See, e.g., Innis et al., 1990, *PCR: A Guide to Methods and Application*, Academic Press, New York. Other nucleic acid amplification procedures such as ligase chain reaction (LCR), ligated activated transcription (LAT) and nucleotide sequence-based amplification (NASBA) may be used.

(166) In one embodiment, the phospholipase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of the phospholipases described or referenced herein (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). In another embodiment, the phospholipase has a mature polypeptide sequence that is a fragment of the any one of the phospholipases described or referenced herein (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length phospholipase (e.g., any one of SEQ ID NOs: 235-242

and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). In other embodiments, the phospholipase may comprise the catalytic domain of any phospholipase described or referenced herein (e.g., the catalytic domain of any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242).

(167) The phospholipase may be a variant of any one of the phospholipases described supra (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). In one embodiment, the phospholipase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the phospholipases described supra (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242).

(168) In one embodiment, the phospholipase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the phospholipases described supra (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). In one embodiment, the phospholipase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the phospholipases described supra (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

(169) The amino acid changes are generally of a minor nature, that is conservative amino acid substitutions or insertions that do not significantly affect the folding and/or activity of the protein; small deletions, typically of one to about 30 amino acids; small amino-terminal or carboxyl-terminal extensions, such as an amino-terminal methionine residue; a small linker peptide of up to about 20-25 residues; or a small extension that facilitates purification by changing net charge or another function, such as a poly-histidine tract, an antigenic epitope or a binding domain.

(170) Examples of conservative substitutions are within the group of basic amino acids (arginine, lysine and histidine), acidic amino acids (glutamic acid and aspartic acid), polar amino acids (glutamine and asparagine), hydrophobic amino acids (leucine, isoleucine and valine), aromatic amino acids (phenylalanine, tryptophan and tyrosine), and small amino acids (glycine, alanine, serine, threonine and methionine). Amino acid substitutions that do not generally alter specific activity are known in the art and are described, for example, by H. Neurath and R. L. Hill, 1979, In, *The Proteins*, Academic Press, New York. The most commonly occurring exchanges are Ala/Ser, Val/Ile, Asp/Glu, Thr/Ser, Ala/Gly, Ala/Thr, Ser/Asn, Ala/Val, Ser/Gly, Tyr/Phe, Ala/Pro, Lys/Arg, Asp/Asn, Leu/Ile, Leu/Val, Ala/Glu, and Asp/Gly.

(171) Alternatively, the amino acid changes are of such a nature that the physico-chemical properties of the polypeptides are altered. For example, amino acid changes may improve the thermal stability of the phospholipase, alter the substrate specificity, change the pH optimum, and the like.

(172) Essential amino acids can be identified according to procedures known in the art, such as site-directed mutagenesis or alanine-scanning mutagenesis (Cunningham and Wells, 1989, *Science* 244: 1081-1085). In the latter technique, single alanine mutations are introduced at every residue in the molecule, and the resultant mutant molecules are tested for activity to identify amino acid residues that are critical to the activity of the molecule. See also, Hilton et al., 1996, *J. Biol. Chem.* 271: 4699-4708. The active site or other biological interaction can also be determined by physical analysis of structure, as determined by such techniques as nuclear magnetic resonance, crystallography, electron diffraction, or photoaffinity labeling, in conjunction with mutation of putative contact site amino acids. See, for example, de Vos et al., 1992, *Science* 255: 306-312;

Smith et al., 1992, *J. Mol. Biol.* 224: 899-904; Wlodaver et al., 1992, *FEBS Lett.* 309: 59-64. The identities of essential amino acids can also be inferred from analysis of identities with other phospholipases that are related to the referenced phospholipase.

(173) Additional guidance on the structure-activity relationship of the polypeptides herein can be determined using multiple sequence alignment (MSA) techniques well-known in the art. Based on the teachings herein, the skilled artisan could make similar alignments with any number of phospholipases described herein or known in the art. Such alignments aid the skilled artisan to determine potentially relevant domains (e.g., binding domains or catalytic domains), as well as which amino acid residues are conserved and not conserved among the different alpha-amylase sequences. It is appreciated in the art that changing an amino acid that is conserved at a particular position between disclosed polypeptides will more likely result in a change in biological activity (Bowie et al., 1990, *Science* 247: 1306-1310: "Residues that are directly involved in protein functions such as binding or catalysis will certainly be among the most conserved"). In contrast, substituting an amino acid that is not highly conserved among the polypeptides will not likely or significantly alter the biological activity.

(174) Even further guidance on the structure-activity relationship for the skilled artisan can be found in published x-ray crystallography studies known in the art.

(175) Single or multiple amino acid substitutions, deletions, and/or insertions can be made and tested using known methods of mutagenesis, recombination, and/or shuffling, followed by a relevant screening procedure, such as those disclosed by Reidhaar-Olson and Sauer, 1988, *Science* 241: 53-57; Bowie and Sauer, 1989, *Proc. Natl. Acad. Sci. USA* 86: 2152-2156; WO 95/17413; or WO 95/22625. Other methods that can be used include error-prone PCR, phage display (e.g., Lowman et al., 1991, *Biochemistry* 30: 10832-10837; U.S. Pat. No. 5,223,409; WO 92/06204), and region-directed mutagenesis (Derbyshire et al., 1986, *Gene* 46: 145; Ner et al., 1988, *DNA* 7:127).

(176) Mutagenesis/shuffling methods can be combined with high-throughput, automated screening methods to detect activity of cloned, mutagenized polypeptides expressed by host cells (Ness et al., 1999, *Nature Biotechnology* 17: 893-896). Mutagenized DNA molecules that encode active alpha-amylases can be recovered from the host cells and rapidly sequenced using standard methods in the art. These methods allow the rapid determination of the importance of individual amino acid residues in a polypeptide.

(177) In some embodiments, the phospholipase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the phospholipase activity of any phospholipase described or referenced herein (e.g., any one of SEQ ID NOs: 235-242 and 252-342, such as any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242) under the same conditions.

(178) In one embodiment, the phospholipase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any phospholipase described or referenced herein (e.g., a coding sequence for the phospholipase of any one of SEQ ID NOs: 235-242 and 252-342; such as the corresponding coding sequence of SEQ ID NO: 244-251 or 343-433, respectively; or the phospholipase of any one of SEQ ID NO: 235, 236, 237, 238, 239, 240, 241 or 242; such as the corresponding coding sequence of SEQ ID NO: 244, 245, 246, 247, 248, 249, 250 or 251, respectively). In one embodiment, the phospholipase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any phospholipase described or referenced herein (e.g., a coding sequence for the phospholipase of any one of SEQ ID NOs: 235-242 and 252-342; such as the corresponding coding sequence of SEQ ID NO: 244-251 or 343-433, respectively; or the phospholipase of any one of SEQ ID NO: 235, 236, 237, 238, 239, 240, 241 or

242; such as the corresponding coding sequence of SEQ ID NO: 244, 245, 246, 247, 248, 249, 250 or 251, respectively).

(179) In one embodiment, the phospholipase comprises the coding sequence of any phospholipase described or referenced herein (e.g., a coding sequence for the phospholipase of any one of SEQ ID NOs: 235-242 and 252-342; such as the corresponding coding sequence of SEQ ID NO: 244-251 or 343-433, respectively; or the phospholipase of any one of SEQ ID NO: 235, 236, 237, 238, 239, 240, 241 or 242; such as the corresponding coding sequence of SEQ ID NO: 244, 245, 246, 247, 248, 249, 250 or 251, respectively). In one embodiment, the phospholipase comprises a coding sequence that is a subsequence of the coding sequence from any phospholipase described or referenced herein, wherein the subsequence encodes a polypeptide having phospholipase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

(180) The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

(181) The phospholipase may be a fused polypeptide or cleavable fusion polypeptide in which another polypeptide is fused at the N-terminus or the C-terminus of the alpha-amylase. A fused polypeptide may be produced by fusing a polynucleotide encoding another polypeptide to a polynucleotide encoding the phospholipase. Techniques for producing fusion polypeptides are known in the art, and include ligating the coding sequences encoding the polypeptides so that they are in frame and that expression of the fused polypeptide is under control of the same promoter(s) and terminator. Fusion proteins may also be constructed using intein technology in which fusions are created post-translationally (Cooper et al., 1993, *EMBO J.* 12: 2575-2583; Dawson et al., 1994, *Science* 266: 776-779).

(182) Alpha-Amylases

(183) The expressed and/or exogenous alpha-amylase may be any alpha-amylase that is suitable for the host cells and/or the methods described herein, such as a naturally occurring alpha-amylase (e.g., a native alpha-amylase from another species or an endogenous alpha-amylase expressed from a modified expression vector) or a variant thereof that retains alpha-amylase activity. Any alpha-amylase contemplated for expression by a fermenting organism described below is also contemplated for aspects of the invention involving exogenous addition of an alpha-amylase.

(184) In some embodiments, the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase, for example, as described in WO2017/087330, the content of which is hereby incorporated by reference. Any alpha-amylase described or referenced herein is contemplated for expression in the fermenting organism.

(185) In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding an alpha-amylase has an increased level of alpha-amylase activity compared to the host cells without the heterologous polynucleotide encoding the alpha-amylase, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of alpha-amylase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the alpha-amylase, when cultivated under the same conditions.

(186) Exemplary alpha-amylases that can be used with the host cells and/or the methods described herein include bacterial, yeast, or filamentous fungal alpha-amylases, e.g., derived from any of the microorganisms described or referenced herein.

(187) The term "bacterial alpha-amylase" means any bacterial alpha-amylase classified under EC 3.2.1.1. A bacterial alpha-amylase used herein may, e.g., be derived from a strain of the genus *Bacillus*, which is sometimes also referred to as the genus *Geobacillus*. In one embodiment, the

Bacillus alpha-amylase is derived from a strain of *Bacillus amyloliquefaciens*, *Bacillus licheniformis*, *Bacillus stearothermophilus*, or *Bacillus subtilis*, but may also be derived from other *Bacillus* sp.

(188) Specific examples of bacterial alpha-amylases include the *Bacillus stearothermophilus* alpha-amylase (BSG) of SEQ ID NO: 3 in WO 99/19467, the *Bacillus amyloliquefaciens* alpha-amylase (BAN) of SEQ ID NO: 5 in WO 99/19467, and the *Bacillus licheniformis* alpha-amylase (BLA) of SEQ ID NO: 4 in WO 99/19467 (all sequences are hereby incorporated by reference). In one embodiment, the alpha-amylase may be an enzyme having a mature polypeptide sequence with a degree of identity of at least 60%, e.g., at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99% to any of the sequences shown in SEQ ID NOs: 3, 4 or 5, in WO 99/19467.

(189) In one embodiment, the alpha-amylase is derived from *Bacillus stearothermophilus*. The *Bacillus stearothermophilus* alpha-amylase may be a mature wild-type or a mature variant thereof. The mature *Bacillus stearothermophilus* alpha-amylases may naturally be truncated during recombinant production. For instance, the *Bacillus stearothermophilus* alpha-amylase may be a truncated at the C-terminal, so that it is from 480-495 amino acids long, such as about 491 amino acids long, e.g., so that it lacks a functional starch binding domain (compared to SEQ ID NO: 3 in WO 99/19467).

(190) The *Bacillus* alpha-amylase may also be a variant and/or hybrid. Examples of such a variant can be found in any of WO 96/23873, WO 96/23874, WO 97/41213, WO 99/19467, WO 00/60059, and WO 02/10355 (each hereby incorporated by reference). Specific alpha-amylase variants are disclosed in U.S. Pat. Nos. 6,093,562, 6,187,576, 6,297,038, and 7,713,723 (hereby incorporated by reference) and include *Bacillus stearothermophilus* alpha-amylase (often referred to as BSG alpha-amylase) variants having a deletion of one or two amino acids at positions R179, G180, I181 and/or G182, preferably a double deletion disclosed in WO 96/23873—see, e.g., page 20, lines 1-10 (hereby incorporated by reference), such as corresponding to deletion of positions 1181 and G182 compared to the amino acid sequence of *Bacillus stearothermophilus* alpha-amylase set forth in SEQ ID NO: 3 disclosed in WO 99/19467 or the deletion of amino acids R179 and G180 using SEQ ID NO: 3 in WO 99/19467 for numbering (which reference is hereby incorporated by reference). In some embodiments, the *Bacillus* alpha-amylases, such as *Bacillus stearothermophilus* alpha-amylases, have a double deletion corresponding to a deletion of positions 181 and 182 and further optionally comprise a N193F substitution (also denoted I181*+G182*+N193F) compared to the wild-type BSG alpha-amylase amino acid sequence set forth in SEQ ID NO: 3 disclosed in WO 99/19467. The bacterial alpha-amylase may also have a substitution in a position corresponding to S239 in the *Bacillus licheniformis* alpha-amylase shown in SEQ ID NO: 4 in WO 99/19467, or a S242 and/or E188P variant of the *Bacillus stearothermophilus* alpha-amylase of SEQ ID NO: 3 in WO 99/19467.

(191) In one embodiment, the variant is a S242A, E or Q variant, e.g., a S242Q variant, of the *Bacillus stearothermophilus* alpha-amylase.

(192) In one embodiment, the variant is a position E188 variant, e.g., E188P variant of the *Bacillus stearothermophilus* alpha-amylase.

(193) The bacterial alpha-amylase may, in one embodiment, be a truncated *Bacillus* alpha-amylase. In one embodiment, the truncation is so that, e.g., the *Bacillus stearothermophilus* alpha-amylase shown in SEQ ID NO: 3 in WO 99/19467, is about 491 amino acids long, such as from 480 to 495 amino acids long, or so it lacks a functional starch bind domain.

(194) The bacterial alpha-amylase may also be a hybrid bacterial alpha-amylase, e.g., an alpha-amylase comprising 445 C-terminal amino acid residues of the *Bacillus licheniformis* alpha-amylase (shown in SEQ ID NO: 4 of WO 99/19467) and the 37 N-terminal amino acid residues of the alpha-amylase derived from *Bacillus amyloliquefaciens* (shown in SEQ ID NO: 5 of WO 99/19467). In one embodiment, this hybrid has one or more, especially all, of the following

substitutions: G48A+T49I+G107A+H156Y+A181T+N190F+I201F+A209V+Q264S (using the *Bacillus licheniformis* numbering in SEQ ID NO: 4 of WO 99/19467). In some embodiments, the variants have one or more of the following mutations (or corresponding mutations in other *Bacillus* alpha-amylases): H154Y, A181T, N190F, A209V and Q264S and/or the deletion of two residues between positions 176 and 179, e.g., deletion of E178 and G179 (using SEQ ID NO: 5 of WO 99/19467 for position numbering).

(195) In one embodiment, the bacterial alpha-amylase is the mature part of the chimeric alpha-amylase disclosed in Richardson et al. (2002), The Journal of Biological Chemistry, Vol. 277, No 29, Issue 19 July, pp. 267501-26507, referred to as BD5088 or a variant thereof. This alpha-amylase is the same as the one shown in SEQ ID NO: 2 in WO 2007/134207. The mature enzyme sequence starts after the initial "Met" amino acid in position 1.

(196) The alpha-amylase may be a thermostable alpha-amylase, such as a thermostable bacterial alpha-amylase, e.g., from *Bacillus stearothermophilus*. In one embodiment, the alpha-amylase used in a process described herein has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂ of at least 10 determined as described in Example 1 of WO2018/098381.

(197) In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 15. In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 20. In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 25. In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 30. In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 40.

(198) In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 50. In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, of at least 60. In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 10-70. In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 15-70. In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 20-70. In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 25-70. In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 30-70. In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 40-70. In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 50-70. In one embodiment, the thermostable alpha-amylase has a $T_{1/2}$ (min) at pH 4.5, 85° C., 0.12 mM CaCl₂, between 60-70.

(199) In one embodiment, the alpha-amylase is a bacterial alpha-amylase, e.g., derived from the genus *Bacillus*, such as a strain of *Bacillus stearothermophilus*, e.g., the *Bacillus stearothermophilus* as disclosed in WO 99/019467 as SEQ ID NO: 3 with one or two amino acids deleted at positions R179, G180, I181 and/or G182, in particular with R179 and G180 deleted, or with I181 and G182 deleted, with mutations in below list of mutations.

(200) In some embodiment, the *Bacillus stearothermophilus* alpha-amylases have double deletion I181+G182, and optional substitution N193F, further comprising one of the following substitutions or combinations of substitutions:

V59A+Q89R+G112D+E129V+K177L+R179E+K220P+N224L+Q254S;

V59A+Q89R+E129V+K177L+R179E+H208Y+K220P+N224L+Q254S;

V59A+Q89R+E129V+K177L+R179E+K220P+N224L+Q254S+D269E+D281N;

V59A+Q89R+E129V+K177L+R179E+K220P+N224L+Q254S+I270L;

V59A+Q89R+E129V+K177L+R179E+K220P+N224L+Q254S+H274K;

V59A+Q89R+E129V+K177L+R179E+K220P+N224L+Q254S+Y276F;

V59A+E129V+R157Y+K177L+R179E+K220P+N224L+S242Q+Q254S;
V59A+E129V+K177L+R179E+H208Y+K220P+N224L+S242Q+Q254S;
V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S;
V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+H274K;
V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+Y276F;
V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+D281N;
V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+M284T;
V59A+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+G416V;
V59A+E129V+K177L+R179E+K220P+N224L+Q254S;
V59A+E129V+K177L+R179E+K220P+N224L+Q254S+M284T;
A91L+M96I+E129V+K177L+R179E+K220P+N224L+S242Q+Q254S; E129V+K177L+R179E;
E129V+K177L+R179E+K220P+N224L+S242Q+Q254S;
E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+Y276F+L427M;
E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+M284T;
E129V+K177L+R179E+K220P+N224L+S242Q+Q254S+N376*+I377*;
E129V+K177L+R179E+K220P+N224L+Q254S;
E129V+K177L+R179E+K220P+N224L+Q254S+M284T; E129V+K177L+R179E+S242Q;
E129V+K177L+R179V+K220P+N224L+S242Q+Q254S; K220P+N224L+S242Q+Q254S;
M284V; V59A+Q89R+E129V+K177L+R179E+Q254S+M284V; and
V59A+E129V+K177L+R179E+Q254S+M284V;

(201) In one embodiment, the alpha-amylase is selected from the group of *Bacillus stearothermophilus* alpha-amylase variants with double deletion I181*+G182*, and optionally substitution N193F, and further one of the following substitutions or combinations of substitutions: E129V+K177L+R179E; V59A+Q89R+E129V+K177L+R179E+H208Y+K220P+N224L+Q254S; V59A+Q89R+E129V+K177L+R179E+Q254S+M284V; V59A+E129V+K177L+R179E+Q254S+M284V; and E129V+K177L+R179E+K220P+N224L+S242Q+Q254S (using SEQ ID NO: 1 herein for numbering).

(202) It should be understood that when referring to *Bacillus stearothermophilus* alpha-amylase and variants thereof they are normally produced in truncated form. In particular, the truncation may be so that the *Bacillus stearothermophilus* alpha-amylase shown in SEQ ID NO: 3 in WO 99/19467, or variants thereof, are truncated in the C-terminal and are typically from 480-495 amino acids long, such as about 491 amino acids long, e.g., so that it lacks a functional starch binding domain.

(203) In one embodiment, the alpha-amylase variant may be an enzyme having a mature polypeptide sequence with a degree of identity of at least 60%, e.g., at least 70%, at least 80%, at least 90%, at least 95%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98% or at least 99%, but less than 100% to the sequence shown in SEQ ID NO: 3 in WO 99/19467.

(204) In one embodiment, the bacterial alpha-amylase, e.g., *Bacillus* alpha-amylase, such as especially *Bacillus stearothermophilus* alpha-amylase, or variant thereof, is dosed to liquefaction in a concentration between 0.01-10 KNU-A/g DS, e.g., between 0.02 and 5 KNU-A/g DS, such as 0.03 and 3 KNU-A, preferably 0.04 and 2 KNU-A/g DS, such as especially 0.01 and 2 KNU-A/g DS. In one embodiment, the bacterial alpha-amylase, e.g., *Bacillus* alpha-amylase, such as especially *Bacillus stearothermophilus* alpha-amylases, or variant thereof, is dosed to liquefaction in a concentration of between 0.0001-1 mg EP (Enzyme Protein)/g DS, e.g., 0.0005-0.5 mg EP/g DS, such as 0.001-0.1 mg EP/g DS.

(205) In one embodiment, the bacterial alpha-amylase is derived from the *Bacillus subtilis* alpha-amylase of SEQ ID NO: 76, the *Bacillus subtilis* alpha-amylase of SEQ ID NO: 82, the *Bacillus subtilis* alpha-amylase of SEQ ID NO: 83, the *Bacillus subtilis* alpha-amylase of SEQ ID NO: 84,

or the *Bacillus licheniformis* alpha-amylase of SEQ ID NO: 85, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 89, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 90, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 91, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 92, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 93, the *Clostridium phytofermentans* alpha-amylase of SEQ ID NO: 94, the *Clostridium thermocellum* alpha-amylase of SEQ ID NO: 95, the *Thermobifida fusca* alpha-amylase of SEQ ID NO: 96, the *Thermobifida fusca* alpha-amylase of SEQ ID NO: 97, the *Anaerocellum thermophilum* of SEQ ID NO: 98, the *Anaerocellum thermophilum* of SEQ ID NO: 99, the *Anaerocellum thermophilum* of SEQ ID NO: 100, the *Streptomyces avermitilis* of SEQ ID NO: 101, or the *Streptomyces avermitilis* of SEQ ID NO: 88.

(206) In one embodiment, the alpha-amylase is derived from *Bacillus amyloliquefaciens*, such as the *Bacillus amyloliquefaciens* alpha-amylase of SEQ ID NO: 231 (e.g., as described in WO2018/002360, or variants thereof as described in WO2017/037614).

(207) In one embodiment, the alpha-amylase is derived from a yeast alpha-amylase, such as the *Saccharomycopsis fibuligera* alpha-amylase of SEQ ID NO: 77, the *Debaryomyces occidentalis* alpha-amylase of SEQ ID NO: 78, the *Debaryomyces occidentalis* alpha-amylase of SEQ ID NO: 79, the *Lipomyces kononenkoae* alpha-amylase of SEQ ID NO: 80, the *Lipomyces kononenkoae* alpha-amylase of SEQ ID NO: 81.

(208) In one embodiment, the alpha-amylase is derived from a filamentous fungal alpha-amylase, such as the *Aspergillus niger* alpha-amylase of SEQ ID NO: 86, or the *Aspergillus niger* alpha-amylase of SEQ ID NO: 87.

(209) Additional alpha-amylases that may be expressed with the fermenting organisms and used with the methods described herein are described in the examples, and include, but are not limited to alpha-amylases shown in Table 2 (or derivatives thereof).

(210) TABLE-US-00002 TABLE 2 Donor Organism SEQ ID NO: (catalytic domain) (mature polypeptide) *Rhizomucor pusillus* 121 *Bacillus licheniformis* 122 *Aspergillus niger* 123 *Aspergillus tamarii* 124 *Acidomyces richmondensis* 125 *Aspergillus bombycis* 126 *Alternaria* sp 127 *Rhizopus microsporus* 128 *Syncephalastrum racemosum* 129 *Rhizomucor pusillus* 130 *Dichotomocladium hesseltinei* 131 *Lichtheimia ramosa* 132 *Penicillium aethiopicum* 133 *Subulispora* sp 134 *Trichoderma paraviridescens* 135 *Byssosascus striatosporus* 136 *Aspergillus brasiliensis* 137 *Penicillium subspinosum* 138 *Penicillium antarcticum* 139 *Penicillium coprophilum* 140 *Penicillium olsonii* 141 *Penicillium vasconiae* 142 *Penicillium* sp 143 *Heterocephalum aurantiacum* 144 *Neosartorya massa* 145 *Penicillium janthinellum* 146 *Aspergillus brasiliensis* 147 *Aspergillus westerdijkiae* 148 *Hamigera avellanea* 149 *Hamigera avellanea* 150 *Meripilus giganteus* 151 *Cerrena unicolor* 152 *Physalacria cryptomeriae* 153 *Lenzites betulinus* 154 *Trametes ljubarskyi* 155 *Bacillus subtilis* 156 *Bacillus subtilis* subsp. *subtilis* 157 *Schwanniomyces occidentalis* 158 *Rhizomucor pusillus* 159 *Aspergillus niger* 160 *Bacillus stearothermophilus* 161 *Bacillus halmapalus* 162 *Aspergillus oryzae* 163 *Bacillus amyloliquefaciens* 164 *Rhizomucor pusillus* 165 *Kionochaeta ivoriensis* 166 *Aspergillus niger* 167 *Aspergillus oryzae* 168 *Penicillium canescens* 169 *Acidomyces acidothermus* 170 *Kionochaeta ivoriensis* 171 *Aspergillus terreus* 172 *Thamnidium elegans* 173 *Meripilus giganteus* 174

(211) Additional alpha-amylases contemplated for use with the present invention can be found in WO2011/153516 (the content of which is incorporated herein).

(212) Additional polynucleotides encoding suitable alpha-amylases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org).

(213) The alpha-amylase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding trehalases from strains of different genera or species, as described supra.

(214) The polynucleotides encoding alpha-amylases may also be identified and obtained from other

sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc) as described supra.

(215) Techniques used to isolate or clone polynucleotides encoding alpha-amylases are described supra.

(216) In one embodiment, the alpha-amylase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of the alpha-amylases described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In another embodiment, the alpha-amylase has a mature polypeptide sequence that is a fragment of the any one of the alpha-amylases described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length alpha-amylase (e.g. any one of SEQ ID NOs: 76-101, 121-174 and 231). In other embodiments, the alpha-amylase may comprise the catalytic domain of any alpha-amylase described or referenced herein (e.g., the catalytic domain of any one of SEQ ID NOs: 76-101, 121-174 and 231).

(217) The alpha-amylase may be a variant of any one of the alpha-amylases described supra (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the alpha-amylase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the alpha-amylases described supra (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231).

(218) In one embodiment, the alpha-amylase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the alpha-amylases described supra (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the alpha-amylase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the alpha-amylases described supra (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

(219) In some embodiments, the alpha-amylase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the alpha-amylase activity of any alpha-amylase described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231) under the same conditions.

(220) In one embodiment, the alpha-amylase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any alpha-amylase described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the alpha-amylase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any alpha-amylase described or referenced herein (e.g., any one of SEQ ID NOs: 76-101, 121-174 and 231).

(221) In one embodiment, the alpha-amylase comprises the coding sequence of any alpha-amylase described or referenced herein (any one of SEQ ID NOs: 76-101, 121-174 and 231). In one embodiment, the alpha-amylase comprises a coding sequence that is a subsequence of the coding sequence from any alpha-amylase described or referenced herein, wherein the subsequence encodes a polypeptide having alpha-amylase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of

the referenced coding sequence.

(222) The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

(223) The alpha-amylase can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

(224) Trehalases

(225) The expressed and/or exogenous trehalase can be any trehalase that is suitable for the fermenting organisms and/or their methods of use described herein, such as a naturally occurring trehalase or a variant thereof that retains trehalase activity. Any trehalase contemplated for expression by a fermenting organism described below is also contemplated for aspects of the invention involving exogenous addition of a trehalase (e.g., added before, during or after liquefaction and/or saccharification).

(226) In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a trehalase has an increased level of trehalase activity compared to the host cells without the heterologous polynucleotide encoding the trehalase, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of trehalase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the trehalase, when cultivated under the same conditions.

(227) Trehalases that may be expressed with the fermenting organisms and used with the methods described herein include, but are not limited to, trehalases shown in Table 3 (or derivatives thereof).

(228) TABLE-US-00003 TABLE 3 Donor Organism SEQ ID NO: (catalytic domain) (mature polypeptide) *Chaetomium megalocarpum* 175 *Lecanicillium psalliotae* 176 *Doratomyces* sp 177 *Mucor moelleri* 178 *Phialophora cyclaminis* 179 *Thielavia arenaria* 180 *Thielavia antarctica* 181 *Chaetomium* sp 182 *Chaetomium nigricolor* 183 *Chaetomium jodhpurens* 184 *Chaetomium piluliferum* 185 *Myceliophthora hinnulea* 186 *Chloridium virescens* 187 *Gelasinospora cratophora* 188 *Acidobacteriaceae bacterium* 189 *Acidobacterium capsulatum* 190 *Acidovorax wautersii* 191 *Xanthomonas arboricola* 192 *Kosakonia sacchari* 193 *Enterobacter* sp 194 *Saitozyma flava* 195 *Phaeotremella skinneri* 196 *Trichoderma asperellum* 197 *Corynascus sepedonium* 198 *Myceliophthora thermophila* 199 *Trichoderma reesei* 200 *Chaetomium virescens* 201 *Rhodothermus marinus* 202 *Myceliophthora sepedonium* 203 *Moelleriella libera* 204 *Acremonium dichrosporium* 205 *Fusarium sambucinum* 206 *Phoma* sp 207 *Lentinus similis* 208 *Diaporthe nobilis* 209 *Solicoccozyma terricola* 210 *Dioszegia cryoxerica* 211 *Talaromyces funiculosus* 212 *Hamigera avellanea* 213 *Talaromyces ruber* 214 *Trichoderma lixii* 215 *Aspergillus cervinus* 216 *Rasamsonia brevistipitata* 217 *Acremonium curvulum* 218 *Talaromyces piceae* 219 *Penicillium* sp 220 *Talaromyces aurantiacus* 221 *Talaromyces pinophilus* 222 *Talaromyces leycettanus* 223 *Talaromyces variabilis* 224 *Aspergillus niger* 225 *Trichoderma reesei* 226

(229) Additional polynucleotides encoding suitable trehalases may be derived from microorganisms of any suitable genus, including those readily available within the UniProtKB database (www.uniprot.org).

(230) The trehalase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding trehalases from strains of different genera or species, as described supra.

(231) The polynucleotides encoding trehalases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc.) as described supra.

(232) Techniques used to isolate or clone polynucleotides encoding trehalases are described supra.

(233) In one embodiment, the trehalase has a mature polypeptide sequence that comprises or

consists of the amino acid sequence of any one of the trehalases described or referenced herein (e.g., any one of SEQ ID NOs: 175-226). In another embodiment, the trehalase has a mature polypeptide sequence that is a fragment of the any one of the trehalases described or referenced herein (e.g., any one of SEQ ID NOs: 175-226). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length trehalase (e.g. any one of SEQ ID NOs: 175-226). In other embodiments, the trehalase may comprise the catalytic domain of any trehalase described or referenced herein (e.g., the catalytic domain of any one of SEQ ID NOs: 175-226).

(234) The trehalase may be a variant of any one of the trehalases described supra (e.g., any one of SEQ ID NOs: 175-226). In one embodiment, the trehalase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the trehalases described supra (e.g., any one of SEQ ID NOs: 175-226).

(235) In one embodiment, the trehalase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the trehalases described supra (e.g., any one of SEQ ID NOs: 175-226). In one embodiment, the trehalase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the trehalases described supra (e.g., any one of SEQ ID NOs: 175-226). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

(236) In some embodiments, the trehalase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the trehalase activity of any trehalase described or referenced herein (e.g., any one of SEQ ID NOs: 175-226) under the same conditions.

(237) In one embodiment, the trehalase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any trehalase described or referenced herein (e.g., any one of SEQ ID NOs: 175-226). In one embodiment, the trehalase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any trehalase described or referenced herein (e.g., any one of SEQ ID NOs: 175-226).

(238) In one embodiment, the trehalase comprises the coding sequence of any trehalase described or referenced herein (any one of SEQ ID NOs: 175-226). In one embodiment, the trehalase comprises a coding sequence that is a subsequence of the coding sequence from any trehalase described or referenced herein, wherein the subsequence encodes a polypeptide having trehalase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

(239) The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

(240) The trehalase can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

(241) Glucoamylases

(242) The expressed and/or exogenous glucoamylase can be any glucoamylase that is suitable for the fermenting organisms and/or their methods of use described herein, such as a naturally

occurring glucoamylase or a variant thereof that retains glucoamylase activity. Any glucoamylase contemplated for expression by a fermenting organism described below is also contemplated for aspects of the invention involving exogenous addition of a glucoamylase (e.g., added before, during or after liquefaction and/or saccharification).

(243) In some embodiments, the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase, for example, as described in WO2017/087330, the content of which is hereby incorporated by reference. Any glucoamylase described or referenced herein is contemplated for expression in the fermenting organism.

(244) In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding an glucoamylase has an increased level of glucoamylase activity compared to the host cells without the heterologous polynucleotide encoding the glucoamylase, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of glucoamylase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the glucoamylase, when cultivated under the same conditions.

(245) Exemplary glucoamylases that can be used with the host cells and/or the methods described herein include bacterial, yeast, or filamentous fungal glucoamylases, e.g., obtained from any of the microorganisms described or referenced herein, as described supra under the sections related to alpha-amylases.

(246) The glucoamylase may be derived from any suitable source, e.g., derived from a microorganism or a plant. Preferred glucoamylases are of fungal or bacterial origin, selected from the group consisting of *Aspergillus* glucoamylases, in particular *Aspergillus niger* G1 or G2 glucoamylase (Boel et al. (1984), EMBO J. 3 (5), p. 1097-1102), or variants thereof, such as those disclosed in WO 92/00381, WO 00/04136 and WO 01/04273 (from Novozymes, Denmark); the *A. awamori* glucoamylase disclosed in WO 84/02921, *Aspergillus oryzae* glucoamylase (Agric. Biol. Chem. (1991), 55 (4), p. 941-949), or variants or fragments thereof. Other *Aspergillus* glucoamylase variants include variants with enhanced thermal stability: G137A and G139A (Chen et al. (1996), Prot. Eng. 9, 499-505); D257E and D293E/Q (Chen et al. (1995), Prot. Eng. 8, 575-582); N182 (Chen et al. (1994), Biochem. J. 301, 275-281); disulphide bonds, A246C (Fierobe et al. (1996), Biochemistry, 35, 8698-8704; and introduction of Pro residues in position A435 and S436 (Li et al. (1997), Protein Eng. 10, 1199-1204).

(247) Other glucoamylases include *Athelia rolfsii* (previously denoted *Corticium rolfsii*) glucoamylase (see U.S. Pat. No. 4,727,026 and (Nagasaka et al. (1998) "Purification and properties of the raw-starch-degrading glucoamylases from *Corticium rolfsii*, Appl Microbiol Biotechnol 50:323-330), *Talaromyces* glucoamylases, in particular derived from *Talaromyces emersonii* (WO 99/28448), *Talaromyces leycettanus* (U.S. Pat. No. Re. 32,153), *Talaromyces duponti*, *Talaromyces thermophilus* (U.S. Pat. No. 4,587,215). In one embodiment, the glucoamylase used during saccharification and/or fermentation is the *Talaromyces emersonii* glucoamylase disclosed in WO 99/28448.

(248) Bacterial glucoamylases contemplated include glucoamylases from the genus *Clostridium*, in particular *C. thermoamylolyticum* (EP 135,138), and *C. thermohydrosulfuricum* (WO 86/01831).

(249) Contemplated fungal glucoamylases include *Trametes cingulate*, *Pachykytospora papyracea*; and *Leucopaxillus giganteus* all disclosed in WO 2006/069289; or *Peniophora rufomarginata* disclosed in WO2007/124285; or a mixture thereof. Also hybrid glucoamylase are contemplated. Examples include the hybrid glucoamylases disclosed in WO 2005/045018.

(250) In one embodiment, the glucoamylase is derived from a strain of the genus *Pycnoporus*, in particular a strain of *Pycnoporus* as described in WO 2011/066576 (SEQ ID NO: 2, 4 or 6 therein), including the *Pycnoporus sanguineus* glucoamylase, or from a strain of the genus *Gloeophyllum*, such as a strain of *Gloeophyllum sepiarium* or *Gloeophyllum trabeum*, in particular a strain of

Gloeophyllum as described in WO 2011/068803 (SEQ ID NO: 2, 4, 6, 8, 10, 12, 14 or 16 therein). In one embodiment, the glucoamylase is SEQ ID NO: 2 in WO 2011/068803 (i.e. *Gloeophyllum sepiarium* glucoamylase). In one embodiment, the glucoamylase is the *Gloeophyllum sepiarium* glucoamylase of SEQ ID NO: 8. In one embodiment, the glucoamylase is the *Pycnoporus sanguineus* glucoamylase of SEQ ID NO: 229.

(251) In one embodiment, the glucoamylase is a *Gloeophyllum trabeum* glucoamylase (disclosed as SEQ ID NO: 3 in WO2014/177546). In another embodiment, the glucoamylase is derived from a strain of the genus *Nigrofomes*, in particular a strain of *Nigrofomes* sp. disclosed in WO 2012/064351 (disclosed as SEQ ID NO: 2 therein).

(252) Also contemplated are glucoamylases with a mature polypeptide sequence which exhibit a high identity to any of the above mentioned glucoamylases, i.e., at least 60%, such as at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or even 100% identity to any one of the mature polypeptide sequences mentioned above.

(253) Glucoamylases may be added to the saccharification and/or fermentation in an amount of 0.0001-20 AGU/g DS, such as 0.001-10 AGU/g DS, 0.01-5 AGU/g DS, or 0.1-2 AGU/g DS.

(254) Glucoamylases may be added to the saccharification and/or fermentation in an amount of 1-1,000 µg EP/g DS, such as 10-500 µg/g DS, or 25-250 µg/g DS.

(255) Glucoamylases may be added to liquefaction in an amount of 0.1-100 µg EP/g DS, such as 0.5-50 µg EP/g DS, 1-25 µg EP/g DS, or 2-12 µg EP/g DS.

(256) In one embodiment, the glucoamylase is added as a blend further comprising an alpha-amylase (e.g., any alpha-amylase described herein). In one embodiment, the alpha-amylase is a fungal alpha-amylase, especially an acid fungal alpha-amylase. The alpha-amylase is typically a side activity.

(257) In one embodiment, the glucoamylase is a blend comprising *Talaromyces emersonii* glucoamylase disclosed in WO 99/28448 as SEQ ID NO: 34 and *Trametes cingulata* glucoamylase disclosed as SEQ ID NO: 2 in WO 06/069289.

(258) In one embodiment, the glucoamylase is a blend comprising *Talaromyces emersonii* glucoamylase disclosed in WO 99/28448, *Trametes cingulata* glucoamylase disclosed as SEQ ID NO: 2 in WO 06/69289, and an alpha-amylase.

(259) In one embodiment, the glucoamylase is a blend comprising *Talaromyces emersonii* glucoamylase disclosed in WO99/28448, *Trametes cingulata* glucoamylase disclosed in WO 06/69289, and *Rhizomucor pusillus* alpha-amylase with *Aspergillus niger* glucoamylase linker and SBD disclosed as V039 in Table 5 in WO 2006/069290.

(260) In one embodiment, the glucoamylase is a blend comprising *Gloeophyllum sepiarium* glucoamylase shown as SEQ ID NO: 2 in WO 2011/068803 and an alpha-amylase, in particular *Rhizomucor pusillus* alpha-amylase with an *Aspergillus niger* glucoamylase linker and starch-binding domain (SBD), disclosed SEQ ID NO: 3 in WO 2013/006756, in particular with the following substitutions: G128D+D143N.

(261) In one embodiment, the alpha-amylase may be derived from a strain of the genus *Rhizomucor*, preferably a strain the *Rhizomucor pusillus*, such as the one shown in SEQ ID NO: 3 in WO2013/006756, or the genus *Meripilus*, preferably a strain of *Meripilus giganteus*. In one embodiment, the alpha-amylase is derived from a *Rhizomucor pusillus* with an *Aspergillus niger* glucoamylase linker and starch-binding domain (SBD), disclosed as V039 in Table 5 in WO 2006/069290.

(262) In one embodiment, the *Rhizomucor pusillus* alpha-amylase or the *Rhizomucor pusillus* alpha-amylase with an *Aspergillus niger* glucoamylase linker and starch-binding domain (SBD) has at least one of the following substitutions or combinations of substitutions: D165M; Y141W; Y141R; K136F; K192R; P224A; P224R; S123H+Y141W; G20S+Y141W; A76G+Y141W; G128D+Y141W; G128D+D143N; P219C+Y141W; N142D+D143N; Y141W+K192R;

Y141W+D143N; Y141W+N383R; Y141W+P219C+A265C; Y141W+N142D+D143N; Y141W+K192R V410A; G128D+Y141W+D143N; Y141W+D143N+P219C; Y141W+D143N+K192R; G128D+D143N+K192R; Y141W+D143N+K192R+P219C; and G128D+Y141W+D143N+K192R; or G128D+Y141W+D143N+K192R+P219C (using SEQ ID NO: 3 in WO 2013/006756 for numbering).

(263) In one embodiment, the glucoamylase blend comprises *Gloeophyllum sepiarium* glucoamylase (e.g., SEQ ID NO: 2 in WO 2011/068803) and *Rhizomucor pusillus* alpha-amylase.

(264) In one embodiment, the glucoamylase blend comprises *Gloeophyllum sepiarium* glucoamylase shown as SEQ ID NO: 2 in WO 2011/068803 and *Rhizomucor pusillus* with an *Aspergillus niger* glucoamylase linker and starch-binding domain (SBD), disclosed SEQ ID NO: 3 in WO 2013/006756 with the following substitutions: G128D+D143N.

(265) Commercially available compositions comprising glucoamylase include AMG 200L; AMG 300 L; SANTM SUPER, SANTM EXTRA L, SPIRIZYME[®] PLUS, SPIRIZYME[®] FUEL, SPIRIZYME[®] B4U, SPIRIZYME[®] ULTRA, SPIRIZYME[®] EXCEL, SPIRIZYME ACHIEVE[®], and AMG[®] E (from Novozymes A/S); OPTIDEXTM 300, GC480, GC417 (from DuPont-Danisco); AMIGASETM and AMIGASETM PLUS (from DSM); G-ZYMETM G900, G-ZYMETM and G990 ZR (from DuPont-Danisco).

(266) In one embodiment, the glucoamylase is derived from the *Debaryomyces occidentalis* glucoamylase of SEQ ID NO: 102. In one embodiment, the glucoamylase is derived from the *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103. In one embodiment, the glucoamylase is derived from the *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 104. In one embodiment, the glucoamylase is derived from the *Saccharomyces cerevisiae* glucoamylase of SEQ ID NO: 105. In one embodiment, the glucoamylase is derived from the *Aspergillus niger* glucoamylase of SEQ ID NO: 106. In one embodiment, the glucoamylase is derived from the *Aspergillus oryzae* glucoamylase of SEQ ID NO: 107. In one embodiment, the glucoamylase is derived from the *Rhizopus oryzae* glucoamylase of SEQ ID NO: 108. In one embodiment, the glucoamylase is derived from the *Clostridium thermocellum* glucoamylase of SEQ ID NO: 109. In one embodiment, the glucoamylase is derived from the *Clostridium thermocellum* glucoamylase of SEQ ID NO: 110. In one embodiment, the glucoamylase is derived from the *Arxula adeninivorans* glucoamylase of SEQ ID NO: 111. In one embodiment, the glucoamylase is derived from the *Hormoconis resinae* glucoamylase of SEQ ID NO: 112. In one embodiment, the glucoamylase is derived from the *Aureobasidium pullulans* glucoamylase of SEQ ID NO: 113.

(267) In one embodiment, the glucoamylase is a *Trichoderma reesei* glucoamylase, such as the *Trichoderma reesei* glucoamylase of SEQ ID NO: 230.

(268) In one embodiment, the glucoamylase has a Relative Activity heat stability at 85° C. of at least 20%, at least 30%, or at least 35% determined as described in Example 4 of WO2018/098381 (heat stability).

(269) In one embodiment, the glucoamylase has a relative activity pH optimum at pH 5.0 of at least 90%, e.g., at least 95%, at least 97%, or 100% determined as described in Example 4 of WO2018/098381 (pH optimum).

(270) In one embodiment, the glucoamylase has a pH stability at pH 5.0 of at least 80%, at least 85%, at least 90% determined as described in Example 4 of WO2018/098381 (pH stability).

(271) In one embodiment, the glucoamylase used in liquefaction, such as a *Penicillium oxalicum* glucoamylase variant, has a thermostability determined as DSC Td at pH 4.0 as described in Example 15 of WO2018/098381 of at least 70° C., preferably at least 75° C., such as at least 80° C., such as at least 81° C., such as at least 82° C., such as at least 83° C., such as at least 84° C., such as at least 85° C., such as at least 86° C., such as at least 87%, such as at least 88° C., such as at least 89° C., such as at least 90° C. In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, has a thermostability determined as DSC Td at pH 4.0 as described in Example 15 of WO2018/098381 in the range between 70° C. and 95° C., such as between 80° C.

and 90° C.

(272) In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, used in liquefaction has a thermostability determined as DSC Td at pH 4.8 as described in Example 15 of WO2018/098381 of at least 70° C., preferably at least 75° C., such as at least 80° C., such as at least 81° C., such as at least 82° C., such as at least 83° C., such as at least 84° C., such as at least 85° C., such as at least 86° C., such as at least 87%, such as at least 88° C., such as at least 89° C., such as at least 90° C., such as at least 91° C. In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, has a thermostability determined as DSC Td at pH 4.8 as described in Example 15 of WO2018/098381 in the range between 70° C. and 95° C., such as between 80° C. and 90° C.

(273) In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, used in liquefaction has a residual activity determined as described in Example 16 of WO2018/098381, of at least 100% such as at least 105%, such as at least 110%, such as at least 115%, such as at least 120%, such as at least 125%. In one embodiment, the glucoamylase, such as a *Penicillium oxalicum* glucoamylase variant, has a thermostability determined as residual activity as described in Example 16 of WO2018/098381, in the range between 100% and 130%.

(274) In one embodiment, the glucoamylase, e.g., of fungal origin such as a filamentous fungi, from a strain of the genus *Penicillium*, e.g., a strain of *Penicillium oxalicum*, in particular the *Penicillium oxalicum* glucoamylase disclosed as SEQ ID NO: 2 in WO 2011/127802 (which is hereby incorporated by reference).

(275) In one embodiment, the glucoamylase has a mature polypeptide sequence of at least 80%, e.g., at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99% or 100% identity to the mature polypeptide shown in SEQ ID NO: 2 in WO 2011/127802.

(276) In one embodiment, the glucoamylase is a variant of the *Penicillium oxalicum* glucoamylase disclosed as SEQ ID NO: 2 in WO 2011/127802, having a K79V substitution. The K79V glucoamylase variant has reduced sensitivity to protease degradation relative to the parent as disclosed in WO 2013/036526 (which is hereby incorporated by reference).

(277) In one embodiment, the glucoamylase is derived from *Penicillium oxalicum*.

(278) In one embodiment, the glucoamylase is a variant of the *Penicillium oxalicum* glucoamylase disclosed as SEQ ID NO: 2 in WO 2011/127802. In one embodiment, the *Penicillium oxalicum* glucoamylase is the one disclosed as SEQ ID NO: 2 in WO 2011/127802 having Val (V) in position 79.

(279) Contemplated *Penicillium oxalicum* glucoamylase variants are disclosed in WO 2013/053801 which is hereby incorporated by reference.

(280) In one embodiment, these variants have reduced sensitivity to protease degradation.

(281) In one embodiment, these variant have improved thermostability compared to the parent.

(282) In one embodiment, the glucoamylase has a K79V substitution (using SEQ ID NO: 2 of WO 2011/127802 for numbering), corresponding to the PE001 variant, and further comprises one of the following alterations or combinations of alterations T65A; Q327F; E501V; Y504T; Y504*; T65A+Q327F; T65A+E501V; T65A+Y504T; T65A+Y504*; Q327F+E501V; Q327F+Y504T; Q327F+Y504*; E501V+Y504T; E501V+Y504*; T65A+Q327F+E501V; T65A+Q327F+Y504T; T65A+E501V+Y504T; Q327F+E501V+Y504T; T65A+Q327F+Y504*; T65A+E501V+Y504*; Q327F+E501V+Y504*; T65A+Q327F+E501V+Y504T; T65A+Q327F+E501V+Y504*; E501V+Y504T; T65A+K161S; T65A+Q405T; T65A+Q327W; T65A+Q327F; T65A+Q327Y; P11F+T65A+Q327F; R1K+D3W+K5Q+G7V+N8S+T10K+P11S+T65A+Q327F; P2N+P4S+P11F+T65A+Q327F; P11F+D26C+K33C+T65A+Q327F; P2N+P4S+P11F+T65A+Q327W+E501V+Y504T; R1E+D3N+P4G+G6R+G7A+N8A+T10D+P11D+T65A+Q327F; P11F+T65A+Q327W; P2N+P4S+P11F+T65A+Q327F+E501V+Y504T; P11F+T65A+Q327W+E501V+Y504T;

T65A+Q327F+E501V+Y504T; T65A+S105P+Q327W; T65A+S105P+Q327F;
T65A+Q327W+S364P; T65A+Q327F+S364P; T65A+S103N+Q327F;
P2N+P4S+P11F+K34Y+T65A+Q327F; P2N+P4S+P11F+T65A+Q327F+D445N+V447S;
P2N+P4S+P11F+T65A+I172V+Q327F; P2N+P4S+P11F+T65A+Q327F+N502*;
P2N+P4S+P11F+T65A+Q327F+N502T+P563S+K571E;
P2N+P4S+P11F+R31S+K33V+T65A+Q327F+N564D+K571S;
P2N+P4S+P11F+T65A+Q327F+S377T; P2N+P4S+P11F+T65A+V325T+Q327W;
P2N+P4S+P11F+T65A+Q327F+D445N+V447S+E501V+Y504T;
P2N+P4S+P11F+T65A+I172V+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+Q327F+S377T+E501V+Y504T;
P2N+P4S+P11F+D26N+K34Y+T65A+Q327F;
P2N+P4S+P11F+T65A+Q327F+I375A+E501V+Y504T;
P2N+P4S+P11F+T65A+K218A+K221D+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+S103N+Q327F+E501V+Y504T;
P2N+P4S+T10D+T65A+Q327F+E501V+Y504T;
P2N+P4S+F12Y+T65A+Q327F+E501V+Y504T; K5A+P11F+T65A+Q327F+E501V+Y504T;
P2N+P4S+T10E+E18N+T65A+Q327F+E501V+Y504T;
P2N+T10E+E18N+T65A+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+Q327F+E501V+Y504T+T568N;
P2N+P4S+P11F+T65A+Q327F+E501V+Y504T+K524T+G526A;
P2N+P4S+P11F+K34Y+T65A+Q327F+D445N+V447S+E501V+Y504T;
P2N+P4S+P11F+R31S+K33V+T65A+Q327F+D445N+V447S+E501V+Y504T;
P2N+P4S+P11F+D26N+K34Y+T65A+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+F80*+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+K112S+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+Q327F+E501V+Y504T+T516P+K524T+G526A;
P2N+P4S+P11F+T65A+Q327F+E501V+N502T+Y504*;
P2N+P4S+P11F+T65A+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+S103N+Q327F+E501V+Y504T;
K5A+P11F+T65A+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+Q327F+E501V+Y504T+T516P+K524T+G526A;
P2N+P4S+P11F+T65A+V79A+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+V79G+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+V79I+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+V79L+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+V79S+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+L72V+Q327F+E501V+Y504T; S255N+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+E74N+V79K+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+G220N+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+Y245N+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+Q253N+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+D279N+Q327F+E501V+Y504T;
P2N+P4S+P11F+T65A+Q327F+S359N+E501V+Y504T;
P2N+P4S+P11F+T65A+Q327F+D370N+E501V+Y504T;
P2N+P4S+P11F+T65A+Q327F+V460S+E501V+Y504T;
P2N+P4S+P11F+T65A+Q327F+V460T+P468T+E501V+Y504T;
P2N+P4S+P11F+T65A+Q327F+T463N+E501V+Y504T;
P2N+P4S+P11F+T65A+Q327F+S465N+E501V+Y504T; and
P2N+P4S+P11F+T65A+Q327F+T477N+E501V+Y504T.

(283) In one embodiment, the *Penicillium oxalicum* glucoamylase variant has a K79V substitution

(using SEQ ID NO: 2 of WO 2011/127802 for numbering), corresponding to the PE001 variant, and further comprises one of the following substitutions or combinations of substitutions: P11F+T65A+Q327F; P2N+P4S+P11F+T65A+Q327F; P11F+D26C+K33C+T65A+Q327F; P2N+P4S+P11F+T65A+Q327W+E501V+Y504T; P2N+P4S+P11F+T65A+Q327F+E501V+Y504T; and P11F+T65A+Q327W+E501V+Y504T.

(284) Additional glucoamylases contemplated for use with the present invention can be found in WO2011/153516 (the content of which is incorporated herein).

(285) Additional polynucleotides encoding suitable glucoamylases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org).

(286) The glucoamylase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding glucoamylases from strains of different genera or species, as described supra.

(287) The polynucleotides encoding glucoamylases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc) as described supra.

(288) Techniques used to isolate or clone polynucleotides encoding glucoamylases are described supra.

(289) In one embodiment, the glucoamylase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of the glucoamylases described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In another embodiment, the glucoamylase has a mature polypeptide sequence that is a fragment of the any one of the glucoamylases described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length glucoamylase (e.g. any one of SEQ ID NOs: 8, 102-113, 229 and 230). In other embodiments, the glucoamylase may comprise the catalytic domain of any glucoamylase described or referenced herein (e.g., the catalytic domain of any one of SEQ ID NOs: 8, 102-113, 229 and 230).

(290) The glucoamylase may be a variant of any one of the glucoamylases described supra (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In one embodiment, the glucoamylase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the glucoamylases described supra (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230).

(291) In one embodiment, the glucoamylase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the glucoamylases described supra (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In one embodiment, the glucoamylase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the glucoamylases described supra (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

(292) In some embodiments, the glucoamylase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the glucoamylase activity of any glucoamylase described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230) under the same conditions.

(293) In one embodiment, the glucoamylase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions,

high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any glucoamylase described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230). In one embodiment, the glucoamylase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any glucoamylase described or referenced herein (e.g., any one of SEQ ID NOs: 8, 102-113, 229 and 230).

(294) In one embodiment, the glucoamylase comprises the coding sequence of any glucoamylase described or referenced herein (any one of SEQ ID NOs: 8, 102-113, 229 and 230). In one embodiment, the glucoamylase comprises a coding sequence that is a subsequence of the coding sequence from any glucoamylase described or referenced herein, wherein the subsequence encodes a polypeptide having glucoamylase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

(295) The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

(296) The glucoamylase can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

(297) Proteases

(298) The expressed and/or exogenous protease can be any protease that is suitable for the fermenting organisms and/or their methods of use described herein, such as a naturally occurring protease or a variant thereof that retains protease activity. Any protease contemplated for expression by a fermenting organism described below is also contemplated for aspects of the invention involving exogenous addition of a protease.

(299) Proteases are classified on the basis of their catalytic mechanism into the following groups: Serine proteases (S), Cysteine proteases (C), Aspartic proteases (A), Metallo proteases (M), and Unknown, or as yet unclassified, proteases (U), see Handbook of Proteolytic Enzymes, A. J. Barrett, N. D. Rawlings, J. F. Woessner (eds), Academic Press (1998), in particular the general introduction part.

(300) Protease activity can be measured using any suitable assay, in which a substrate is employed, that includes peptide bonds relevant for the specificity of the protease in question. Assay-pH and assay-temperature are likewise to be adapted to the protease in question. Examples of assay-pH-values are pH 6, 7, 8, 9, 10, or 11. Examples of assay-temperatures are 30, 35, 37, 40, 45, 50, 55, 60, 65, 70 or 80° C.

(301) In some aspects, the fermenting organism comprising a heterologous polynucleotide encoding a protease has an increased level of protease activity compared to the fermenting organism without the heterologous polynucleotide encoding the protease, when cultivated under the same conditions. In some aspects, the fermenting organism has an increased level of protease activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the protease, when cultivated under the same conditions.

(302) Exemplary proteases that may be expressed with the fermenting organisms and used with the methods described herein include, but are not limited to, proteases shown in Table 4 (or derivatives thereof).

(303) TABLE-US-00004 TABLE 4 Donor Organism SEQ ID NO: (catalytic domain) (mature polypeptide) Family *Aspergillus niger* 9 A1 *Trichoderma reesei* 10 *Thermoascus aurantiacus* 11

M35 *Dichomitus squalens* 12 S53 *Nocardiosis prasina* 13 S1 *Penicillium simplicissimum* 14 S10 *Aspergillus niger* 15 *Meriphilus giganteus* 16 S53 *Lecanicillium* sp. WMM742 17 S53 *Talaromyces proteolyticus* 18 S53 *Penicillium ranomafanaense* 19 A1A *Aspergillus oryzae* 20 S53 *Talaromyces liani* 21 S10 *Thermoascus thermophilus* 22 S53 *Pyrococcus furiosus* 23 *Trichoderma reesei* 24 *Rhizomucor miehei* 25 *Lenzites betulinus* 26 S53 *Neolentinus lepideus* 27 S53 *Thermococcus* sp. 28 S8 *Thermococcus* sp. 29 S8 *Thermomyces lanuginosus* 30 S53 *Thermococcus thioeducens* 31 S53 *Polyporus arcularius* 32 S53 *Ganoderma lucidum* 33 S53 *Ganoderma lucidum* 34 S53 *Ganoderma lucidum* 35 S53 *Trametes* sp. AH28-2 36 S53 *Cinereomyces lindbladii* 37 S53 *Trametes versicolor* O82DDP 38 S53 *Paecilomyces hepiali* 39 S53 *Isaria tenuipes* 40 S53 *Aspergillus tamarii* 41 S53 *Aspergillus brasiliensis* 42 S53 *Aspergillus iizukae* 43 S53 *Penicillium* sp-72364 44 S10 *Aspergillus denticulatus* 45 S10 *Hamigera* sp. t184-6 46 S10 *Penicillium janthinellum* 47 S10 *Penicillium vasconiae* 48 S10 *Hamigera paravellanea* 49 S10 *Talaromyces variabilis* 50 S10 *Penicillium arenicola* 51 S10 *Nocardiosis kunsanensis* 52 S1 *Streptomyces parvulus* 53 S1 *Saccharopolyspora endophytica* 54 S1 *luteus* cellwall enrichments K 55 S1 *Saccharothrix australiensis* 56 S1 *Nocardiosis baichengensis* 57 S1 *Streptomyces* sp. SM15 58 S1 *Actinobolus spitiensis* 59 S1 *Byssoscleromyces verrucosa* 60 M35 *Hamigera terricola* 61 M35 *Aspergillus tamarii* 62 M35 *Aspergillus niveus* 63 M35 *Penicillium sclerotiorum* 64 A1 *Penicillium bilaiae* 65 A1 *Penicillium antarcticum* 66 A1 *Penicillium sumatrense* 67 A1 *Trichoderma lixii* 68 A1 *Trichoderma brevicompactum* 69 A1 *Penicillium cinnamomipurpureum* 70 A1 *Bacillus licheniformis* 71 S8 *Bacillus subtilis* 72 S8 *Trametes* cf *versicol* 73 S53

(304) Additional polynucleotides encoding suitable proteases may be derived from microorganisms of any suitable genus, including those readily available within the UniProtKB database (www.uniprot.org).

(305) In one embodiment, the protease is derived from *Aspergillus*, such as the *Aspergillus niger* protease of SEQ ID NO: 9, the *Aspergillus tamarii* protease of SEQ ID NO: 41, or the *Aspergillus denticulatus* protease of SEQ ID NO: 45. In one embodiment, the protease is derived from *Dichomitus*, such as the *Dichomitus squalens* protease of SEQ ID NO: 12. In one embodiment, the protease is derived from *Penicillium*, such as the *Penicillium simplicissimum* protease of SEQ ID NO: 14, the *Penicillium antarcticum* protease of SEQ ID NO: 66, or the *Penicillium sumatrense* protease of SEQ ID NO: 67. In one aspect, the protease is derived from *Meriphilus*, such as the *Meriphilus giganteus* protease of SEQ ID NO: 16. In one aspect, the protease is derived from *Talaromyces*, such as the *Talaromyces liani* protease of SEQ ID NO: 21. In one aspect, the protease is derived from *Thermoascus*, such as the *Thermoascus thermophilus* protease of SEQ ID NO: 22. In one aspect, the protease is derived from *Ganoderma*, such as the *Ganoderma lucidum* protease of SEQ ID NO: 33. In one aspect, the protease is derived from *Hamigera*, such as the *Hamigera terricola* protease of SEQ ID NO: 61. In one aspect, the protease is derived from *Trichoderma*, such as the *Trichoderma brevicompactum* protease of SEQ ID NO: 69.

(306) The protease coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding proteases from strains of different genera or species, as described supra.

(307) The polynucleotides encoding proteases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc) as described supra.

(308) Techniques used to isolate or clone polynucleotides encoding proteases are described supra.

(309) In one embodiment, the protease has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69). In another embodiment, the protease has a mature polypeptide sequence that is a fragment of the protease of any one of SEQ ID NOs: 9-73 (e.g., wherein the fragment has protease activity). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least

80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length protease (e.g. any one of SEQ ID NOs: 9-73). In other embodiments, the protease may comprise the catalytic domain of any protease described or referenced herein (e.g., the catalytic domain of any one of SEQ ID NOs: 9-73).

(310) The protease may be a variant of any one of the proteases described supra (e.g., any one of SEQ ID NOs: 9-73). In one embodiment, the protease has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the proteases described supra (e.g., any one of SEQ ID NOs: 9-73).

(311) In one embodiment, the protease has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the proteases described supra (e.g., any one of SEQ ID NOs: 9-73). In one embodiment, the protease has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the proteases described supra (e.g., any one of SEQ ID NOs: 9-73). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

(312) In one embodiment, the protease coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any protease described or referenced herein (e.g., any one of SEQ ID NOs: 9-73). In one embodiment, the protease coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any protease described or referenced herein (e.g., any one of SEQ ID NOs: 9-73).

(313) In one embodiment, the protease comprises the coding sequence of any protease described or referenced herein (any one of SEQ ID NOs: 9-73). In one embodiment, the protease comprises a coding sequence that is a subsequence of the coding sequence from any protease described or referenced herein, wherein the subsequence encodes a polypeptide having protease activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

(314) The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

(315) The protease can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

(316) In one embodiment, the protease used according to a process described herein is a Serine proteases. In one particular embodiment, the protease is a serine protease belonging to the family 53, e.g., an endo-protease, such as S53 protease from *Meripilus giganteus*, *Dichomitus squalens*, *Trametes versicolor*, *Polyporus arcularius*, *Lenzites betulinus*, *Ganoderma lucidum*, *Neolentinus lepideus*, or *Bacillus* sp. 19138, in a process for producing ethanol from a starch-containing material, the ethanol yield was improved, when the S53 protease was present/or added during saccharification and/or fermentation of either gelatinized or un-gelatinized starch. In one embodiment, the proteases is selected from: (a) proteases belonging to the EC 3.4.21 enzyme group; and/or (b) proteases belonging to the EC 3.4.14 enzyme group; and/or (c) Serine proteases of the peptidase family S53 that comprises two different types of peptidases: tripeptidyl aminopeptidases (exo-type) and endo-peptidases; as described in 1993, *Biochem. J.* 290:205-218 and in MEROPS protease database, release, 9.4 (31 Jan. 2011) (www.merops.ac.uk). The database

is described in Rawlings, N. D., Barrett, A. J. and Bateman, A., 2010, "MEROPS: the peptidase database", *Nucl. Acids Res.* 38: D227-D233.

(317) For determining whether a given protease is a Serine protease, and a family S53 protease, reference is made to the above Handbook and the principles indicated therein. Such determination can be carried out for all types of proteases, be it naturally occurring or wild-type proteases; or genetically engineered or synthetic proteases.

(318) Peptidase family S53 contains acid-acting endopeptidases and tripeptidyl-peptidases. The residues of the catalytic triad are Glu, Asp, Ser, and there is an additional acidic residue, Asp, in the oxyanion hole. The order of the residues is Glu, Asp, Asp, Ser. The Ser residue is the nucleophile equivalent to Ser in the Asp, His, Ser triad of subtilisin, and the Glu of the triad is a substitute for the general base, His, in subtilisin.

(319) The peptidases of the S53 family tend to be most active at acidic pH (unlike the homologous subtilisins), and this can be attributed to the functional importance of carboxylic residues, notably Asp in the oxyanion hole. The amino acid sequences are not closely similar to those in family S8 (i.e. serine endopeptidase subtilisins and homologues), and this, taken together with the quite different active site residues and the resulting lower pH for maximal activity, provides for a substantial difference to that family. Protein folding of the peptidase unit for members of this family resembles that of subtilisin, having the clan type SB.

(320) In one embodiment, the protease used according to a process described herein is a Cysteine proteases.

(321) In one embodiment, the protease used according to a process described herein is a Aspartic proteases. Aspartic acid proteases are described in, for example, Hand-book of Proteolytic Enzymes, Edited by A. J. Barrett, N. D. Rawlings and J. F. Woessner, Academic Press, San Diego, 1998, Chapter 270). Suitable examples of aspartic acid protease include, e.g., those disclosed in R. M. Berka et al. *Gene*, 96, 313 (1990)); (R. M. Berka et al. *Gene*, 125, 195-198 (1993)); and Gomi et al. *Biosci. Biotech. Biochem.* 57, 1095-1100 (1993), which are hereby incorporated by reference.

(322) The protease also may be a metalloprotease, which is defined as a protease selected from the group consisting of: (a) proteases belonging to EC 3.4.24 (metalloendopeptidases); preferably EC 3.4.24.39 (acid metallo proteinases); (b) metalloproteases belonging to the M group of the above Handbook; (c) metalloproteases not yet assigned to clans (designation: Clan MX), or belonging to either one of clans MA, MB, MC, MD, ME, MF, MG, MH (as defined at pp. 989-991 of the above Handbook); (d) other families of metalloproteases (as defined at pp. 1448-1452 of the above Handbook); (e) metalloproteases with a HEXXH motif; (f) metalloproteases with an HEFTH motif; (g) metalloproteases belonging to either one of families M3, M26, M27, M32, M34, M35, M36, M41, M43, or M47 (as defined at pp. 1448-1452 of the above Handbook); (h) metalloproteases belonging to the M28E family; and (i) metalloproteases belonging to family M35 (as defined at pp. 1492-1495 of the above Handbook).

(323) In other particular embodiments, metalloproteases are hydrolases in which the nucleophilic attack on a peptide bond is mediated by a water molecule, which is activated by a divalent metal cation. Examples of divalent cations are zinc, cobalt or manganese. The metal ion may be held in place by amino acid ligands. The number of ligands may be five, four, three, two, one or zero. In a particular embodiment the number is two or three, preferably three.

(324) There are no limitations on the origin of the metalloprotease used in a process of the invention. In an embodiment the metalloprotease is classified as EC 3.4.24, preferably EC 3.4.24.39. In one embodiment, the metalloprotease is an acid-stable metalloprotease, e.g., a fungal acid-stable metalloprotease, such as a metalloprotease derived from a strain of the genus *Thermoascus*, preferably a strain of *Thermoascus aurantiacus*, especially *Thermoascus aurantiacus* CGMCC No. 0670 (classified as EC 3.4.24.39). In another embodiment, the metalloprotease is derived from a strain of the genus *Aspergillus*, preferably a strain of *Aspergillus oryzae*.

(325) In one embodiment the metalloprotease has a degree of sequence identity to amino acids

–178 to 177, –159 to 177, or preferably amino acids 1 to 177 (the mature polypeptide) of SEQ ID NO: 1 of WO 2010/008841 (a *Thermoascus aurantiacus* metalloprotease) of at least 80%, at least 82%, at least 85%, at least 90%, at least 95%, or at least 97%; and which have metalloprotease activity. In particular embodiments, the metalloprotease consists of an amino acid sequence with a degree of identity to SEQ ID NO: 1 as mentioned above.

(326) The *Thermoascus aurantiacus* metalloprotease is a preferred example of a metalloprotease suitable for use in a process of the invention. Another metalloprotease is derived from *Aspergillus oryzae* and comprises the sequence of SEQ ID NO: 11 disclosed in WO 2003/048353, or amino acids –23-353; –23-374; –23-397; 1-353; 1-374; 1-397; 177-353; 177-374; or 177-397 thereof, and SEQ ID NO: 10 disclosed in WO 2003/048353.

(327) Another metalloprotease suitable for use in a process of the invention is the *Aspergillus oryzae* metalloprotease comprising SEQ ID NO: 5 of WO 2010/008841, or a metalloprotease is an isolated polypeptide which has a degree of identity to SEQ ID NO: 5 of at least about 80%, at least 82%, at least 85%, at least 90%, at least 95%, or at least 97%; and which have metalloprotease activity. In particular embodiments, the metalloprotease consists of the amino acid sequence of SEQ ID NO: 5 of WO 2010/008841.

(328) In a particular embodiment, a metalloprotease has an amino acid sequence that differs by forty, thirty-five, thirty, twenty-five, twenty, or by fifteen amino acids from amino acids –178 to 177, –159 to 177, or +1 to 177 of the amino acid sequences of the *Thermoascus aurantiacus* or *Aspergillus oryzae* metalloprotease.

(329) In another embodiment, a metalloprotease has an amino acid sequence that differs by ten, or by nine, or by eight, or by seven, or by six, or by five amino acids from amino acids –178 to 177, –159 to 177, or +1 to 177 of the amino acid sequences of these metalloproteases, e.g., by four, by three, by two, or by one amino acid.

(330) In particular embodiments, the metalloprotease a) comprises or b) consists of i) the amino acid sequence of amino acids –178 to 177, –159 to 177, or +1 to 177 of SEQ ID NO: 1 of WO 2010/008841; ii) the amino acid sequence of amino acids –23-353, –23-374, –23-397, 1-353, 1-374, 1-397, 177-353, 177-374, or 177-397 of SEQ ID NO: 3 of WO 2010/008841; iii) the amino acid sequence of SEQ ID NO: 5 of WO 2010/008841; or allelic variants, or fragments, of the sequences of i), ii), and iii) that have protease activity.

(331) A fragment of amino acids –178 to 177, –159 to 177, or +1 to 177 of SEQ ID NO: 1 of WO 2010/008841 or of amino acids –23-353, –23-374, –23-397, 1-353, 1-374, 1-397, 177-353, 177-374, or 177-397 of SEQ ID NO: 3 of WO 2010/008841; is a polypeptide having one or more amino acids deleted from the amino and/or carboxyl terminus of these amino acid sequences. In one embodiment a fragment contains at least 75 amino acid residues, or at least 100 amino acid residues, or at least 125 amino acid residues, or at least 150 amino acid residues, or at least 160 amino acid residues, or at least 165 amino acid residues, or at least 170 amino acid residues, or at least 175 amino acid residues.

(332) To determine whether a given protease is a metalloprotease or not, reference is made to the above “Handbook of Proteolytic Enzymes” and the principles indicated therein. Such determination can be carried out for all types of proteases, be it naturally occurring or wild-type proteases; or genetically engineered or synthetic proteases.

(333) The protease may be a variant of, e.g., a wild-type protease, having thermostability properties defined herein. In one embodiment, the thermostable protease is a variant of a metalloprotease. In one embodiment, the thermostable protease used in a process described herein is of fungal origin, such as a fungal metalloprotease, such as a fungal metalloprotease derived from a strain of the genus *Thermoascus*, preferably a strain of *Thermoascus aurantiacus*, especially *Thermoascus aurantiacus* CGMCC No. 0670 (classified as EC 3.4.24.39).

(334) In one embodiment, the thermostable protease is a variant of the mature part of the metalloprotease shown in SEQ ID NO: 2 disclosed in WO 2003/048353 or the mature part of SEQ ID NO:

1 in WO 2010/008841 further with one of the following substitutions or combinations of substitutions: S5*+D79L+S87P+A112P+D142L; D79L+S87P+A112P+T124V+D142L; S5*+N26R+D79L+S87P+A112P+D142L; N26R+T46R+D79L+S87P+A112P+D142L; T46R+D79L+S87P+T116V+D142L; D79L+P81 R+S87P+A112P+D142L; A27K+D79L+S87P+A112P+T124V+D142L; D79L+Y82F+S87P+A112P+T124V+D142L; D79L+Y82F+S87P+A112P+T124V+D142L; D79L+S87P+A112P+T124V+A126V+D142L; D79L+S87P+A112P+D142L; D79L+Y82F+S87P+A112P+D142L; S38T+D79L+S87P+A112P+A126V+D142L; D79L+Y82F+S87P+A112P+A126V+D142L; A27K+D79L+S87P+A112P+A126V+D142L; D79L+S87P+N98C+A112P+G135C+D142L; D79L+S87P+A112P+D142L+T141C+M161C; S36P+D79L+S87P+A112P+D142L; A37P+D79L+S87P+A112P+D142L; S49P+D79L+S87P+A112P+D142L; S50P+D79L+S87P+A112P+D142L; D79L+S87P+D104P+A112P+D142L; D79L+Y82F+S87G+A112P+D142L; S70V+D79L+Y82F+S87G+Y97W+A112P+D142L; D79L+Y82F+S87G+Y97W+D104P+A112P+D142L; S70V+D79L+Y82F+S87G+A112P+D142L; D79L+Y82F+S87G+D104P+A112P+D142L; D79L+Y82F+S87G+A112P+A126V+D142L; Y82F+S87G+S70V+D79L+D104P+A112P+D142L; Y82F+S87G+D79L+D104P+A112P+A126V+D142L; A27K+D79L+Y82F+S87G+D104P+A112P+A126V+D142L; A27K+Y82F+S87G+D104P+A112P+A126V+D142L; A27K+D79L+Y82F+D104P+A112P+A126V+D142L; A27K+Y82F+D104P+A112P+A126V+D142L; A27K+D79L+S87P+A112P+D142L; and D79L+S87P+D142L.

(335) In one embodiment, the thermostable protease is a variant of the metallo protease disclosed as the mature part of SEQ ID NO: 2 disclosed in WO 2003/048353 or the mature part of SEQ ID NO: 1 in WO 2010/008841 with one of the following substitutions or combinations of substitutions: D79L+S87P+A112P+D142L; D79L+S87P+D142L; and A27K+D79L+Y82F+S87G+D104P+A112P+A126V+D142L.

(336) In one embodiment, the protease variant has at least 75% identity preferably at least 80%, more preferably at least 85%, more preferably at least 90%, more preferably at least 91%, more preferably at least 92%, even more preferably at least 93%, most preferably at least 94%, and even most preferably at least 95%, such as even at least 96%, at least 97%, at least 98%, at least 99%, but less than 100% identity to the mature part of the polypeptide of SEQ ID NO: 2 disclosed in WO 2003/048353 or the mature part of SEQ ID NO: 1 in WO 2010/008841.

(337) The thermostable protease may also be derived from any bacterium as long as the protease has the thermostability properties.

(338) In one embodiment, the thermostable protease is derived from a strain of the bacterium *Pyrococcus*, such as a strain of *Pyrococcus furiosus* (pfu protease).

(339) In one embodiment, the protease is one shown as SEQ ID NO: 1 in U.S. Pat. No. 6,358,726-E1 (Takara Shuzo Company).

(340) In one embodiment, the thermostable protease is a protease having a mature polypeptide sequence of at least 80% identity, such as at least 85%, such as at least 90%, such as at least 95%, such as at least 96%, such as at least 97%, such as at least 98%, such as at least 99% identity to SEQ ID NO: 1 in U.S. Pat. No. 6,358,726-B1. The *Pyrococcus furiosus* protease can be purchased from Takara Bio, Japan.

(341) The *Pyrococcus furiosus* protease may be a thermostable protease as described in SEQ ID NO: 13 of WO2018/098381. This protease (PfuS) was found to have a thermostability of 110% (80° C./70° C.) and 103% (90° C./70° C.) at pH 4.5 determined.

(342) In one embodiment a thermostable protease used in a process described herein has a thermostability value of more than 20% determined as Relative Activity at 80° C./70° C. determined as described in Example 2 of WO2018/098381.

(343) In one embodiment, the protease has a thermostability of more than 30%, more than 40%, more than 50%, more than 60%, more than 70%, more than 80%, more than 90%, more than 100%, such as more than 105%, such as more than 110%, such as more than 115%, such as more than 120% determined as Relative Activity at 80° C./70° C.

(344) In one embodiment, protease has a thermostability of between 20 and 50%, such as between 20 and 40%, such as 20 and 30% determined as Relative Activity at 80° C./70° C. In one embodiment, the protease has a thermostability between 50 and 115%, such as between 50 and 70%, such as between 50 and 60%, such as between 100 and 120%, such as between 105 and 115% determined as Relative Activity at 80° C./70° C.

(345) In one embodiment, the protease has a thermostability value of more than 10% determined as Relative Activity at 85° C./70° C. determined as described in Example 2 of WO2018/098381.

(346) In one embodiment, the protease has a thermostability of more than 10%, such as more than 12%, more than 14%, more than 16%, more than 18%, more than 20%, more than 30%, more than 40%, more than 50%, more than 60%, more than 70%, more than 80%, more than 90%, more than 100%, more than 110% determined as Relative Activity at 85° C./70° C.

(347) In one embodiment, the protease has a thermostability of between 10% and 50%, such as between 10% and 30%, such as between 10% and 25% determined as Relative Activity at 85° C./70° C.

(348) In one embodiment, the protease has more than 20%, more than 30%, more than 40%, more than 50%, more than 60%, more than 70%, more than 80%, more than 90% determined as Remaining Activity at 80° C.; and/or the protease has more than 20%, more than 30%, more than 40%, more than 50%, more than 60%, more than 70%, more than 80%, more than 90% determined as Remaining Activity at 84° C.

(349) Determination of “Relative Activity” and “Remaining Activity” is done as described in Example 2 of WO2018/098381.

(350) In one embodiment, the protease may have a thermostability for above 90, such as above 100 at 85° C. as determined using the Zein-BCA assay as disclosed in Example 3 of WO2018/098381.

(351) In one embodiment, the protease has a thermostability above 60%, such as above 90%, such as above 100%, such as above 110% at 85° C. as determined using the Zein-BCA assay of WO2018/098381.

(352) In one embodiment, protease has a thermostability between 60-120, such as between 70-120%, such as between 80-120%, such as between 90-120%, such as between 100-120%, such as 110-120% at 85° C. as determined using the Zein-BCA assay of WO2018/098381.

(353) In one embodiment, the thermostable protease has at least 20%, such as at least 30%, such as at least 40%, such as at least 50%, such as at least 60%, such as at least 70%, such as at least 80%, such as at least 90%, such as at least 95%, such as at least 100% of the activity of the JTP196 protease variant or Protease Pfu determined by the AZCL-casein assay of WO2018/098381, and described herein.

(354) In one embodiment, the thermostable protease has at least 20%, such as at least 30%, such as at least 40%, such as at least 50%, such as at least 60%, such as at least 70%, such as at least 80%, such as at least 90%, such as at least 95%, such as at least 100% of the protease activity of the Protease 196 variant or Protease Pfu determined by the AZCL-casein assay of WO2018/098381, and described herein.

(355) Pullulanases

(356) In some embodiments, a pullulanase is present and/or added in liquefaction step and/or saccharification step, or simultaneous saccharification and fermentation (SSF).

(357) Pullulanases (E.C. 3.2.1.41, pullulan 6-glucano-hydrolase), are debranching enzymes characterized by their ability to hydrolyze the alpha-1,6-glycosidic bonds in, for example, amylopectin and pullulan.

(358) In some embodiments, the fermenting organism comprises a heterologous polynucleotide

encoding a pullulanase. Any pullulanase described or referenced herein is contemplated for expression in the fermenting organism.

(359) The pullulanase may be any pullulanase that is suitable for the host cells and/or the methods described herein, such as a naturally occurring pullulanase or a variant thereof that retains pullulanase activity.

(360) In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a pullulanase has an increased level of pullulanase activity compared to the host cells without the heterologous polynucleotide encoding the pullulanase, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of pullulanase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the pullulanase, when cultivated under the same conditions.

(361) Exemplary pullulanases that can be used with the host cells and/or the methods described herein include bacterial, yeast, or filamentous fungal pullulanases, e.g., obtained from any of the microorganisms described or referenced herein, as described supra under the sections related to alpha-amylases.

(362) Contemplated pullulanases include the pullulanases from *Bacillus amyloclavus* disclosed in U.S. Pat. No. 4,560,651 (hereby incorporated by reference), the pullulanase disclosed as SEQ ID NO: 2 in WO 01/151620 (hereby incorporated by reference), the *Bacillus deramificans* disclosed as SEQ ID NO: 4 in WO 01/151620 (hereby incorporated by reference), and the pullulanase from *Bacillus acidopullulyticus* disclosed as SEQ ID NO: 6 in WO 01/151620 (hereby incorporated by reference) and also described in FEMS Mic. Let. (1994) 115, 97-106.

(363) Additional pullulanases contemplated include the pullulanases from *Pyrococcus woesei*, specifically from *Pyrococcus woesei* DSM No. 3773 disclosed in WO92/02614.

(364) In one embodiment, the pullulanase is a family GH57 pullulanase. In one embodiment, the pullulanase includes an X47 domain as disclosed in U.S. 61/289,040 published as WO 2011/087836 (which are hereby incorporated by reference). More specifically the pullulanase may be derived from a strain of the genus *Thermococcus*, including *Thermococcus litoralis* and *Thermococcus hydrothermalis*, such as the *Thermococcus hydrothermalis* pullulanase truncated at site X4 right after the X47 domain (i.e., amino acids 1-782). The pullulanase may also be a hybrid of the *Thermococcus litoralis* and *Thermococcus hydrothermalis* pullulanases or a *T. hydrothermalis/T. litoralis* hybrid enzyme with truncation site X4 disclosed in U.S. 61/289,040 published as WO 2011/087836 (which is hereby incorporated by reference).

(365) In another embodiment, the pullulanase is one comprising an X46 domain disclosed in WO 2011/076123 (Novozymes).

(366) The pullulanase may be added in an effective amount which include the preferred amount of about 0.0001-10 mg enzyme protein per gram DS, preferably 0.0001-0.10 mg enzyme protein per gram DS, more preferably 0.0001-0.010 mg enzyme protein per gram DS. Pullulanase activity may be determined as NPUN. An Assay for determination of NPUN is described in WO2018/098381.

(367) Suitable commercially available pullulanase products include PROMOZYME D, PROMOZYME™ D2 (Novozymes A/S, Denmark), OPTIMAX L-300 (DuPont-Danisco, USA), and AMANO 8 (Amano, Japan).

(368) In one embodiment, the pullulanase is derived from the *Bacillus subtilis* pullulanase of SEQ ID NO: 114. In one embodiment, the pullulanase is derived from the *Bacillus licheniformis* pullulanase of SEQ ID NO: 115. In one embodiment, the pullulanase is derived from the *Oryza sativa* pullulanase of SEQ ID NO: 116. In one embodiment, the pullulanase is derived from the *Triticum aestivum* pullulanase of SEQ ID NO: 117. In one embodiment, the pullulanase is derived from the *Clostridium phytofermentans* pullulanase of SEQ ID NO: 118. In one embodiment, the pullulanase is derived from the *Streptomyces avermitilis* pullulanase of SEQ ID NO: 119. In one

embodiment, the pullulanase is derived from the *Klebsiella pneumoniae* pullulanase of SEQ ID NO: 120.

(369) Additional pullulanases contemplated for use with the present invention can be found in WO2011/153516 (the content of which is incorporated herein).

(370) Additional polynucleotides encoding suitable pullulanases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org).

(371) The pullulanase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding pullulanases from strains of different genera or species, as described supra.

(372) The polynucleotides encoding pullulanases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc) as described supra.

(373) Techniques used to isolate or clone polynucleotides encoding pullulanases are described supra.

(374) In one embodiment, the pullulanase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any one of the pullulanases described or referenced herein (e.g., any one of SEQ ID NOs: 114-120). In another embodiment, the pullulanase has a mature polypeptide sequence that is a fragment of the any one of the pullulanases described or referenced herein (e.g., any one of SEQ ID NOs: 114-120). In one embodiment, the number of amino acid residues in the fragment is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of amino acid residues in referenced full length pullulanase. In other embodiments, the pullulanase may comprise the catalytic domain of any pullulanase described or referenced herein (e.g., any one of SEQ ID NOs: 114-120).

(375) The pullulanase may be a variant of any one of the pullulanases described supra (e.g., any one of SEQ ID NOs: 114-120). In one embodiment, the pullulanase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to any one of the pullulanases described supra (e.g., any one of SEQ ID NOs: 114-120).

(376) In one embodiment, the pullulanase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of the pullulanases described supra (e.g., any one of SEQ ID NOs: 114-120). In one embodiment, the pullulanase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) of amino acid sequence of any one of the pullulanases described supra (e.g., any one of SEQ ID NOs: 114-120). In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

(377) In some embodiments, the pullulanase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the pullulanase activity of any pullulanase described or referenced herein under the same conditions (e.g., any one of SEQ ID NOs: 114-120).

(378) In one embodiment, the pullulanase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any pullulanase described or referenced herein (e.g., any one of SEQ ID NOs: 114-120). In one embodiment, the pullulanase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at

least 99%, or 100% sequence identity with the coding sequence from any pullulanase described or referenced herein (e.g., any one of SEQ ID NOs: 114-120).

(379) In one embodiment, the pullulanase comprises the coding sequence of any pullulanase described or referenced herein (e.g., any one of SEQ ID NOs: 114-120). In one embodiment, the pullulanase comprises a coding sequence that is a subsequence of the coding sequence from any pullulanase described or referenced herein, wherein the subsequence encodes a polypeptide having pullulanase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

(380) The referenced coding sequence of any related aspect or embodiment described herein can be the native coding sequence or a degenerate sequence, such as a codon-optimized coding sequence designed for use in a particular host cell (e.g., optimized for expression in *Saccharomyces cerevisiae*).

(381) The pullulanase can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

(382) Methods Using a Cellulosic-Containing Material

(383) In some aspects, the methods described herein produce a fermentation product from a cellulosic-containing material. The predominant polysaccharide in the primary cell wall of biomass is cellulose, the second most abundant is hemicellulose, and the third is pectin. The secondary cell wall, produced after the cell has stopped growing, also contains polysaccharides and is strengthened by polymeric lignin covalently cross-linked to hemicellulose. Cellulose is a homopolymer of anhydrocellobiose and thus a linear beta-(1-4)-D-glucan, while hemicelluloses include a variety of compounds, such as xylans, xyloglucans, arabinoxylans, and mannans in complex branched structures with a spectrum of substituents. Although generally polymorphous, cellulose is found in plant tissue primarily as an insoluble crystalline matrix of parallel glucan chains. Hemicelluloses usually hydrogen bond to cellulose, as well as to other hemicelluloses, which help stabilize the cell wall matrix.

(384) Cellulose is generally found, for example, in the stems, leaves, hulls, husks, and cobs of plants or leaves, branches, and wood of trees. The cellulosic-containing material can be, but is not limited to, agricultural residue, herbaceous material (including energy crops), municipal solid waste, pulp and paper mill residue, waste paper, and wood (including forestry residue) (see, for example, Wiselogle et al., 1995, in Handbook on Bioethanol (Charles E. Wyman, editor), pp. 105-118, Taylor & Francis, Washington D.C.; Wyman, 1994, *Bioresource Technology* 50: 3-16; Lynd, 1990, *Applied Biochemistry and Biotechnology* 24/25: 695-719; Mosier et al., 1999, Recent Progress in Bioconversion of Lignocellulosics, in *Advances in Biochemical Engineering/Biotechnology*, T. Scheper, managing editor, Volume 65, pp. 23-40, Springer-Verlag, New York). It is understood herein that the cellulose may be in the form of lignocellulose, a plant cell wall material containing lignin, cellulose, and hemicellulose in a mixed matrix. In one embodiment, the cellulosic-containing material is any biomass material. In another embodiment, the cellulosic-containing material is lignocellulose, which comprises cellulose, hemicelluloses, and lignin.

(385) In one embodiment, the cellulosic-containing material is agricultural residue, herbaceous material (including energy crops), municipal solid waste, pulp and paper mill residue, waste paper, or wood (including forestry residue).

(386) In another embodiment, the cellulosic-containing material is *arundo*, bagasse, bamboo, corn cob, corn fiber, corn stover, *miscanthus*, rice straw, switchgrass, or wheat straw.

(387) In another embodiment, the cellulosic-containing material is aspen, *eucalyptus*, fir, pine, poplar, spruce, or willow.

(388) In another embodiment, the cellulosic-containing material is algal cellulose, bacterial cellulose, cotton linter, filter paper, microcrystalline cellulose (e.g., AVICEL®), or phosphoric-acid

treated cellulose.

(389) In another embodiment, the cellulosic-containing material is an aquatic biomass. As used herein the term “aquatic biomass” means biomass produced in an aquatic environment by a photosynthesis process. The aquatic biomass can be algae, emergent plants, floating-leaf plants, or submerged plants.

(390) The cellulosic-containing material may be used as is or may be subjected to pretreatment, using conventional methods known in the art, as described herein. In a preferred embodiment, the cellulosic-containing material is pretreated.

(391) The methods of using cellulosic-containing material can be accomplished using methods conventional in the art. Moreover, the methods of can be implemented using any conventional biomass processing apparatus configured to carry out the processes.

(392) Cellulosic Pretreatment

(393) In one embodiment the cellulosic-containing material is pretreated before saccharification.

(394) In practicing the processes described herein, any pretreatment process known in the art can be used to disrupt plant cell wall components of the cellulosic-containing material (Chandra et al., 2007, *Adv. Biochem. Engin./Biotechnol.* 108: 67-93; Galbe and Zacchi, 2007, *Adv. Biochem. Engin./Biotechnol.* 108: 41-65; Hendriks and Zeeman, 2009, *Bioresource Technology* 100: 10-18; Mosier et al., 2005, *Bioresource Technology* 96: 673-686; Taherzadeh and Karimi, 2008, *Int. J. Mol. Sci.* 9: 1621-1651; Yang and Wyman, 2008, *Biofuels Bioproducts and Biorefining-Biofpr.* 2: 26-40).

(395) The cellulosic-containing material can also be subjected to particle size reduction, sieving, pre-soaking, wetting, washing, and/or conditioning prior to pretreatment using methods known in the art.

(396) Conventional pretreatments include, but are not limited to, steam pretreatment (with or without explosion), dilute acid pretreatment, hot water pretreatment, alkaline pretreatment, lime pretreatment, wet oxidation, wet explosion, ammonia fiber explosion, organosolv pretreatment, and biological pretreatment. Additional pretreatments include ammonia percolation, ultrasound, electroporation, microwave, supercritical CO₂, supercritical H₂O, ozone, ionic liquid, and gamma irradiation pretreatments.

(397) In a one embodiment, the cellulosic-containing material is pretreated before saccharification (i.e., hydrolysis) and/or fermentation. Pretreatment is preferably performed prior to the hydrolysis. Alternatively, the pretreatment can be carried out simultaneously with enzyme hydrolysis to release fermentable sugars, such as glucose, xylose, and/or cellobiose. In most cases the pretreatment step itself results in some conversion of biomass to fermentable sugars (even in absence of enzymes).

(398) In one embodiment, the cellulosic-containing material is pretreated with steam. In steam pretreatment, the cellulosic-containing material is heated to disrupt the plant cell wall components, including lignin, hemicellulose, and cellulose to make the cellulose and other fractions, e.g., hemicellulose, accessible to enzymes. The cellulosic-containing material is passed to or through a reaction vessel where steam is injected to increase the temperature to the required temperature and pressure and is retained therein for the desired reaction time. Steam pretreatment is preferably performed at 140-250° C., e.g., 160-200° C. or 170-190° C., where the optimal temperature range depends on optional addition of a chemical catalyst. Residence time for the steam pretreatment is preferably 1-60 minutes, e.g., 1-30 minutes, 1-20 minutes, 3-12 minutes, or 4-10 minutes, where the optimal residence time depends on the temperature and optional addition of a chemical catalyst. Steam pretreatment allows for relatively high solids loadings, so that the cellulosic-containing material is generally only moist during the pretreatment. The steam pretreatment is often combined with an explosive discharge of the material after the pretreatment, which is known as steam explosion, that is, rapid flashing to atmospheric pressure and turbulent flow of the material to increase the accessible surface area by fragmentation (Duff and Murray, 1996, *Bioresource Technology* 855: 1-33; Galbe and Zacchi, 2002, *Appl. Microbiol. Biotechnol.* 59: 618-628; U.S.

Patent Application No. 2002/0164730). During steam pretreatment, hemicellulose acetyl groups are cleaved and the resulting acid autocatalyzes partial hydrolysis of the hemicellulose to monosaccharides and oligosaccharides. Lignin is removed to only a limited extent.

(399) In one embodiment, the cellulosic-containing material is subjected to a chemical pretreatment. The term “chemical treatment” refers to any chemical pretreatment that promotes the separation and/or release of cellulose, hemicellulose, and/or lignin. Such a pretreatment can convert crystalline cellulose to amorphous cellulose. Examples of suitable chemical pretreatment processes include, for example, dilute acid pretreatment, lime pretreatment, wet oxidation, ammonia fiber/freeze expansion (AFEX), ammonia percolation (APR), ionic liquid, and organosolv pretreatments.

(400) A chemical catalyst such as H₂SO₄ or SO₂ (typically 0.3 to 5% w/w) is sometimes added prior to steam pretreatment, which decreases the time and temperature, increases the recovery, and improves enzymatic hydrolysis (Ballesteros et al., 2006, *Appl. Biochem. Biotechnol.* 129-132: 496-508; Varga et al., 2004, *Appl. Biochem. Biotechnol.* 113-116: 509-523; Sassner et al., 2006, *Enzyme Microb. Technol.* 39: 756-762). In dilute acid pretreatment, the cellulosic-containing material is mixed with dilute acid, typically H₂SO₄, and water to form a slurry, heated by steam to the desired temperature, and after a residence time flashed to atmospheric pressure. The dilute acid pretreatment can be performed with a number of reactor designs, e.g., plug-flow reactors, counter-current reactors, or continuous counter-current shrinking bed reactors (Duff and Murray, 1996, *Bioresource Technology* 85: 1-33; Schell et al., 2004, *Bioresource Technology* 91: 179-188; Lee et al., 1999, *Adv. Biochem. Eng. Biotechnol.* 65: 93-115). In a specific embodiment the dilute acid pretreatment of cellulosic-containing material is carried out using 4% w/w sulfuric acid at 180° C. for 5 minutes.

(401) Several methods of pretreatment under alkaline conditions can also be used. These alkaline pretreatments include, but are not limited to, sodium hydroxide, lime, wet oxidation, ammonia percolation (APR), and ammonia fiber/freeze expansion (AFEX) pretreatment. Lime pretreatment is performed with calcium oxide or calcium hydroxide at temperatures of 85-150° C. and residence times from 1 hour to several days (Wyman et al., 2005, *Bioresource Technology* 96: 1959-1966; Mosier et al., 2005, *Bioresource Technology* 96: 673-686). WO 2006/110891, WO 2006/110899, WO 2006/110900, and WO 2006/110901 disclose pretreatment methods using ammonia.

(402) Wet oxidation is a thermal pretreatment performed typically at 180-200° C. for 5-15 minutes with addition of an oxidative agent such as hydrogen peroxide or over-pressure of oxygen (Schmidt and Thomsen, 1998, *Bioresource Technology* 64: 139-151; Palonen et al., 2004, *Appl. Biochem. Biotechnol.* 117: 1-17; Varga et al., 2004, *Biotechnol. Bioeng.* 88: 567-574; Martin et al., 2006, *J. Chem. Technol. Biotechnol.* 81: 1669-1677). The pretreatment is performed preferably at 1-40% dry matter, e.g., 2-30% dry matter or 5-20% dry matter, and often the initial pH is increased by the addition of alkali such as sodium carbonate.

(403) A modification of the wet oxidation pretreatment method, known as wet explosion (combination of wet oxidation and steam explosion) can handle dry matter up to 30%. In wet explosion, the oxidizing agent is introduced during pretreatment after a certain residence time. The pretreatment is then ended by flashing to atmospheric pressure (WO 2006/032282).

(404) Ammonia fiber expansion (AFEX) involves treating the cellulosic-containing material with liquid or gaseous ammonia at moderate temperatures such as 90-150° C. and high pressure such as 17-20 bar for 5-10 minutes, where the dry matter content can be as high as 60% (Gollapalli et al., 2002, *Appl. Biochem. Biotechnol.* 98: 23-35; Chundawat et al., 2007, *Biotechnol. Bioeng.* 96: 219-231; Alizadeh et al., 2005, *Appl. Biochem. Biotechnol.* 121: 1133-1141; Teymouri et al., 2005, *Bioresource Technology* 96: 2014-2018). During AFEX pretreatment cellulose and hemicelluloses remain relatively intact. Lignin-carbohydrate complexes are cleaved.

(405) Organosolv pretreatment delignifies the cellulosic-containing material by extraction using aqueous ethanol (40-60% ethanol) at 160-200° C. for 30-60 minutes (Pan et al., 2005, *Biotechnol.*

Bioeng. 90: 473-481; Pan et al., 2006, *Biotechnol. Bioeng.* 94: 851-861; Kurabi et al., 2005, *Appl. Biochem. Biotechnol.* 121: 219-230). Sulphuric acid is usually added as a catalyst. In organosolv pretreatment, the majority of hemicellulose and lignin is removed.

(406) Other examples of suitable pretreatment methods are described by Schell et al., 2003, *Appl. Biochem. Biotechnol.* 105-108: 69-85, and Mosier et al., 2005, *Bioresource Technology* 96: 673-686, and U.S. Published Application 2002/0164730.

(407) In one embodiment, the chemical pretreatment is carried out as a dilute acid treatment, and more preferably as a continuous dilute acid treatment. The acid is typically sulfuric acid, but other acids can also be used, such as acetic acid, citric acid, nitric acid, phosphoric acid, tartaric acid, succinic acid, hydrogen chloride, or mixtures thereof. Mild acid treatment is conducted in the pH range of preferably 1-5, e.g., 1-4 or 1-2.5. In one aspect, the acid concentration is in the range from preferably 0.01 to 10 wt. % acid, e.g., 0.05 to 5 wt. % acid or 0.1 to 2 wt. % acid. The acid is contacted with the cellulosic-containing material and held at a temperature in the range of preferably 140-200° C., e.g., 165-190° C., for periods ranging from 1 to 60 minutes.

(408) In another embodiment, pretreatment takes place in an aqueous slurry. In preferred aspects, the cellulosic-containing material is present during pretreatment in amounts preferably between 10-80 wt. %, e.g., 20-70 wt. % or 30-60 wt. %, such as around 40 wt. %. The pretreated cellulosic-containing material can be unwashed or washed using any method known in the art, e.g., washed with water.

(409) In one embodiment, the cellulosic-containing material is subjected to mechanical or physical pretreatment. The term “mechanical pretreatment” or “physical pretreatment” refers to any pretreatment that promotes size reduction of particles. For example, such pretreatment can involve various types of grinding or milling (e.g., dry milling, wet milling, or vibratory ball milling).

(410) The cellulosic-containing material can be pretreated both physically (mechanically) and chemically. Mechanical or physical pretreatment can be coupled with steaming/steam explosion, hydrothermolysis, dilute or mild acid treatment, high temperature, high pressure treatment, irradiation (e.g., microwave irradiation), or combinations thereof. In one aspect, high pressure means pressure in the range of preferably about 100 to about 400 psi, e.g., about 150 to about 250 psi. In another aspect, high temperature means temperature in the range of about 100 to about 300° C., e.g., about 140 to about 200° C. In a preferred aspect, mechanical or physical pretreatment is performed in a batch-process using a steam gun hydrolyzer system that uses high pressure and high temperature as defined above, e.g., a Sunds Hydrolyzer available from Sunds Defibrator AB, Sweden. The physical and chemical pretreatments can be carried out sequentially or simultaneously, as desired.

(411) Accordingly, in one embodiment, the cellulosic-containing material is subjected to physical (mechanical) or chemical pretreatment, or any combination thereof, to promote the separation and/or release of cellulose, hemicellulose, and/or lignin.

(412) In one embodiment, the cellulosic-containing material is subjected to a biological pretreatment. The term “biological pretreatment” refers to any biological pretreatment that promotes the separation and/or release of cellulose, hemicellulose, and/or lignin from the cellulosic-containing material. Biological pretreatment techniques can involve applying lignin-solubilizing microorganisms and/or enzymes (see, for example, Hsu, T.-A., 1996, Pretreatment of biomass, in *Handbook on Bioethanol: Production and Utilization*, Wyman, C. E., ed., Taylor & Francis, Washington, DC, 179-212; Ghosh and Singh, 1993, *Adv. Appl. Microbiol.* 39: 295-333; McMillan, J. D., 1994, Pretreating lignocellulosic biomass: a review, in *Enzymatic Conversion of Biomass for Fuels Production*, Himmel, M. E., Baker, J. O., and Overend, R. P., eds., ACS Symposium Series 566, American Chemical Society, Washington, DC, chapter 15; Gong, C. S., Cao, N. J., Du, J., and Tsao, G. T., 1999, Ethanol production from renewable resources, in *Advances in Biochemical Engineering/Biotechnology*, Scheper, T., ed., Springer-Verlag Berlin Heidelberg, Germany, 65: 207-241; Olsson and Hahn-Hagerdal, 1996, *Enz. Microb. Tech.* 18: 312-

331; and Vallander and Eriksson, 1990, *Adv. Biochem. Eng./Biotechnol.* 42: 63-95).

(413) Saccharification and Fermentation of Cellulosic-Containing Material

(414) Saccharification (i.e., hydrolysis) and fermentation, separate or simultaneous, include, but are not limited to, separate hydrolysis and fermentation (SHF); simultaneous saccharification and fermentation (SSF); simultaneous saccharification and co-fermentation (SSCF); hybrid hydrolysis and fermentation (HHF); separate hydrolysis and co-fermentation (SHCF); hybrid hydrolysis and co-fermentation (HHCF).

(415) SHF uses separate process steps to first enzymatically hydrolyze the cellulosic-containing material to fermentable sugars, e.g., glucose, cellobiose, and pentose monomers, and then ferment the fermentable sugars to ethanol. In SSF, the enzymatic hydrolysis of the cellulosic-containing material and the fermentation of sugars to ethanol are combined in one step (Philippidis, G. P., 1996, *Cellulose bioconversion technology*, in *Handbook on Bioethanol: Production and Utilization*, Wyman, C. E., ed., Taylor & Francis, Washington, DC, 179-212). SSCF involves the co-fermentation of multiple sugars (Sheehan and Himmel, 1999, *Biotechnol. Prog.* 15: 817-827). HHF involves a separate hydrolysis step, and in addition a simultaneous saccharification and hydrolysis step, which can be carried out in the same reactor. The steps in an HHF process can be carried out at different temperatures, i.e., high temperature enzymatic saccharification followed by SSF at a lower temperature that the fermentation organism can tolerate. It is understood herein that any method known in the art comprising pretreatment, enzymatic hydrolysis (saccharification), fermentation, or a combination thereof, can be used in the practicing the processes described herein.

(416) A conventional apparatus can include a fed-batch stirred reactor, a batch stirred reactor, a continuous flow stirred reactor with ultrafiltration, and/or a continuous plug-flow column reactor (de Castilhos Corazza et al., 2003, *Acta Scientiarum. Technology* 25: 33-38; Gusakov and Sinitsyn, 1985, *Enz. Microb. Technol.* 7: 346-352), an attrition reactor (Ryu and Lee, 1983, *Biotechnol. Bioeng.* 25: 53-65). Additional reactor types include fluidized bed, upflow blanket, immobilized, and extruder type reactors for hydrolysis and/or fermentation.

(417) In the saccharification step (i.e., hydrolysis step), the cellulosic and/or starch-containing material, e.g., pretreated, is hydrolyzed to break down cellulose, hemicellulose, and/or starch to fermentable sugars, such as glucose, cellobiose, xylose, xylulose, arabinose, mannose, galactose, and/or soluble oligosaccharides. The hydrolysis is performed enzymatically e.g., by a cellulolytic enzyme composition. The enzymes of the compositions can be added simultaneously or sequentially.

(418) Enzymatic hydrolysis may be carried out in a suitable aqueous environment under conditions that can be readily determined by one skilled in the art. In one aspect, hydrolysis is performed under conditions suitable for the activity of the enzyme(s), i.e., optimal for the enzyme(s). The hydrolysis can be carried out as a fed batch or continuous process where the cellulosic and/or starch-containing material is fed gradually to, for example, an enzyme containing hydrolysis solution.

(419) The saccharification is generally performed in stirred-tank reactors or fermentors under controlled pH, temperature, and mixing conditions. Suitable process time, temperature and pH conditions can readily be determined by one skilled in the art. For example, the saccharification can last up to 200 hours, but is typically performed for preferably about 12 to about 120 hours, e.g., about 16 to about 72 hours or about 24 to about 48 hours. The temperature is in the range of preferably about 25° C. to about 70° C., e.g., about 30° C. to about 65° C., about 40° C. to about 60° C., or about 50° C. to about 55° C. The pH is in the range of preferably about 3 to about 8, e.g., about 3.5 to about 7, about 4 to about 6, or about 4.5 to about 5.5. The dry solids content is in the range of preferably about 5 to about 50 wt. %, e.g., about 10 to about 40 wt. % or about 20 to about 30 wt. %.

(420) Saccharification in may be carried out using a cellulolytic enzyme composition. Such enzyme

compositions are described below in the “Cellulolytic Enzyme Composition”-section below. The cellulolytic enzyme compositions can comprise any protein useful in degrading the cellulosic-containing material. In one aspect, the cellulolytic enzyme composition comprises or further comprises one or more (e.g., several) proteins selected from the group consisting of a cellulase, an AA9 (GH61) polypeptide, a hemicellulase, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin.

(421) In another embodiment, the cellulase is preferably one or more (e.g., several) enzymes selected from the group consisting of an endoglucanase, a cellobiohydrolase, and a beta-glucosidase.

(422) In another embodiment, the hemicellulase is preferably one or more (e.g., several) enzymes selected from the group consisting of an acetylmannan esterase, an acetylxylan esterase, an arabinanase, an arabinofuranosidase, a coumaric acid esterase, a feruloyl esterase, a galactosidase, a glucuronidase, a glucuronoyl esterase, a mannanase, a mannosidase, a xylanase, and a xylosidase. In another embodiment, the oxidoreductase is one or more (e.g., several) enzymes selected from the group consisting of a catalase, a laccase, and a peroxidase. The enzymes or enzyme compositions used in a processes of the present invention may be in any form suitable for use, such as, for example, a fermentation broth formulation or a cell composition, a cell lysate with or without cellular debris, a semi-purified or purified enzyme preparation, or a host cell as a source of the enzymes. The enzyme composition may be a dry powder or granulate, a non-dusting granulate, a liquid, a stabilized liquid, or a stabilized protected enzyme. Liquid enzyme preparations may, for instance, be stabilized by adding stabilizers such as a sugar, a sugar alcohol or another polyol, and/or lactic acid or another organic acid according to established processes.

(423) In one embodiment, an effective amount of cellulolytic or hemicellulolytic enzyme composition to the cellulosic-containing material is about 0.5 to about 50 mg, e.g., about 0.5 to about 40 mg, about 0.5 to about 25 mg, about 0.75 to about 20 mg, about 0.75 to about 15 mg, about 0.5 to about 10 mg, or about 2.5 to about 10 mg per g of the cellulosic-containing material.

(424) In one embodiment, such a compound is added at a molar ratio of the compound to glucosyl units of cellulose of about $10^{\text{sup.}-6}$ to about 10^0 , e.g., about $10^{\text{sup.}-6}$ to about 7.5 , about $10^{\text{sup.}-6}$ to about 5 , about $10^{\text{sup.}-6}$ to about 2.5 , about $10^{\text{sup.}-6}$ to about 1 , about $10^{\text{sup.}-5}$ to about 1 , about $10^{\text{sup.}-5}$ to about $10^{\text{sup.}-1}$, about $10^{\text{sup.}-4}$ to about $10^{\text{sup.}-1}$, about $10^{\text{sup.}-3}$ to about $10^{\text{sup.}-1}$, or about $10^{\text{sup.}-3}$ to about $10^{\text{sup.}-2}$. In another aspect, an effective amount of such a compound is about 0.1 μM to about 1 M, e.g., about 0.5 μM to about 0.75 M, about 0.75 μM to about 0.5 M, about 1 μM to about 0.25 M, about 1 μM to about 0.1 M, about 5 μM to about 50 mM, about 10 μM to about 25 mM, about 50 μM to about 25 mM, about 10 μM to about 10 mM, about 5 μM to about 5 mM, or about 0.1 mM to about 1 mM.

(425) The term “liquor” means the solution phase, either aqueous, organic, or a combination thereof, arising from treatment of a lignocellulose and/or hemicellulose material in a slurry, or monosaccharides thereof, e.g., xylose, arabinose, mannose, etc., under conditions as described in WO 2012/021401, and the soluble contents thereof. A liquor for cellulolytic enhancement of an AA9 polypeptide (GH61 polypeptide) can be produced by treating a lignocellulose or hemicellulose material (or feedstock) by applying heat and/or pressure, optionally in the presence of a catalyst, e.g., acid, optionally in the presence of an organic solvent, and optionally in combination with physical disruption of the material, and then separating the solution from the residual solids. Such conditions determine the degree of cellulolytic enhancement obtainable through the combination of liquor and an AA9 polypeptide during hydrolysis of a cellulosic substrate by a cellulolytic enzyme preparation. The liquor can be separated from the treated material using a method standard in the art, such as filtration, sedimentation, or centrifugation.

(426) In one embodiment, an effective amount of the liquor to cellulose is about $10^{\text{sup.}-6}$ to about 10^0 g per g of cellulose, e.g., about $10^{\text{sup.}-6}$ to about 7.5 g, about $10^{\text{sup.}-6}$ to about 5 g, about $10^{\text{sup.}-6}$ to about 2.5 g, about $10^{\text{sup.}-6}$ to about 1 g, about $10^{\text{sup.}-5}$ to about 1 g, about $10^{\text{sup.}}$

–5 to about 10.sup.–1 g, about 10.sup.–4 to about 10.sup.–1 g, about 10.sup.–3 to about 10.sup.–1 g, or about 10.sup.–3 to about 10.sup.–2 g per g of cellulose.

(427) In the fermentation step, sugars, released from the cellulosic-containing material, e.g., as a result of the pretreatment and enzymatic hydrolysis steps, are fermented to ethanol, by a fermenting organism, such as yeast described herein. Hydrolysis (saccharification) and fermentation can be separate or simultaneous.

(428) Any suitable hydrolyzed cellulosic-containing material can be used in the fermentation step in practicing the processes described herein. Such feedstocks include, but are not limited to carbohydrates (e.g., lignocellulose, xylans, cellulose, starch, etc.). The material is generally selected based on economics, i.e., costs per equivalent sugar potential, and recalcitrance to enzymatic conversion.

(429) Production of ethanol by a fermenting organism using cellulosic-containing material results from the metabolism of sugars (monosaccharides). The sugar composition of the hydrolyzed cellulosic-containing material and the ability of the fermenting organism to utilize the different sugars has a direct impact in process yields. Prior to Applicant's disclosure herein, strains known in the art utilize glucose efficiently but do not (or very limitedly) metabolize pentoses like xylose, a monosaccharide commonly found in hydrolyzed material.

(430) Compositions of the fermentation media and fermentation conditions depend on the fermenting organism and can easily be determined by one skilled in the art. Typically, the fermentation takes place under conditions known to be suitable for generating the fermentation product. In some embodiments, the fermentation process is carried out under aerobic or microaerophilic (i.e., where the concentration of oxygen is less than that in air), or anaerobic conditions. In some embodiments, fermentation is conducted under anaerobic conditions (i.e., no detectable oxygen), or less than about 5, about 2.5, or about 1 mmol/L/h oxygen. In the absence of oxygen, the NADH produced in glycolysis cannot be oxidized by oxidative phosphorylation. Under anaerobic conditions, pyruvate or a derivative thereof may be utilized by the host cell as an electron and hydrogen acceptor in order to generate NAD⁺.

(431) The fermentation process is typically run at a temperature that is optimal for the recombinant fungal cell. For example, in some embodiments, the fermentation process is performed at a temperature in the range of from about 25° C. to about 42° C. Typically the process is carried out at a temperature that is less than about 38° C., less than about 35° C., less than about 33° C., or less than about 38° C., but at least about 20° C., 22° C., or 25° C.

(432) A fermentation stimulator can be used in a process described herein to further improve the fermentation, and in particular, the performance of the fermenting organism, such as, rate enhancement and product yield (e.g., ethanol yield). A “fermentation stimulator” refers to stimulators for growth of the fermenting organisms, in particular, yeast. Preferred fermentation stimulators for growth include vitamins and minerals. Examples of vitamins include multivitamins, biotin, pantothenate, nicotinic acid, meso-inositol, thiamine, pyridoxine, para-aminobenzoic acid, folic acid, riboflavin, and Vitamins A, B, C, D, and E. See, for example, Alfenore et al., Improving ethanol production and viability of *Saccharomyces cerevisiae* by a vitamin feeding strategy during fed-batch process, Springer-Verlag (2002), which is hereby incorporated by reference. Examples of minerals include minerals and mineral salts that can supply nutrients comprising P, K, Mg, S, Ca, Fe, Zn, Mn, and Cu.

(433) Cellulolytic Enzymes and Compositions

(434) A cellulolytic enzyme or cellulolytic enzyme composition may be present and/or added during saccharification. A cellulolytic enzyme composition is an enzyme preparation containing one or more (e.g., several) enzymes that hydrolyze cellulosic-containing material. Such enzymes include endoglucanase, cellobiohydrolase, beta-glucosidase, and/or combinations thereof.

(435) In some embodiments, the fermenting organism comprises one or more (e.g., several) heterologous polynucleotides encoding enzymes that hydrolyze cellulosic-containing material (e.g.,

an endoglucanase, cellobiohydrolase, beta-glucosidase or combinations thereof). Any enzyme described or referenced herein that hydrolyzes cellulosic-containing material is contemplated for expression in the fermenting organism.

(436) The cellulolytic enzyme may be any cellulolytic enzyme that is suitable for the host cells and/or the methods described herein (e.g., an endoglucanase, cellobiohydrolase, beta-glucosidase), such as a naturally occurring cellulolytic enzyme or a variant thereof that retains cellulolytic enzyme activity.

(437) In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a cellulolytic enzyme has an increased level of cellulolytic enzyme activity (e.g., increased endoglucanase, cellobiohydrolase, and/or beta-glucosidase) compared to the host cells without the heterologous polynucleotide encoding the cellulolytic enzyme, when cultivated under the same conditions. In some embodiments, the fermenting organism has an increased level of cellulolytic enzyme activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the fermenting organism without the heterologous polynucleotide encoding the cellulolytic enzyme, when cultivated under the same conditions.

(438) Exemplary cellulolytic enzymes that can be used with the host cells and/or the methods described herein include bacterial, yeast, or filamentous fungal cellulolytic enzymes, e.g., obtained from any of the microorganisms described or referenced herein, as described supra under the sections related to proteases.

(439) The cellulolytic enzyme may be of any origin. In an embodiment the cellulolytic enzyme is derived from a strain of *Trichoderma*, such as a strain of *Trichoderma reesei*; a strain of *Humicola*, such as a strain of *Humicola insolens*, and/or a strain of *Chrysosporium*, such as a strain of *Chrysosporium lucknowense*. In a preferred embodiment the cellulolytic enzyme is derived from a strain of *Trichoderma reesei*.

(440) The cellulolytic enzyme composition may further comprise one or more of the following polypeptides, such as enzymes: AA9 polypeptide (GH61 polypeptide) having cellulolytic enhancing activity, beta-glucosidase, xylanase, beta-xylosidase, CBH I, CBH II, or a mixture of two, three, four, five or six thereof.

(441) The further polypeptide(s) (e.g., AA9 polypeptide) and/or enzyme(s) (e.g., beta-glucosidase, xylanase, beta-xylosidase, CBH I and/or CBH II) may be foreign to the cellulolytic enzyme composition producing organism (e.g., *Trichoderma reesei*).

(442) In an embodiment the cellulolytic enzyme composition comprises an AA9 polypeptide having cellulolytic enhancing activity and a beta-glucosidase.

(443) In another embodiment the cellulolytic enzyme composition comprises an AA9 polypeptide having cellulolytic enhancing activity, a beta-glucosidase, and a CBH I.

(444) In another embodiment the cellulolytic enzyme composition comprises an AA9 polypeptide having cellulolytic enhancing activity, a beta-glucosidase, a CBH I and a CBH II.

(445) Other enzymes, such as endoglucanases, may also be comprised in the cellulolytic enzyme composition.

(446) As mentioned above the cellulolytic enzyme composition may comprise a number of different polypeptides, including enzymes.

(447) In one embodiment, the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Thermoascus aurantiacus* AA9 (GH61A) polypeptide having cellulolytic enhancing activity (e.g., WO 2005/074656), and *Aspergillus oryzae* beta-glucosidase fusion protein (e.g., one disclosed in WO 2008/057637, in particular shown as SEQ ID NOs: 59 and 60).

(448) In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Thermoascus aurantiacus* AA9 (GH61A) polypeptide having cellulolytic enhancing activity (e.g., SEQ ID NO: 2 in WO 2005/074656), and

Aspergillus fumigatus beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499).

(449) In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO 2011/041397, and *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499).

(450) In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO 2011/041397, and *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499) or a variant disclosed in WO 2012/044915 (hereby incorporated by reference), in particular one comprising one or more such as all of the following substitutions: F100D, S283G, N456E, F512Y.

(451) In an embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic composition, further comprising an AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one derived from a strain of *Penicillium emersonii* (e.g., SEQ ID NO: 2 in WO 2011/041397), *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 in WO 2005/047499) variant with one or more, in particular all of the following substitutions: F100D, S283G, N456E, F512Y and disclosed in WO 2012/044915; *Aspergillus fumigatus* Cel7A CBH1, e.g., the one disclosed as SEQ ID NO: 6 in WO2011/057140 and *Aspergillus fumigatus* CBH II, e.g., the one disclosed as SEQ ID NO: 18 in WO 2011/057140.

(452) In a preferred embodiment the cellulolytic enzyme composition is a *Trichoderma reesei*, cellulolytic enzyme composition, further comprising a hemicellulase or hemicellulolytic enzyme composition, such as an *Aspergillus fumigatus* xylanase and *Aspergillus fumigatus* beta-xylosidase.

(453) In an embodiment the cellulolytic enzyme composition also comprises a xylanase (e.g., derived from a strain of the genus *Aspergillus*, in particular *Aspergillus aculeatus* or *Aspergillus fumigatus*; or a strain of the genus *Talaromyces*, in particular *Talaromyces leycettanus*) and/or a beta-xylosidase (e.g., derived from *Aspergillus*, in particular *Aspergillus fumigatus*, or a strain of *Talaromyces*, in particular *Talaromyces emersonii*).

(454) In an embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Thermoascus aurantiacus* AA9 (GH61A) polypeptide having cellulolytic enhancing activity (e.g., WO 2005/074656), *Aspergillus oryzae* beta-glucosidase fusion protein (e.g., one disclosed in WO 2008/057637, in particular as SEQ ID NOs: 59 and 60), and *Aspergillus aculeatus* xylanase (e.g., Xyl II in WO 94/21785).

(455) In another embodiment the cellulolytic enzyme composition comprises a *Trichoderma reesei* cellulolytic preparation, further comprising *Thermoascus aurantiacus* GH61A polypeptide having cellulolytic enhancing activity (e.g., SEQ ID NO: 2 in WO 2005/074656), *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499) and *Aspergillus aculeatus* xylanase (Xyl II disclosed in WO 94/21785).

(456) In another embodiment the cellulolytic enzyme composition comprises a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Thermoascus aurantiacus* AA9 (GH61A) polypeptide having cellulolytic enhancing activity (e.g., SEQ ID NO: 2 in WO 2005/074656), *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499) and *Aspergillus aculeatus* xylanase (e.g., Xyl II disclosed in WO 94/21785).

(457) In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO 2011/041397, *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499) and *Aspergillus fumigatus* xylanase (e.g., Xyl III in WO 2006/078256).

(458) In another embodiment the cellulolytic enzyme composition comprises a *Trichoderma reesei*

cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO 2011/041397, *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499), *Aspergillus fumigatus* xylanase (e.g., Xyl III in WO 2006/078256), and CBH I from *Aspergillus fumigatus*, in particular Cel7A CBH1 disclosed as SEQ ID NO: 2 in WO2011/057140.

(459) In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO 2011/041397, *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499), *Aspergillus fumigatus* xylanase (e.g., Xyl III in WO 2006/078256), CBH I from *Aspergillus fumigatus*, in particular Cel7A CBH1 disclosed as SEQ ID NO: 2 in WO 2011/057140, and CBH II derived from *Aspergillus fumigatus* in particular the one disclosed as SEQ ID NO: 4 in WO 2013/028928.

(460) In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition, further comprising *Penicillium emersonii* AA9 (GH61A) polypeptide having cellulolytic enhancing activity, in particular the one disclosed in WO 2011/041397, *Aspergillus fumigatus* beta-glucosidase (e.g., SEQ ID NO: 2 of WO 2005/047499) or variant thereof with one or more, in particular all, of the following substitutions: F100D, S283G, N456E, F512Y; *Aspergillus fumigatus* xylanase (e.g., Xyl III in WO 2006/078256), CBH I from *Aspergillus fumigatus*, in particular Cel7A CBH I disclosed as SEQ ID NO: 2 in WO 2011/057140, and CBH II derived from *Aspergillus fumigatus*, in particular the one disclosed in WO 2013/028928.

(461) In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising the CBH I (GENSEQP Accession No. AZY49536 (WO2012/103293)); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288)); a beta-glucosidase variant (GENSEQP Accession No. AZU67153 (WO 2012/44915)), in particular with one or more, in particular all, of the following substitutions: F100D, S283G, N456E, F512Y; and AA9 (GH61 polypeptide) (GENSEQP Accession No. BAL61510 (WO 2013/028912)).

(462) In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising a CBH I (GENSEQP Accession No. AZY49536 (WO2012/103293)); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288)); a GH10 xylanase (GENSEQP Accession No. BAK46118 (WO 2013/019827)); and a beta-xylosidase (GENSEQP Accession No. AZI04896 (WO 2011/057140)).

(463) In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising a CBH I (GENSEQP Accession No. AZY49536 (WO2012/103293)); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288)); and an AA9 (GH61 polypeptide; GENSEQP Accession No. BAL61510 (WO 2013/028912)).

(464) In another embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising a CBH I (GENSEQP Accession No. AZY49536 (WO2012/103293)); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288)), an AA9 (GH61 polypeptide; GENSEQP Accession No. BAL61510 (WO 2013/028912)), and a catalase (GENSEQP Accession No. BAC11005 (WO 2012/130120)).

(465) In an embodiment the cellulolytic enzyme composition is a *Trichoderma reesei* cellulolytic enzyme composition comprising a CBH I (GENSEQP Accession No. AZY49446 (WO2012/103288)); a CBH II (GENSEQP Accession No. AZY49446 (WO2012/103288)), a beta-glucosidase variant (GENSEQP Accession No. AZU67153 (WO 2012/44915)), with one or more, in particular all, of the following substitutions: F100D, S283G, N456E, F512Y; an AA9 (GH61 polypeptide; GENSEQP Accession No. BAL61510 (WO 2013/028912)), a GH10 xylanase (GENSEQP Accession No. BAK46118 (WO 2013/019827)), and a beta-xylosidase (GENSEQP Accession No. AZI04896 (WO 2011/057140)).

(466) In an embodiment the cellulolytic composition is a *Trichoderma reesei* cellulolytic enzyme preparation comprising an EG I (Swissprot Accession No. P07981), EG II (EMBL Accession No. M19373), CBH I (supra); CBH II (supra); beta-glucosidase variant (supra) with the following substitutions: F100D, S283G, N456E, F512Y; an AA9 (GH61 polypeptide; supra), GH10 xylanase (supra); and beta-xylosidase (supra).

(467) All cellulolytic enzyme compositions disclosed in WO 2013/028928 are also contemplated and hereby incorporated by reference.

(468) The cellulolytic enzyme composition comprises or may further comprise one or more (several) proteins selected from the group consisting of a cellulase, a AA9 (i.e., GH61) polypeptide having cellulolytic enhancing activity, a hemicellulase, an expansin, an esterase, a laccase, a ligninolytic enzyme, a pectinase, a peroxidase, a protease, and a swollenin.

(469) In one embodiment the cellulolytic enzyme composition is a commercial cellulolytic enzyme composition. Examples of commercial cellulolytic enzyme compositions suitable for use in a process of the invention include: CELLIC® CTec (Novozymes A/S), CELLIC® CTec2 (Novozymes A/S), CELLIC® CTec3 (Novozymes A/S), CELLUCLAST™ (Novozymes A/S), SPEZYME™ CP (Genencor Int.), ACCELLERASE™ 1000, ACCELLERASE 1500, ACCELLERASE™ TRIO (DuPont), FILTRASE® NL (DSM); METHAPLUS® S/L 100 (DSM), ROHAMENT™ 7069 W (Röhm GmbH), or ALTERNAFUEL® CMAX3™ (Dyadic International, Inc.). The cellulolytic enzyme composition may be added in an amount effective from about 0.001 to about 5.0 wt. % of solids, e.g., about 0.025 to about 4.0 wt. % of solids or about 0.005 to about 2.0 wt. % of solids.

(470) Additional enzymes, and compositions thereof can be found in WO2011/153516 and WO2016/045569 (the contents of which are incorporated herein).

(471) Additional polynucleotides encoding suitable cellulolytic enzymes may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org).

(472) The cellulolytic enzyme coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding cellulolytic enzymes from strains of different genera or species, as described supra.

(473) The polynucleotides encoding cellulolytic enzymes may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc.) as described supra.

(474) Techniques used to isolate or clone polynucleotides encoding cellulolytic enzymes are described supra.

(475) In one embodiment, the cellulolytic enzyme has a mature polypeptide sequence of at least 60%, e.g., at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to any cellulolytic enzyme described or referenced herein (e.g., any endoglucanase, cellobiohydrolase, or beta-glucosidase). In one aspect, the cellulolytic enzyme has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from any cellulolytic enzyme described or referenced herein. In one embodiment, the cellulolytic enzyme has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any cellulolytic enzyme described or referenced herein, allelic variant, or a fragment thereof having cellulolytic enzyme activity. In one embodiment, the cellulolytic enzyme has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) amino acids. In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

(476) In some embodiments, the cellulolytic enzyme has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the cellulolytic enzyme activity of any cellulolytic enzyme described or referenced herein (e.g., any endoglucanase, cellobiohydrolase, or beta-glucosidase) under the same conditions.

(477) In one embodiment, the cellulolytic enzyme coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any cellulolytic enzyme described or referenced herein (e.g., any endoglucanase, cellobiohydrolase, or beta-glucosidase). In one embodiment, the cellulolytic enzyme coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any cellulolytic enzyme described or referenced herein.

(478) In one embodiment, the polynucleotide encoding the cellulolytic enzyme comprises the coding sequence of any cellulolytic enzyme described or referenced herein (e.g., any endoglucanase, cellobiohydrolase, or beta-glucosidase). In one embodiment, the polynucleotide encoding the cellulolytic enzyme comprises a subsequence of the coding sequence from any cellulolytic enzyme described or referenced herein, wherein the subsequence encodes a polypeptide having cellulolytic enzyme activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

(479) The cellulolytic enzyme can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

(480) Xylose Metabolism

(481) In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a xylose isomerase (XI). The xylose isomerase may be any xylose isomerase that is suitable for the host cells and the methods described herein, such as a naturally occurring xylose isomerase or a variant thereof that retains xylose isomerase activity. In one embodiment, the xylose isomerase is present in the cytosol of the host cells.

(482) In some embodiments, the fermenting organism comprising a heterologous polynucleotide encoding a xylose isomerase has an increased level of xylose isomerase activity compared to the host cells without the heterologous polynucleotide encoding the xylose isomerase, when cultivated under the same conditions. In some embodiments, the fermenting organisms have an increased level of xylose isomerase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the host cells without the heterologous polynucleotide encoding the xylose isomerase, when cultivated under the same conditions.

(483) Exemplary xylose isomerases that can be used with the recombinant host cells and methods of use described herein include, but are not limited to, XIs from the fungus *Piromyces* sp.

(WO2003/062430) or other sources (Madhavan et al., 2009, *Appl Microbiol Biotechnol.* 82(6), 1067-1078) have been expressed in *S. cerevisiae* host cells. Still other XIs suitable for expression in yeast have been described in US 2012/0184020 (an XI from *Ruminococcus flavefaciens*), WO2011/078262 (several XIs from *Reticulitermes speratus* and *Mastotermes darwiniensis*) and WO2012/009272 (constructs and fungal cells containing an XI from *Abiotrophia defectiva*). U.S. Pat. No. 8,586,336 describes a *S. cerevisiae* host cell expressing an XI obtained by bovine rumen fluid (shown herein as SEQ ID NO: 74).

(484) Additional polynucleotides encoding suitable xylose isomerases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org). In one embodiment, the xylose isomerases is a bacterial, a yeast, or a

filamentous fungal xylose isomerase, e.g., obtained from any of the microorganisms described or referenced herein, as described supra.

(485) The xylose isomerase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding xylose isomerases from strains of different genera or species, as described supra.

(486) The polynucleotides encoding xylose isomerases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc) as described supra.

(487) Techniques used to isolate or clone polynucleotides encoding xylose isomerases are described supra.

(488) In one embodiment, the xylose isomerase has a mature polypeptide sequence of having at least 60%, e.g., at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74). In one aspect, the xylose isomerase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74). In one embodiment, the xylose isomerase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74), allelic variant, or a fragment thereof having xylose isomerase activity. In one embodiment, the xylose isomerase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) amino acids. In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

(489) In some embodiments, the xylose isomerase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the xylose isomerase activity of any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74) under the same conditions.

(490) In one embodiment, the xylose isomerase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74). In one embodiment, the xylose isomerase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74).

(491) In one embodiment, the heterologous polynucleotide encoding the xylose isomerase comprises the coding sequence of any xylose isomerase described or referenced herein (e.g., the xylose isomerase of SEQ ID NO: 74). In one embodiment, the heterologous polynucleotide encoding the xylose isomerase comprises a subsequence of the coding sequence from any xylose isomerase described or referenced herein, wherein the subsequence encodes a polypeptide having xylose isomerase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

(492) The xylose isomerases can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

(493) In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a xylulokinase (XK). A xylulokinase, as used herein, provides enzymatic activity for converting D-xylulose to xylulose 5-phosphate. The xylulokinase may be any xylulokinase that is suitable for the host cells and the methods described herein, such as a naturally occurring xylulokinase or a variant thereof that retains xylulokinase activity. In one embodiment, the xylulokinase is present in the cytosol of the host cells.

(494) In some embodiments, the fermenting organisms comprising a heterologous polynucleotide encoding a xylulokinase have an increased level of xylulokinase activity compared to the host cells without the heterologous polynucleotide encoding the xylulokinase, when cultivated under the same conditions. In some embodiments, the host cells have an increased level of xylose isomerase activity of at least 5%, e.g., at least 10%, at least 15%, at least 20%, at least 25%, at least 50%, at least 100%, at least 150%, at least 200%, at least 300%, or at 500% compared to the host cells without the heterologous polynucleotide encoding the xylulokinase, when cultivated under the same conditions.

(495) Exemplary xylulokinases that can be used with the fermenting organisms and methods of use described herein include, but are not limited to, the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75. Additional polynucleotides encoding suitable xylulokinases may be obtained from microorganisms of any genus, including those readily available within the UniProtKB database (www.uniprot.org). In one embodiment, the xylulokinases is a bacterial, a yeast, or a filamentous fungal xylulokinase, e.g., obtained from any of the microorganisms described or referenced herein, as described supra.

(496) The xylulokinase coding sequences can also be used to design nucleic acid probes to identify and clone DNA encoding xylulokinases from strains of different genera or species, as described supra.

(497) The polynucleotides encoding xylulokinases may also be identified and obtained from other sources including microorganisms isolated from nature (e.g., soil, composts, water, etc.) or DNA samples obtained directly from natural materials (e.g., soil, composts, water, etc) as described supra.

(498) Techniques used to isolate or clone polynucleotides encoding xylulokinases are described supra.

(499) In one embodiment, the xylulokinase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity to any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75). In one embodiment, the xylulokinase has a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75). In one embodiment, the xylulokinase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75), allelic variant, or a fragment thereof having xylulokinase activity. In one embodiment, the xylulokinase has an amino acid substitution, deletion, and/or insertion of one or more (e.g., two, several) amino acids. In some embodiments, the total number of amino acid substitutions, deletions and/or insertions is not more than 10, e.g., not more than 9, 8, 7, 6, 5, 4, 3, 2, or 1.

(500) In some embodiments, the xylulokinase has at least 20%, e.g., at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of the xylulokinase activity of any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75) under the

same conditions.

(501) In one embodiment, the xylulokinase coding sequence hybridizes under at least low stringency conditions, e.g., medium stringency conditions, medium-high stringency conditions, high stringency conditions, or very high stringency conditions with the full-length complementary strand of the coding sequence from any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75). In one embodiment, the xylulokinase coding sequence has at least 65%, e.g., at least 70%, at least 75%, at least 80%, at least 85%, at least 85%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% sequence identity with the coding sequence from any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75).

(502) In one embodiment, the heterologous polynucleotide encoding the xylulokinase comprises the coding sequence of any xylulokinase described or referenced herein (e.g., the *Saccharomyces cerevisiae* xylulokinase of SEQ ID NO: 75). In one embodiment, the heterologous polynucleotide encoding the xylulokinase comprises a subsequence of the coding sequence from any xylulokinase described or referenced herein, wherein the subsequence encodes a polypeptide having xylulokinase activity. In one embodiment, the number of nucleotides residues in the subsequence is at least 75%, e.g., at least 80%, 85%, 90%, or 95% of the number of the referenced coding sequence.

(503) The xylulokinases can also include fused polypeptides or cleavable fusion polypeptides, as described supra.

(504) In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a ribulose 5 phosphate 3-epimerase (RPE1). A ribulose 5 phosphate 3-epimerase, as used herein, provides enzymatic activity for converting L-ribulose 5-phosphate to L-xylulose 5-phosphate (EC 5.1.3.22). The RPE1 may be any RPE1 that is suitable for the host cells and the methods described herein, such as a naturally occurring RPE1 or a variant thereof that retains RPE1 activity. In one embodiment, the RPE1 is present in the cytosol of the host cells.

(505) In one embodiment, the recombinant cell comprises a heterologous polynucleotide encoding a ribulose 5 phosphate 3-epimerase (RPE1), wherein the RPE1 is *Saccharomyces cerevisiae* RPE1, or an RPE1 having at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to a *Saccharomyces cerevisiae* RPE1.

(506) In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a ribulose 5 phosphate isomerase (RKI1). A ribulose 5 phosphate isomerase, as used herein, provides enzymatic activity for converting ribose-5-phosphate to ribulose 5-phosphate. The RKI1 may be any RKI1 that is suitable for the host cells and the methods described herein, such as a naturally occurring RKI1 or a variant thereof that retains RKI1 activity. In one embodiment, the RKI1 is present in the cytosol of the host cells.

(507) In one embodiment, the fermenting organism comprises a heterologous polynucleotide encoding a ribulose 5 phosphate isomerase (RKI1), wherein the RKI1 is a *Saccharomyces cerevisiae* RKI1, or an RKI1 having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to a *Saccharomyces cerevisiae* RKI1.

(508) In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a transketolase (TKL1). The TKL1 may be any TKL1 that is suitable for the host cells and the methods described herein, such as a naturally occurring TKL1 or a variant thereof that retains TKL1 activity. In one embodiment, the TKL1 is present in the cytosol of the host cells.

(509) In one embodiment, the fermenting organism comprises a heterologous polynucleotide encoding a transketolase (TKL1), wherein the TKL1 is a *Saccharomyces cerevisiae* TKL1, or a TKL1 having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%,

85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to a *Saccharomyces cerevisiae* TKL1.

(510) In one aspect, the fermenting organism (e.g., yeast cell) further comprises a heterologous polynucleotide encoding a transaldolase (TAL1). The TAL1 may be any TAL1 that is suitable for the host cells and the methods described herein, such as a naturally occurring TAL1 or a variant thereof that retains TAL1 activity. In one embodiment, the TAL1 is present in the cytosol of the host cells.

(511) In one embodiment, the fermenting organism comprises a heterologous polynucleotide encoding a transketolase (TAL1), wherein the TAL1 is a *Saccharomyces cerevisiae* TAL1, or a TAL1 having a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to a *Saccharomyces cerevisiae* TAL1.

(512) Fermentation Products

(513) A fermentation product can be any substance derived from the fermentation. The fermentation product can be, without limitation, an alcohol (e.g., arabinitol, n-butanol, isobutanol, ethanol, glycerol, methanol, ethylene glycol, 1,3-propanediol [propylene glycol], butanediol, glycerin, sorbitol, and xylitol); an alkane (e.g., pentane, hexane, heptane, octane, nonane, decane, undecane, and dodecane), a cycloalkane (e.g., cyclopentane, cyclohexane, cycloheptane, and cyclooctane), an alkene (e.g., pentene, hexene, heptene, and octene); an amino acid (e.g., aspartic acid, glutamic acid, glycine, lysine, serine, and threonine); a gas (e.g., methane, hydrogen (H.sub.2), carbon dioxide (CO.sub.2), and carbon monoxide (CO)); isoprene; a ketone (e.g., acetone); an organic acid (e.g., acetic acid, acetic acid, adipic acid, ascorbic acid, citric acid, 2,5-diketo-D-gluconic acid, formic acid, fumaric acid, glucaric acid, gluconic acid, glucuronic acid, glutaric acid, 3-hydroxypropionic acid, itaconic acid, lactic acid, malic acid, malonic acid, oxalic acid, oxaloacetic acid, propionic acid, succinic acid, and xylonic acid); and polyketide.

(514) In one aspect, the fermentation product is an alcohol. The term “alcohol” encompasses a substance that contains one or more hydroxyl moieties. The alcohol can be, but is not limited to, n-butanol, isobutanol, ethanol, methanol, arabinitol, butanediol, ethylene glycol, glycerin, glycerol, 1,3-propanediol, sorbitol, xylitol. See, for example, Gong et al., 1999, Ethanol production from renewable resources, in *Advances in Biochemical Engineering/Biotechnology*, Scheper, T., ed., Springer-Verlag Berlin Heidelberg, Germany, 65: 207-241; Silveira and Jonas, 2002, *Appl. Microbiol. Biotechnol.* 59: 400-408; Nigam and Singh, 1995, *Process Biochemistry* 30(2): 117-124; Ezeji et al., 2003, *World Journal of Microbiology and Biotechnology* 19(6): 595-603. In one embodiment, the fermentation product is ethanol.

(515) In another aspect, the fermentation product is an alkane. The alkane may be an unbranched or a branched alkane. The alkane can be, but is not limited to, pentane, hexane, heptane, octane, nonane, decane, undecane, or dodecane.

(516) In another aspect, the fermentation product is a cycloalkane. The cycloalkane can be, but is not limited to, cyclopentane, cyclohexane, cycloheptane, or cyclooctane.

(517) In another aspect, the fermentation product is an alkene. The alkene may be an unbranched or a branched alkene. The alkene can be, but is not limited to, pentene, hexene, heptene, or octene. In another aspect, the fermentation product is an amino acid. The organic acid can be, but is not limited to, aspartic acid, glutamic acid, glycine, lysine, serine, or threonine. See, for example, Richard and Margaritis, 2004, *Biotechnology and Bioengineering* 87(4): 501-515.

(518) In another aspect, the fermentation product is a gas. The gas can be, but is not limited to, methane, H.sub.2, CO.sub.2, or CO. See, for example, Kataoka et al., 1997, *Water Science and Technology* 36(6-7): 41-47; and Gunaseelan, 1997, *Biomass and Bioenergy* 13(1-2): 83-114.

(519) In another aspect, the fermentation product is isoprene.

(520) In another aspect, the fermentation product is a ketone. The term “ketone” encompasses a substance that contains one or more ketone moieties. The ketone can be, but is not limited to, acetone.

(521) In another aspect, the fermentation product is an organic acid. The organic acid can be, but is not limited to, acetic acid, acetonetic acid, adipic acid, ascorbic acid, citric acid, 2,5-diketo-D-gluconic acid, formic acid, fumaric acid, glucaric acid, gluconic acid, glucuronic acid, glutaric acid, 3-hydroxypropionic acid, itaconic acid, lactic acid, malic acid, malonic acid, oxalic acid, propionic acid, succinic acid, or xylonic acid. See, for example, Chen and Lee, 1997, *Appl. Biochem. Biotechnol.* 63-65: 435-448.

(522) In another aspect, the fermentation product is polyketide.

(523) In some aspects, the fermenting organism (or processes thereof), provide higher yield of fermentation product (e.g., ethanol) when compared to the same process using an identical cell without the heterologous polynucleotide encoding the phospholipase under the same conditions (e.g., at about or after 54 hours fermentation, such as the conditions described in Example 3 or 4). In some embodiments, the process results in at least 0.25%, such as 0.5%, 0.75%, 1.0%, 1.25%, 1.5%, 1.75%, 2%, 3% or 5% higher yield of the fermentation product (e.g., ethanol).

(524) Recovery

(525) The fermentation product, e.g., ethanol, can optionally be recovered from the fermentation medium using any method known in the art including, but not limited to, chromatography, electrophoretic procedures, differential solubility, distillation, or extraction. For example, alcohol is separated from the fermented cellulosic material and purified by conventional methods of distillation. Ethanol with a purity of up to about 96 vol. % can be obtained, which can be used as, for example, fuel ethanol, drinking ethanol, i.e., potable neutral spirits, or industrial ethanol.

(526) In some aspects of the methods, the fermentation product after being recovered is substantially pure. With respect to the methods herein, “substantially pure” intends a recovered preparation that contains no more than 15% impurity, wherein impurity intends compounds other than the fermentation product (e.g., ethanol). In one variation, a substantially pure preparation is provided wherein the preparation contains no more than 25% impurity, or no more than 20% impurity, or no more than 10% impurity, or no more than 5% impurity, or no more than 3% impurity, or no more than 1% impurity, or no more than 0.5% impurity.

(527) Suitable assays to test for the production of ethanol and contaminants, and sugar consumption can be performed using methods known in the art. For example, ethanol product, as well as other organic compounds, can be analyzed by methods such as HPLC (High Performance Liquid Chromatography), GC-MS (Gas Chromatography Mass Spectroscopy) and LC-MS (Liquid Chromatography-Mass Spectroscopy) or other suitable analytical methods using routine procedures well known in the art. The release of ethanol in the fermentation broth can also be tested with the culture supernatant. Byproducts and residual sugar in the fermentation medium (e.g., glucose or xylose) can be quantified by HPLC using, for example, a refractive index detector for glucose and alcohols, and a UV detector for organic acids (Lin et al., *Biotechnol. Bioeng.* 90:775-779 (2005)), or using other suitable assay and detection methods well known in the art.

(528) The invention may further be described in the following numbered paragraphs: Paragraph [1]. A method of producing a fermentation product from a starch-containing or cellulosic-containing material comprising: (a) saccharifying the starch-containing or cellulosic-containing material; and (b) fermenting the saccharified material of step (a) with a fermenting organism; wherein the fermenting organism comprises a heterologous polynucleotide encoding a phospholipase. Paragraph [2]. The method of paragraph [1], wherein the phospholipase is a Phospholipase A or a Phospholipase C. Paragraph [3]. The method of paragraph [1] or [2], wherein the phospholipase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). Paragraph [4]. The method of any one of paragraphs [1]-[3], wherein the heterologous polynucleotide encodes a phospholipase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than

four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). Paragraph [5]. The method of any one of paragraphs [1]-[4], wherein the heterologous polynucleotide encodes a phospholipase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). Paragraph [6]. The method of any one of paragraphs [1]-[5], wherein saccharification of step (a) occurs on a starch-containing material, and wherein the starch-containing material is either gelatinized or ungelatinized starch. Paragraph [7]. The method of paragraph [6], comprising liquefying the starch-containing material by contacting the material with an alpha-amylase prior to saccharification. Paragraph [8]. The method of paragraph [7], wherein liquefying the starch-containing material and/or saccharifying the starch-containing material is conducted in presence of exogenously added protease. Paragraph [9]. The method of any one of paragraphs [1]-[8], wherein fermentation is performed under reduced nitrogen conditions (e.g., less than 1000 ppm urea or ammonium hydroxide, such as less than 750 ppm, less than 500 ppm, less than 400 ppm, less than 300 ppm, less than 250 ppm, less than 200 ppm, less than 150 ppm, less than 100 ppm, less than 75 ppm, less than 50 ppm, less than 25 ppm, or less than 10 ppm). Paragraph [10]. The method of any one of paragraphs [1]-[9], wherein fermentation and saccharification are performed simultaneously in a simultaneous saccharification and fermentation (SSF). Paragraph [11]. The method of any one of paragraphs [1]-[9], wherein fermentation and saccharification are performed sequentially (SHF). Paragraph [12]. The method of any one of paragraphs paragraph [1]-[11], comprising recovering the fermentation product from the fermentation. Paragraph [13]. The method of paragraph [12], wherein recovering the fermentation product from the from the fermentation comprises distillation. Paragraph [14]. The method of any one of paragraphs [1]-[13], wherein the fermentation product is ethanol. Paragraph [15]. The method of any one of paragraphs [1]-[14], wherein the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase. Paragraph [16]. The method of paragraph [15], wherein the glucoamylase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of a *Pycnoporus* glycoamylase (e.g., a *Pycnoporus sanguineus* glucoamylase of SEQ ID NO: 229), a *Gloeophyllum* glucoamylase (e.g. a *Gloeophyllum sepiarium* of SEQ ID NO: 8), or a glucoamylase of any one of SEQ ID NOs: 102-113 (e.g., a *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103 or 104, or a *Trichoderma reesei* glucoamylase of SEQ ID NO: 230). Paragraph [17]. The method of any one of paragraphs [1]-[16], wherein the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase. Paragraph [18]. The method of paragraph [17], wherein the alpha-amylase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 76-101, 121-174 and 231. Paragraph [19]. The method of any one of paragraphs [1]-[18], wherein the fermenting organism comprises a heterologous polynucleotide encoding a protease. Paragraph [20]. The method of paragraph [19], wherein the protease has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69). Paragraph [21]. The method of any one of paragraphs [1]-[20], wherein the fermenting organism comprises a heterologous polynucleotide encoding a trehalase. Paragraph [22]. The method of paragraph [21], wherein the trehalase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 175-226. Paragraph [23]. The method of any one of paragraphs [1]-[22], wherein saccharification of step occurs on a cellulosic-containing material,

and wherein the cellulosic-containing material is pretreated. Paragraph [24]. The method of paragraph [23], wherein the pretreatment is a dilute acid pretreatment. Paragraph [25]. The method of any one of paragraphs [1]-[24], wherein saccharification occurs on a cellulosic-containing material, and wherein the enzyme composition comprises one or more enzymes selected from a cellulase, an AA9 polypeptide, a hemicellulase, a CIP, an esterase, an expansin, a ligninolytic enzyme, an oxidoreductase, a pectinase, a protease, and a swollenin. Paragraph [26]. The method of paragraph [25], wherein the cellulase is one or more enzymes selected from an endoglucanase, a cellobiohydrolase, and a beta-glucosidase. Paragraph [27]. The method of paragraph [25] or [26], wherein the hemicellulase is one or more enzymes selected a xylanase, an acetylxyloxy esterase, a feruloyl esterase, an arabinofuranosidase, a xylosidase, and a glucuronidase. Paragraph [28]. The method of any one of paragraphs [1]-[27], wherein the fermenting organism is a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell. Paragraph [29]. The method of any one of paragraphs [1]-[28], wherein the fermenting organism is a *Saccharomyces cerevisiae* cell. Paragraph [30]. The method of any one of paragraphs [1]-[29], wherein the method results in higher yield of fermentation product and/or reduced foam accumulation when compared to the same process using an identical cell without the heterologous polynucleotide encoding the phospholipase under the same conditions (e.g., at about or after 54 hours fermentation, such as the conditions described in Examples 3 or 4). Paragraph [31]. The method of paragraph [30], wherein the method results in at least 0.25% (e.g., 0.5%, 0.75%, 1.0%, 1.25%, 1.5%, 1.75%, 2%, 3% or 5%) higher yield of fermentation product. Paragraph [32]. A recombinant yeast cell comprising a heterologous polynucleotide encoding a phospholipase. Paragraph [33]. The recombinant yeast cell of paragraph [32], wherein the phospholipase is a Phospholipase A or a Phospholipase C. Paragraph [34]. The recombinant yeast cell of paragraph [32] or [33], wherein the phospholipase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). Paragraph [35]. The recombinant yeast cell of any one of paragraphs [32]-[34], wherein the heterologous polynucleotide encodes a phospholipase having a mature polypeptide sequence that differs by no more than ten amino acids, e.g., by no more than five amino acids, by no more than four amino acids, by no more than three amino acids, by no more than two amino acids, or by one amino acid from the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). Paragraph [36]. The recombinant yeast cell of any one of paragraphs [32]-[35], wherein the heterologous polynucleotide encodes a phospholipase having a mature polypeptide sequence comprising or consisting of the amino acid sequence of any one of SEQ ID NOs: 235-242 and 252-342 (e.g., any one of SEQ ID NOs: 235, 236, 237, 238, 239, 240, 241 and 242). Paragraph [37]. The recombinant yeast cell of any one of paragraphs [32]-[36], wherein the fermenting organism comprises a heterologous polynucleotide encoding a glucoamylase. Paragraph [38]. The recombinant yeast cell of paragraph [37], wherein the glucoamylase has a mature polypeptide sequence with 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity, to the amino acid sequence of a *Pycnoporus* glycoamylase (e.g., a *Pycnoporus sanguineus* glucoamylase of SEQ ID NO: 229), a *Gloeophyllum* glucoamylase (e.g. a *Gloeophyllum sepiarium* of SEQ ID NO: 8), or a glucoamylase of any one of SEQ ID NOs: 102-113 (e.g., a *Saccharomycopsis fibuligera* glucoamylase of SEQ ID NO: 103 or 104, or a *Trichoderma reesei* glucoamylase of SEQ ID NO: 230). Paragraph [39]. The recombinant yeast cell of any one of paragraphs [32]-[38], wherein the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase. Paragraph [40]. The recombinant yeast cell of paragraph [39], wherein the alpha-amylase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID

NOs: 76-101, 121-174 and 231. Paragraph [41]. The recombinant yeast cell of any one of paragraphs [32]-[40], wherein the fermenting organism comprises a heterologous polynucleotide encoding a protease. Paragraph [42]. The recombinant yeast cell of paragraph [41], wherein the protease has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 9-73 (e.g., any one of SEQ ID NOs: 9, 14, 16, 21, 22, 33, 41, 45, 61, 62, 66, 67, and 69; such as any one of SEQ NOs: 9, 14, 16, and 69). Paragraph [43]. The recombinant yeast cell of any one of paragraphs [32]-[42], wherein the fermenting organism comprises a heterologous polynucleotide encoding a trehalase. Paragraph [44]. The recombinant yeast cell of paragraph [43], wherein the trehalase has a mature polypeptide sequence of at least 60%, e.g., at least 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100% sequence identity to the amino acid sequence of any one of SEQ ID NOs: 175-226. Paragraph [45]. The recombinant yeast of any one of paragraphs [32]-[44], wherein the cell is a *Saccharomyces*, *Rhodotorula*, *Schizosaccharomyces*, *Kluyveromyces*, *Pichia*, *Hansenula*, *Rhodospiridium*, *Candida*, *Yarrowia*, *Lipomyces*, *Cryptococcus*, or *Dekkera* sp. cell. Paragraph [46]. The recombinant yeast of paragraph [45], wherein the cell is a *Saccharomyces cerevisiae* cell. Paragraph [47]. The recombinant cell of any one of paragraphs [32]-[46], wherein the cell is capable of higher yield of fermentation product and/or reduced foam accumulation when compared to fermentation using the same process and an identical cell without the heterologous polynucleotide encoding the phospholipase under the same conditions (e.g., at about or after 54 hours fermentation, such as the conditions described in Examples 3 or 4). Paragraph [48]. The recombinant cell of paragraph [47], wherein the cell is capable of at least 0.25% (e.g., 0.5%, 0.75%, 1.0%, 1.25%, 1.5%, 1.75%, 2%, 3% or 5%) higher yield of fermentation product.

(529) The invention described and claimed herein is not to be limited in scope by the specific aspects herein disclosed, since these aspects are intended as illustrations of several aspects of the invention. Any equivalent aspects are intended to be within the scope of this invention. Indeed, various modifications of the invention in addition to those shown and described herein will become apparent to those skilled in the art from the foregoing description. Such modifications are also intended to fall within the scope of the appended claims. In the case of conflict, the present disclosure including definitions will control. All references are specifically incorporated by reference for that which is described.

(530) The following examples are offered to illustrate certain aspects of the present invention, but not in any way intended to limit the scope of the invention as claimed.

EXAMPLES

Materials and Methods

(531) Chemicals used as buffers and substrates were commercial products of at least reagent grade.

(532) ETHANOL RED® (“ER”): *Saccharomyces cerevisiae* yeast available from Fermentis/Lesaffre, USA.

(533) YPD+clonNAT plates were composed of 10 g of yeast extract, 20 g of peptone, 20 g bacto agar, and deionized water to 960 ml, followed by autoclave treatment. 40 mL sterile 50% glucose and 1 mL clonNAT stock solution was added, followed by mixing and pouring.

(534) clonNAT stock solution was composed of 2 g nourseothricin sulfate and deionized water to 20 ml.

Example 1: Construction of Yeast Strains Expressing a Heterologous Phospholipase

(535) This example describes the construction of yeast cells containing a heterologous phospholipase under control of a *S. cerevisiae* TDH3, TEF2, PGK1, ADH1 or RPL18B promoter. Three pieces of DNA containing the promoter, gene and terminator were designed to allow for homologous recombination between the three DNA fragments and into the X-3 locus of the yeast MHCT-484 (WO2018/222990). The resulting strain has one promoter-containing fragment (left fragment), one gene-containing fragment (middle fragment) and one PRM9 terminator fragment

(right fragment) integrated into the *S. cerevisiae* genome at the X-3 locus.

(536) Construction of the Promoter Containing Fragments (Left Fragments)

(537) Synthetic linear uncloned DNA containing 60 bp homology to the X-3 site, *S. cerevisiae* promoter TDH3, TEF2, PGK1, ADH1 or RPL18B (SEQ ID NOs: 1, 2, 4, 5, and 6, respectively) and coding sequence for the *S. cerevisiae* MF1 α signal peptide (SEQ ID NO: 7) were synthesized by Thermo Fisher Scientific (Waltham, MA). To generate additional linear DNA for transformation into yeast, each of the five linear DNAs containing the left cassette above was PCR amplified.

(538) Construction of the Terminator Containing Fragment (Right Fragment)

(539) Synthetic linear uncloned DNA containing the *S. cerevisiae* PRM9 terminator (SEQ ID NO: 243) and 300 bp homology to the X-3 site was synthesized by Thermo Fisher Scientific.

(540) Construction of the Gene Containing Fragment (Middle Fragment)

(541) Synthetic linear uncloned DNA containing coding sequence for the *S. cerevisiae* MF1 α signal peptide, coding sequence for the mature polypeptide and 50 bp PRM9 terminator was synthesized by Thermo Fisher Scientific.

(542) Integration of the Left, Middle and Right-Hand Fragments to Generate Yeast Strains with a Heterologous Phospholipase

(543) The yeast MHCT-484 (WO2018/222990) was transformed with the left, middle and right integration fragments described above. In each transformation pool a fixed left fragment and right fragment were used as well as a fixed middle fragment containing the phospholipase gene with 100 ng of each fragment. To aid homologous recombination of the left, middle and right fragments at the genomic X-3 sites a plasmid containing Cas9 and guide RNA specific to X-3 (pMcTs442) was also used in the transformation. These four components were transformed into the *S. cerevisiae* strain MHCT-484 following a yeast electroporation protocol. Transformants were selected on YPD+cloNAT to select for transformants that contain the CRISPR/Cas9 plasmid pMcTs442. Transformants were picked using a Q-pix Colony Picking System (Molecular Devices; San Jose, CA) to inoculate one well of 96-well plate containing YPD+cloNAT media. The plates were grown for 2 days then glycerol was added to 20% final concentration and the plates were stored at -80°C . until needed. Integration of specific phospholipase construct was verified by PCR with locus specific primers and subsequent sequencing. The strains generated in this example are shown in Table 5.

(544) TABLE-US-00005

TABLE 5	SEQ ID NO:	Strain	mature Donor Organism Name	Promoter	polypeptide)	(catalytic domain)
HP21-B03	pADH1v1 236	<i>Talaromyces leycettanus</i>	HP21-C03	pADH1v1 236	<i>Talaromyces leycettanus</i>	HP21-D04
pTEF2 236	<i>Talaromyces leycettanus</i>	HP21-D05	pTDH3 236	<i>Talaromyces leycettanus</i>	HP21-E01	pPGK1 236
<i>Talaromyces leycettanus</i>	HP21-F01	pPGK1 236	<i>Talaromyces leycettanus</i>	HP21-G06	pRPL18B 236	<i>Talaromyces leycettanus</i>
HP21-A02	pADH1v1 237	<i>Penicillium emersonii</i>	HP21-F04	pTDH3 237	<i>Penicillium emersonii</i>	HP21-F05
pRPL18B 237	<i>Penicillium emersonii</i>	HP21-A04	pTEF2 240	<i>Kionochaeta</i> sp.	HP21-B01	pPGK1 240
<i>Kionochaeta</i> sp.	HP21-C05	pTDH3 240	<i>Kionochaeta</i> sp.	HP21-G02	pADH1v1 240	<i>Kionochaeta</i> sp.
HP21-B05	pTDH3 241	<i>Mariannaea pinicola</i>	HP21-B06	pRPL18B 241	<i>Mariannaea pinicola</i>	HP21-C06
pRPL18B 241	<i>Mariannaea pinicola</i>	HP21-F02	pADH1v1 241	<i>Mariannaea pinicola</i>	HP21-H03	pTEF2 241
<i>Mariannaea pinicola</i>	HP21-G04	pTDH3 239	<i>Pseudomonas</i> sp. 62186	HP21-G05	pRPL18B 239	<i>Pseudomonas</i> sp. 62186
HP21-H05	pRPL18B 239	<i>Pseudomonas</i> sp. 62186	HP21-A03	pADH1v1 238	<i>Bacillus thuringiensis</i>	HP21-B04
pTEF2 238	<i>Bacillus thuringiensis</i>	HP21-C01	pPGK1 238	<i>Bacillus thuringiensis</i>	HP21-C04	pTEF2 238
<i>Bacillus thuringiensis</i>	HP21-D01	pPGK1 238	<i>Bacillus thuringiensis</i>	HP21-D06	pRPL18B 238	<i>Bacillus thuringiensis</i>
HP21-E06	pRPL18B 238	<i>Bacillus thuringiensis</i>	HP21-F06	pRPL18B 238	<i>Bacillus thuringiensis</i>	HP21-H02
pADH1v1 238	<i>Bacillus thuringiensis</i>	HP21-D03	pADH1v1 235	<i>Thermomyces lanuginosus</i>	HP21-E04	pTEF2 235
<i>Thermomyces lanuginosus</i>	HP21-E05	pTDH3 235	<i>Thermomyces lanuginosus</i>	HP21-G01	pPGK1 235	<i>Thermomyces lanuginosus</i>
HP21-H06	pRPL18B 235	<i>Thermomyces lanuginosus</i>	HP21-A01	pPGK1 242	<i>Fictibacillus macauensis</i>	HP21-

A05 pTDH3 242 *Fictibacillus macauensis* HP21-A06 pRPL18B 242 *Fictibacillus macauensis*
HP21-F03 pTEF2 242 *Fictibacillus macauensis* HP21-G03 pTEF2 242 *Fictibacillus macauensis*
HP21-H04 pTDH3 242 *Fictibacillus macauensis*

Example 2: Activity Assay and Small-Scale Fermentation of Yeast Strains Expressing Phospholipase

(545) Preparation of Yeast Cells for Activity Assays

(546) Yeast strains were cultivated for 24 hours in standard YPD media containing 2% glucose. After the cultivation, samples were centrifuged, and the supernatants assayed for enzyme activity as described below.

(547) PLA (Phospholipase A) Activity Assay

(548) PLA activity was detected by using the EnzChek® PLA1 Kit from Invitrogen (Carlsbad, CA). The EnzChek® Direct Phospholipase A Assay Kit measures phosphatidylcholine-specific phospholipase C (PC-PLC) activity. PLA1 hydrolyzes the ester linkage of phospholipids and fatty acids. The EnzChek® Phospholipase A1 substrate (PED-A1) is a dye-labeled glycerophosphoethanolamines with BODIPY® FL dye-labeled acyl chain. PLA assay measures the release of the dye at a fluorescence emission at 515 nm. Reaction conditions are described in Table 6.

(549) Initial Preparation of Solutions for PLA1 Activity Assay:

(550) 1. Reaction Buffer: 100 mM MOPS+0.5 mM Zn pH 7 2. PLA substrate: a. 40 ul DMSO (Comp E) to one vial PLA sub (Comp A) b. Protected from light c. Sufficient volume for 100 rxns 3. 500 U/mL stock of positive control PLA (Comp D) a. Dissolved Comp D vial in 100 uL of reaction buffer 4. 10 mM DOPC (Comp B) a. Dissolved Comp B vial in 100 ul of ETOH 5. 10 mM DOPG (Comp C) a. Dissolved Comp C vial in 100 ul of ETOH

Final Preparation of Solutions and Samples for PLA1 Activity Assay: 1. 10 U/mL positive control made by diluting 500 U/mL stock positive control in reaction buffer a. Example: added 20 ul of 500 U/mL positive control to 980 mL of Reaction buffer 2. Serial dilutions of positive control to obtain 8-point standard curve with initial concentration beginning at 10 U/ml a. Final positive control concentration two-fold lower (5 U/ml) when substrate added 3. When necessary, samples diluted in reaction buffer 4. Lipid Mix: 30 ul 10 mM DOPC, 30 ul 10 mM DOPG, 30 ul 1 mM PLA substrate 5. PLA substrate a. 50 ul lipid mixed slowly to 5 mL reaction buffer in a smaller beaker containing a stir bar. b. Stirred for ~2-5 minutes

Assay Protocol for PLA1 Activity Assay: 1. 50 ul of 8-point standard curve to columns 10, 11, 12 a. last row left blank for buffer 2. Added 50 ul of samples to remaining wells 3. Added 50 ul of PLA-lipid substrate to wells containing samples and controls. Mixed well without introducing bubbles. 4. Read a T0 at 505EX/515EM (450EX/515EM is ok as well) 5. Covered plate and incubate protected from light 6. Read after 5 hours of incubation

(551) TABLE-US-00006 TABLE 6 PLA1 Activity Assay Condition Amount of yeast supernatant 50 µl Amount of PLA-lipid substrate 50 µl Substrate PLA1 substrate from Kit Buffer 100 mM MOPS + 0.5 mM Zn pH 7.0 ± 0.05 Incubation temperature 22° C. (room Temperature) Reaction time 5 hrs Wavelength 505EX/515EM

PLC (Phospholipase C) Activity Assay

(552) PLC activity was detected by using EnzChek® PLC Kit from Invitrogen. The EnzChek® Phospholipase A Assay Kit measures phosphatidylcholine-specific phospholipase C (PC-PLC) activity by measuring the amount of starch degraded through enzymatic hydrolysis of starch. The assay uses a proprietary substrate (glycero-phosphoethanolamine with a dye-labeled sn-2 acyl chain) to detect PLC activity. Substrate cleavage by PLC releases the dye-labeled diacylglycerol, which produces a fluorescence signal that can be measure at 516 nm emission. Reaction conditions are described in Table 7.

(553) Initial Preparation of Solutions for PLC Activity Assay:

(554) 1. Reaction Buffer: 100 mM NaAc+0.5 mM Zn pH 5 2. 200× stock of PLC substrate a. 100

ul DMSO (Comp B) added to one vial PLC sub b. Protected from light c. Sufficient volume for 125 rxns 3. 40 U/mL stock of positive control PC-PLC (Comp E) a. Dissolved Comp E vial in 200 uL of reaction buffer

Final Secondary Preparation of Solutions and Samples for PLC Activity Assay 1. 1 U/mL positive control made by diluting 40 U/mL stock positive control 40-fold a. Example: added 25 ul of 40 U/mL positive control to 0.975 mL of Reaction buffer 2. Serial dilutions of positive control to obtain 8-point standard curve with initial concentration beginning at 0.125 U/ml a. Final positive control concentration two-fold lower (0.0625 U/ml) when substrate is added 3. When necessary, samples diluted in reaction buffer 4. PLC substrate: Added 40 ul of lecithin (Comp D) and 100 ul of PLC Substrate 200× stock solution (prepared in earlier step) to 9.86 mL of reaction buffer.

Assay Protocol for PLC Activity Assay: 1. 75 ul of 8-point standard curve to columns 10, 11, 12 a. last row left blank for buffer 2. Added 75 ul of samples to remaining wells 3. Added 75 ul of PLC substrate to wells containing samples and controls. Mixed well without introducing bubbles. 4. Read a T0 at 509EX/516EM (490EX/520EM is ok as well) 5. Covered plate and incubate protected from light 6. Read after 5 hours

(555) TABLE-US-00007 TABLE 7 PLC Activity Assay Condition Amount of yeast supernatant 75 µl Amount of PLA-lipid substrate 75 µl Substrate PLC substrate from Kit Buffer 100 mM NaAc + 0.5 mM Zn pH 5.0 ± 0.05 Incubation temperature 22° C. (room Temperature) Reaction time 5 hrs Wavelength 509EX/516EM

Preparation of Yeast Culture for Microtiter Plate Fermentations

(556) Simultaneous saccharification and fermentation (SSF) was performed via mini-scale fermentations using industrial corn mash (Avantec® Amp; Novozymes A/S). Yeast strains were cultivated overnight in YPD media with 6% glucose for 24 hours at 30° C. and 300 rpm. The corn mash was supplemented with 250 ppm of urea and dosed with 0.45 AGU/g-DS of an exogenous glucoamylase enzyme product (Spirizyme® Excel; Novozymes A/S). Approximately 0.6 mL of corn mash was dispensed per well to 96 well microtiter plates, followed by the addition of approximately 10⁸ yeast cells/g of corn mash from the overnight culture. Plates were incubated at 32° C. without shaking. Fermentation was stopped by the addition of 100 µL of 8% H.sub.2SO.sub.4, followed by centrifugation at 3000 rpm for 10 min. The supernatant was analyzed for ethanol using HPLC. Fermentation reaction conditions are summarized in Table 8. SSF and activity assay results of yeast strains expressing a phospholipase are in Table 9.

(557) TABLE-US-00008 TABLE 8 Microtiter plate fermentation reaction conditions Substrate Avantec ® Amp corn mash Yeast pitch 10⁸ cells/g corn mash Supplementary urea 250 ppm Exogenous glucoamylase product dose 0.15 AGU/g-DS pH 5.0 ± 0.05 Incubation temperature 32° C. Reaction time 48 hours

(558) TABLE-US-00009 TABLE 9 Phospholipase yeast strains enzyme activity measurements and ethanol titers from SSF SEQ ID NO: PLC PLA1 Mean (mature activity activity Ethanol Strain ID Promoter polypeptide) units units (g/L) MHCT-484 none none 99 0 107.31 HP21-D04 TEF2 236 776 23 110.67 HP21-G06 RPL18B 236 793 0 110.6105 HP21-B03 ADH1v1 236 863 43 110.2833 HP21-D05 TDH3 236 688 29 110.0453 HP21-C03 ADH1v1 236 878 47 108.885 HP21-E01 PGK1 236 710 11 108.2008 HP21-F01 PGK1 236 680 0 106.1183 HP21-F04 TDH3 237 207 0 110.67 HP21-F05 RPL18B 237 174 0 109.1528 HP21-A02 ADH1v1 237 273 0 109.0933 HP21-B01 PGK1 240 452 0 94.0695 HP21-A04 TEF2 240 451 0 83.27025 HP21-G02 ADH1v1 240 410 0 82.19925 HP21-C05 TDH3 240 426 0 81.36625 HP21-C06 RPL18B 241 236 0 112.455 HP21-B06 RPL18B 241 201 0 107.8735 HP21-B05 TDH3 241 356 0 100.436 HP21-H03 TEF2 241 275 0 99.3055 HP21-F02 ADH1v1 241 364 0 97.58 HP21-G04 TDH3 239 85 0 107.1893 HP21-G05 RPL18B 239 251 0 84.7875 HP21-H05 RPL18B 239 220 0 84.609 HP21-E06 RPL18B 238 156 0 109.599 HP21-B04 TEF2 238 152 0 108.6768 HP21-D01 PGK1 238 99 0 108.0223 HP21-C01 PGK1 238 116 0 107.5463 HP21-F06 RPL18B 238 142 0 105.077 HP21-A03 ADH1v1 238 99 0 104.5118 HP21-H02 ADH1v1 238 93 0 102.102 HP21-C04 TEF2 238 146 0 101.9235 HP21-D06

RPL18B 235 141 0 82.25875 HP21-E05 TDH3 235 846 31 110.5808 HP21-H06 RPL18B 235 685
 56 109.0635 HP21-E04 TEF2 235 500 139 100.5253 HP21-G01 PGK1 235 428 123 96.06275
 HP21-D03 ADH1v1 235 462 142 88.08975 HP21-F03 TEF2 242 115 0 108.171 HP21-A05 TDH3
 242 126 0 108.0818 HP21-A06 RPL18B 242 135 0 106.2373 HP21-G03 TEF2 242 113 0 105.791
 HP21-H04 TDH3 242 102 0 104.6903 HP21-A01 PGK1 242 91 0 104.4523

Example 3: Fermentation of Yeast Strains Expressing Phospholipase

(559) The purpose of this experiment was to determine if expressing phospholipase in yeast would enhance ethanol yields at the end of fermentation. Yeast strains (HP21-H06, HP21-D05, HP21-G12, HP21-A08, HP21-F05 and HP21-F04) were propagated overnight, and then used to dose a fermentation of industrially liquified mash. The fermented mash was sampled at the end of fermentation and the supernatant was filtered and analyzed by HPLC to determine the final ethanol titers and residual sugars.

(560) Yeast Propagation

(561) Yeast strains were received as glycerol stocks and were kept frozen at -80°C . until used. The yeast strains were propagated according to the following procedure: 1. Cryo-vials of yeast were removed from the -80°C . freezer. 2. For each sample, $2 \times 50\text{ mL}$ sterile Erlenmeyer flasks were obtained and labeled. 3. The flasks were placed into a sterile hood along with 6% YPD media, pipettes and tips, and cryovials. 4. Using sterile technique, $\sim 50\text{ mL}$ of 6% YPD media was poured into each bottle. 5. 150 uL of the appropriate yeast sample was added to each flask using a sterile 200 uL pipette and sterile tips. 6. The flasks were capped after each addition to maintain a sterile solution. 7. The flasks were placed into a 32°C . shaking incubator and mixed at $\sim 150\text{ rpm}$ overnight.

Cell Counting

(562) The propagation slurries described above were then tested to determine the number of yeast cells present for dosing, according to the following procedure: 1. The propagated yeast samples were removed from the incubator. 2. The 50 mL samples were poured into 50 mL flip top falcon tubes. 3. The samples were centrifuged $3,500\text{ rpm}$ for 7 min . 4. The supernatant was discarded into the old 50 mL flasks to be autoclaved. 5. 10 mL of deionized water was added to each tube. 6. The tubes were vortex mixed to suspend the pellet. 7. The replicate samples were combined into a single 50 mL falcon tube. 8. The samples were centrifuged $3,500\text{ rpm}$ for 7 min . 9. The supernatant was discarded into the original 50 mL flasks to be autoclaved. 10. 10 mL of deionized water was added to each remaining tube. 11. The tubes were vortex mixed to suspend the pellet. a. This slurry was saved and will be used to dose into the fermentations. 12. The yeast slurries were diluted $100\times$ into 15 mL falcon tubes by adding 100 uL to 9.9 mL of deionized water. 13. The samples subjected to cell count using a NucleoCounter® YC-100 (Chemometec A/S) as follows: a. 50 uL of the $100\times$ slurry dilution was added to 450 uL of Lysis buffer. b. The samples were vortexed and allowed to sit $\sim 5\text{ min}$ for the cell lysis to occur. c. A nucleo-cassette was used to sample the lysed solution, then placed into the NucleoCounter® YC-100. d. The NucleoCounter® YC-100 analyzed the slurry and produced a cell count number. e. The resulting number is multiplied by 1000 to get the final cell counts for use in the spreadsheet. 14. The yeasts used in this experiment, and the yeast counts that were obtained according to the procedure above are displayed in the Table 10 below.

(563) TABLE-US-00010 TABLE 10 Average Strain ID YCL-1 YCL-2 total cells MHCT-484
 1.26E+09 1.28E+09 1.27E+09 HP21-H06 1.23E+09 1.16E+09 1.19E+09 HP21-D05 1.38E+09
 1.43E+09 1.40E+09 HP21-G12 1.14E+09 1.12E+09 1.13E+09 HP21-A08 1.38E+09 1.29E+09
 1.33E+09 HP21-F05 1.28E+09 1.28E+09 1.28E+09 HP21-F04 1.41E+09 1.24E+09 1.33E+09 15.
 The calculation below was used to determine the number of mL of yeast slurry to add to each tube.

(564) $\frac{10,000,000\text{cells}}{\text{mLmash}} \times \frac{1\text{mLmash}}{1.15\text{gmash}} \times \text{gmash} \times \frac{\text{mL}}{\text{Xcells}} \times \frac{1000.\text{Math.L}}{\text{mL}} = .\text{Math.Lyeasttoadd}$

Fermentation

(565) Fermentations of industrially liquified mash were conducted according to the following procedure: 1. Industrially liquefied mash was acquired and stored frozen for future analysis. The

mash used in this experiment was from a plant using Avantec® Amp and running a hydro-heater or jet cooker during liquefaction stage. 2. Two liters of the mash was thawed for approximately 2 hours prior to starting this study. 3. The mash was adjusted to 500 ppm urea and 3 ppm penicillin using stock solutions of 200 g/L urea and 1 g/L penicillin. 4. The mash was adjusted to pH 5 using 40% v/v H.sub.2SO.sub.4 and the dry solids content were measured on a Mettler-Toledo moisture balance. 5. 50.0+/-0.05 g of prepared mash was weighed into 250 mL media bottles. 6. The bottles were capped and stored in a refrigerator overnight. 7. The bottles were dosed with enzymes and yeast in the morning using a repeater pipette. 8. The conditions tested include Achieve® glucoamylase (0.42 AGU/gDS), trehalase (2 µg/gDS), and yeast dosed to approximately 7,000,000 cells per fermentation. 9. The bottles were capped with hole drilled caps. The holes were covered with labeling tape. A 0.24-gauge syringe needle was used to pierce a uniform vent hole to allow for CO.sub.2 gas to escape during fermentation. 10. The bottles were placed into a 32° C. air shaker set to 150 rpm to begin fermentation. 11. Fermentation was run for 54 hours.

Sampling of Fermentations 1. After 52 hours of fermentation, 5 g of fermented mash were pipetted into a 15 mL conical falcon tube with 5 mL pipette equipped with a cut tip. 2. 50 µL of 40% H.sub.2SO.sub.4 were added to each of the tubes to stop fermentation and enzymatic hydrolysis. 3. The tubes were briefly mixed with a vortex mixer and centrifuged at 3,500 rpm for 7 minutes. 4. The supernatant was poured into a syringe equipped with a 0.45 µm filter and the plunger of the syringe was used to push the sample through the filter into prenumbered HPLC vials. 5. The HPLC vials were capped and the samples were submitted to HPLC for analysis of the ethanol and sugars produced during fermentation. The HPLC setup was as shown in Table 11.

(566) TABLE-US-00011 TABLE 11 HPLC system Agilent's 1100/1200 series with Chem station software Degasser Quaternary Pump Auto-Sampler Column Compartment/w Heater Refractive Index Detector (RI) Column Bio-Rad HPX- 87H Ion Exclusion Column 300 mm × 7.8 mm parts# 125-0140 Bio-Rad guard cartridge cation H parts# 125- 0129, Holder parts# 125-0131 Method 0.005M H.sub.2SO.sub.4 mobile phase Flow rate of 0.6 ml/min Column temperature - 65° C. RI detector temperature - 55° C.

(567) The method quantifies analytes using calibration standards for dextrans (DP4+), maltotriose, maltose, glucose, fructose, acetic acid, lactic acid, glycerol and ethanol. A 4-point calibration including the origin was used.

(568) Ethanol improvement for tested strains is shown in FIG. 1. Strains HP21-F05 and HP21-F04 show approximately 1.1% improvement in ethanol over control strain MHCT-484, while strain HP21-A08 showed approximately 0.5% improvement in ethanol titer of the control.

Example 4: Fermentation of Yeast Strains Expressing Phospholipase at Varying Urea Levels

(569) The purpose of this experiment was to examine the effect of nitrogen loading (urea level) on ethanol yields of phospholipase-expressing yeasts. Yeast strains (HP21-H08, HP21-F04 and HP21-F05) were propagated overnight, and then used to dose a fermentation of industrially liquified mash. The fermented mash was sampled at the end of fermentation and the supernatant was filtered and run on HPLC to determine the final ethanol titers and residual sugars.

(570) Yeast Propagation

(571) Yeast strains were propagated as described above in Example 3.

(572) Cell Counting

(573) The propagation slurries described above were then tested to determine the number of yeast cells present for dosing according to the procedure described above in Example 3. The yeasts used in this experiment, and the corresponding yeast counts are displayed in the Table 12.

(574) TABLE-US-00012 TABLE 12 Average Strain ID YCL-1 YCL-2 total cells MHCT-484 1.48E+09 1.44E+09 1.46E+09 HP21-A08 1.24E+09 1.24E+09 1.24E+09 HP21-F05 1.50E+09 1.48E+09 1.49E+09 HP21-F04 1.37E+09 1.40E+09 1.38E+09

Fermentation

(575) Fermentations of industrially liquified mash were conducted according to the following

procedure: 1. Industrially liquefied mash was acquired and stored frozen for future analysis. The two mashes used in this experiment were from a plant using Avantec® Amp (Novozymes A/S) and a plant running another product Liquozyme® Pro (Novozymes A/S). Both mashes were from plants running a hydro-heater or jet cooker during liquefaction stage. 2. Two liters of each mash was thawed for approximately 2 hours prior to starting this study. 3. The mash was dosed with 3 ppm penicillin using a stock solution 1 g/L penicillin. Urea was added to the fermentation during enzyme dosing using stock solutions of 100 and 20 g/L. 4. The mash was adjusted to pH 5 using 40% v/v H.sub.2SO.sub.4 and the dry solids content were measured on a Mettler-Toledo moisture balance. 5. ~5 g of the prepared mash was added to preweighed 15 mL tubes 6. The tubes were reweighed and the weight of mash was used in calculating enzymes, yeast and urea dosing. 7. The tubes were capped and stored in a refrigerator overnight. 8. The tubes were dosed with enzymes, yeast and urea in the morning using a Biomek FX liquid handling robot. 9. The conditions tested in this experiment are displayed in Table 13 below. Yeast dose in this experiment was 7,000,000 cells per ferm. 10. The tubes were capped with holes to allow released CO.sub.2 from fermentation. 11. Tubes were placed into a 32° C. incubator to begin fermentation. 12. Tubes were mixed with a vortex mixer twice a day for 52 hours.

(576) TABLE-US-00013 TABLE 13 Treatment # Yeast# Glucoamylase Dose Units Trehalase Dose Units PLC Dose Units Urea Dose Units MHCT-484- 1 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 0 ppm Control MHCT-484- 2 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 5 µg/g DS Urea 0 ppm 5 ug PLC MHCT-484- 3 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 20 µg/g DS Urea 0 ppm 20 ug PLC HP21-A08 4 2 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 0 ppm HP21-F05 5 3 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 0 ppm HP21-F04 6 4 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 0 ppm MHCT-484- 7 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 150 ppm Control MHCT-484- 8 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 5 µg/g DS Urea 150 ppm 5 ug PLC MHCT-484- 9 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 20 µg/g DS Urea 150 ppm 20 ug PLC HP21-A08 10 2 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 150 ppm HP21-F05 11 3 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 150 ppm HP21-F04 12 4 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 150 ppm MHCT-484- 13 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 500 ppm Control MHCT-484- 14 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 5 µg/g DS Urea 500 ppm 5 ug PLC MHCT-484- 15 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 20 µg/g DS Urea 500 ppm 20 ug PLC HP21-A08 16 2 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 500 ppm HP21-F05 17 3 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 500 ppm HP21-F04 18 4 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 500 ppm MHCT-484- 19 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 0 ppm Control MHCT-484- 20 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 5 µg/g DS Urea 0 ppm 5 ug PLC MHCT-484- 21 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 20 µg/g DS Urea 0 ppm 20 ug PLC HP21-A08 22 2 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 0 ppm HP21-F05 23 3 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 0 ppm HP21-F04 24 4 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 0 ppm MHCT-484- 25 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 300 ppm Control MHCT-484- 26 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 5 µg/g DS Urea 300 ppm 5 ug PLC MHCT-484- 27 1 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 20 µg/g DS Urea 300 ppm 20 ug PLC HP21-A08 28 2 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 300 ppm HP21-F05 29 3 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 300 ppm HP21-F04 30 4 Achieve 0.42 AGU/gDS Trehalase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 300 ppm MHCT-484- 31 1 Achieve

0.42 AGU/gDS Trehelase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 1000 ppm Control MHCT-484-32 1 Achieve 0.42 AGU/gDS Trehelase 1 ug/gDS *P. emersonii* 5 µg/g DS Urea 1000 ppm 5 ug PLC MHCT-484-33 1 Achieve 0.42 AGU/gDS Trehelase 1 ug/gDS *P. emersonii* 20 µg/g DS Urea 1000 ppm 20 ug PLC HP21-A08 34 2 Achieve 0.42 AGU/gDS Trehelase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 1000 ppm HP21-F05 35 3 Achieve 0.42 AGU/gDS Trehelase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 1000 ppm HP21-F04 36 4 Achieve 0.42 AGU/gDS Trehelase 1 ug/gDS *P. emersonii* 0 µg/g DS Urea 1000 ppm

Sampling of Fermentations 1. After 52 hours of fermentation, 50 uL of 40% H.sub.2SO.sub.4 were added to each of the tubes to stop fermentation and enzymatic hydrolysis. 2. The tubes were briefly mixed with a vortex mixer and centrifuged at 3,500 rpm for 7 minutes. 3. The tubes were placed onto a Biomek liquid handler which was used to pipette 200 uL of supernatant from the tubes into a 96 well 0.22 um filter plate. 4. The filter plate was placed on top of a round bottom 96 well polypropylene plate. 5. The filter and plate were placed into a floor centrifuge with bioseal caps on the buckets and spun at 3,500 rpm for 7 minutes or until liquid had passed through all the wells of the 96 well filter plate. 6. The plates were removed from the centrifuge and the round bottom plate with the filtered sample was heat sealed and submitted to HPLC for analysis of the ethanol and sugars produced during fermentation. The HPLC setup was as shown in Table 14.

(577) TABLE-US-00014 TABLE 14 HPLC system Agilent's 1100/1200 series with Chem station software Degasser Quaternary Pump Auto-Sampler Column Compartment/w Heater Refractive Index Detector (RI) Column Bio-Rad HPX- 87H Ion Exclusion Column 300 mm × 7.8 mm parts# 125-0140 Bio-Rad guard cartridge cation H parts# 125- 0129, Holder parts# 125-0131 Method 0.005M H.sub.2SO.sub.4 mobile phase Flow rate of 0.6 ml/min Column temperature - 65° C. RI detector temperature - 55° C.

(578) The method quantifies analytes using calibration standards for dextrans (DP4+), maltotriose, maltose, glucose, fructose, acetic acid, lactic acid, glycerol and ethanol. A 4-point calibration including the origin was used. Normalized mean ethanol values after 54 hours of fermentation of AMP mash at 0 and 150 ppm urea are shown in FIG. 2.

(579) All tested strains expressing phospholipase showed greater increases in ethanol titer with no urea loading. Normalized mean ethanol values after 54 hours of fermentation of non-AMP mash at 0 and 150 ppm urea are shown in FIG. 3. All tested strains expressing phospholipase showed greater increases in ethanol titer with no urea loading.

Example 5: Construction of Additional Yeast Strains Expressing a Heterologous Phospholipase

(580) This example describes the construction of yeast cells containing the remaining heterologous phospholipases of Table 1 under the control of an *S. cerevisiae* PGK1 promoter. Three pieces of DNA containing the promoter, gene and terminator were designed to allow for homologous recombination between the four DNA fragments and into the XII-2 locus of the strain MeJi797 (a derivative of MBG5012 expressing both alpha-amylase and glucoamylase; WO2019/161227). The resulting strain has one promoter-containing fragment (left fragment), one gene-containing fragment (middle fragment) and one PRM9 terminator fragment (right fragment) integrated into the *S. cerevisiae* genome at the XII-2 locus.

(581) Construction of the Promoter-Containing Fragment (Left Fragment)

(582) Linear DNA containing 500 bp homology to the XII-2 site and the *S. cerevisiae* pPGK1 promoter was PCR amplified from HP27 plasmid DNA with primers 1229945 (5'-TCTTT TCGCG CCCTG GAAAG G-3'; SEQ ID NO: 434) and 1227122 (5'-TGTTT TATAT TTGTT GTAAA AAGTA GATAA TTA CT TCCTT GATGA TCTG-3'; SEQ ID NO: 435). Fifty pmoles each of forward and reverse primer was used in a PCR reaction containing 5 ng of plasmid DNA as template, 0.1 mM each dATP, dGTP, dCTP, dTTP, 1× Phusion HF Buffer (Thermo Fisher Scientific), and 2 units Phusion Hot Start DNA polymerase in a final volume of 50 µL. The PCR was performed in a T100™ Thermal Cycler (Bio-Rad Laboratories, Inc.) programmed for one cycle at 98° C. for 30 seconds followed by 32 cycles each at 98° C. for 10 seconds, 59° C. for 20

seconds, and 72° C. for 40 seconds with a final extension at 72° C. for 10 minutes. Following thermocycling, the PCR reaction products gel isolated and cleaned up using the NucleoSpin Gel and PCR clean-up kit (Machery-Nagel).

(583) The linear DNA containing 243 bp homology to the *S. cerevisiae* pPGK1 promoter (SEQ ID NO: 4) and the MF1 α signal peptide (SEQ ID NO: 7) was PCR amplified from DNA synthesized by GeneArt with primers 1229946 (5'-GTGAC AACAA CAGCC TGTTC TC-3'; SEQ ID NO: 436) and 1222995 (5'-AGCTA ATGCG GAGGA TGCTG C-3'; SEQ ID NO: 437). Fifty pmoles each of forward and reverse primer was used in a PCR reaction containing 5 ng of plasmid DNA as template, 0.1 mM each dATP, dGTP, dCTP, dTTP, 1 $\times$ Phusion HF Buffer (Thermo Fisher Scientific), and 2 units Phusion Hot Start DNA polymerase in a final volume of 50 μ L. The PCR was performed in a T100™ Thermal Cycler (Bio-Rad Laboratories, Inc.) programmed for one cycle at 98° C. for 30 seconds followed by 32 cycles each at 98° C. for 10 seconds, 59° C. for 20 seconds, and 72° C. for 20 seconds with a final extension at 72° C. for 5 minutes. Following thermocycling, the PCR reaction products gel isolated and cleaned up using the NucleoSpin Gel and PCR clean-up kit (Machery-Nagel).

(584) Construction of the Terminator-Containing Fragment (Right Fragment)

(585) The DNA containing 250 bp of the PRM9 terminator and 500 bp of the XII-2 3'-end homology was PCR amplified from TH12 plasmid DNA (GeneArt) with primers 1221473 (5'-ACAGA AGACG GGAGA CACTA GC-3'; SEQ ID NO: 438) and 1229949 (5'-GGGGT CGCAA CTTTT CCC-3'; SEQ ID NO: 439). Fifty pmoles each of forward and reverse primer was used in a PCR reaction containing 5 ng of plasmid DNA as template, 0.1 mM each dATP, dGTP, dCTP, dTTP, 1 $\times$ Phusion HF Buffer (Thermo Fisher Scientific), and 2 units Phusion Hot Start DNA polymerase in a final volume of 50 μ L. The PCR was performed in a T100™ Thermal Cycler (Bio-Rad Laboratories, Inc.) programmed for one cycle at 98° C. for 30 seconds followed by 32 cycles each at 98° C. for 10 seconds, 59° C. for 20 seconds, and 72° C. for 40 seconds with a final extension at 72° C. for 10 minutes. Following thermocycling, the PCR reaction products gel isolated and cleaned up using the NucleoSpin Gel and PCR clean-up kit (Machery-Nagel).

(586) Construction of the Phospholipase-Containing Fragments (Middle Fragments)

(587) Synthetic linear uncloned DNA containing the MF1 α signal peptide, phospholipase gene and 50 bp of the PRM9 terminator were obtained from Geneart or Twist Bioscience.

(588) Integration of the Left, Middle and Right-Hand Fragments to Generate Yeast Strains with a Heterologous Phospholipase

(589) The yeast MeJi797 was transformed with the left, middle and right integration fragments described above. In each transformation pool a fixed left fragment and right fragment with 100 ng of each fragment was used. The middle fragment consisted of a signal peptide and phospholipase gene with ~100 ng of each fragment (700 ng total). To aid homologous recombination of the left, middle and right fragments at the genomic XII-2 sites a plasmid containing MAD7 and guide RNA specific to XII-2 (pMIBa638) was also used in the transformation. These four components were transformed into the *S. cerevisiae* strain MeJi797 following a yeast electroporation protocol. Transformants were selected on YPD+cloNAT to select for transformants that contain the Mad7 plasmid pMIBa638. Transformants were picked using a Q-pix Colony Picking System (Molecular Devices) to inoculate one well of 96-well plate containing YPD+cloNAT media. The plates were grown for two days then glycerol was added to 20% final concentration and the plates were stored at -80° C. until needed. Integration of specific phospholipase construct was verified by PCR with locus specific primers and subsequent sequencing. The strains generated were used in the following examples.

Example 6: Corn Mash Fermentations of Phospholipase-Expressing Yeast Strains

(590) The strains described in Example 5 were tested for ethanol production and residual glucose using a 96-well corn mash fermentation described below. Propagation plates were prepared by inoculating 10 μ L of each strain into a 96-well seed plate containing 150 μ L YP+2% glucose

medium per well. Plates were incubated at 30° C. and 300 RPM overnight. The following day, 10 µL of the seed culture was transferred to 96-deep well plates containing 500 µL of East Kansas Agri-Energy Liquefact Amp corn mash supplemented with 100 ppm urea and 0.42 AGU/g Spirizyme Excel. Plates were sealed with EnzyScreen plate covers and tightly clamped to limit oxygen transfer. Corn mash plates were incubated statically at 32° C. for 56 hours. After fermentation was completed, the plates were placed at –80 C for about 10 minutes before 100 µL of 8% H.sub.2SO.sub.4 was added to each well of the 96-deep well corn mash plates. Plates were sealed and mixed by inversion and centrifuged at 3000 rpm for 10 minutes. Supernatants were removed and diluted to 12× in sterile deionized water prior to HPLC analysis. The average ethanol and residual glucose per integrated heterologous phospholipase gene are shown in Table 15.

(591) TABLE-US-00015

TABLE 15	SEQ ID NO:	SEQ	(mature mean mean Donor Organism ID NO: poly- ethanol glucose (catalytic domain) (coding) peptide) (g/L) (g/L) N/A (MeJi797 control strain) — —
128.84	4.82	<i>Penicillium cinnamopurpureum</i>	423 332 138.52 3.30
<i>Talaromyces rugulosus</i>	407 316	138.39	1.28
<i>Aspergillus turcosus</i>	384 293	136.06	1.42
<i>Aspergillus egyptiacus</i>	376 285	135.98	2.97
<i>Bacillus mycoides</i>	381 290	135.88	5.16
<i>Penicillium swiecickii</i>	392 301	134.40	5.30
<i>Lysinibacillus xylanilyticus</i>	388 297	134.23	1.94
<i>Listeria seeligeri</i>	391 300	133.80	3.99
<i>Penicillium spikei</i>	410 319	133.37	1.68
<i>Talaromyces leycettanus</i>	402 311	133.29	1.02
<i>Bacillus toyonensis</i>	404 313	133.10	2.35
<i>Bacillus mycoides</i>	373 282	132.84	5.32
<i>Bacillus mycoides</i>	397 306	132.72	5.88
<i>Rasamsonia argillacea</i>	356 265	132.72	0.96
<i>Bacillus toyonensis</i>	427 336	132.66	4.17
<i>Rasamsonia byssochlamydoides</i>	348 257	132.53	1.34
<i>Bacillus mycoides</i>	414 323	132.42	3.18
<i>Listeria innocua</i>	375 284	132.36	5.22
<i>Penicillium simplicissimum</i>	430 339	132.26	2.57
<i>Bacillus mycoides</i>	380 289	132.26	2.40
<i>Talaromyces boninensis</i>	393 302	132.05	1.08
<i>Hamigera paravellanea</i>	401 310	131.88	2.04
<i>Penicillium vasconiae</i>	416 325	131.76	1.08
<i>Rasamsonia brevistipitata</i>	352 261	131.76	2.52
<i>Bacillus mycoides</i>	422 331	131.68	4.36
<i>Bacillus pseudomycoides</i>	372 281	131.66	3.57
<i>Bacillus sp.</i>	428 337	131.60	2.88
<i>Bacillus acidicer</i>	419 328	131.50	4.78
<i>Bacillus manliponensis</i>	429 338	131.36	2.44
<i>Hamigera avellanea</i>	409 318	131.23	1.49
<i>Penicillium flavescens</i>	358 267	131.22	4.68
<i>Bacillus sp.</i>	420 329	130.98	5.96
<i>Talaromyces bacillisporus</i>	367 276	130.86	3.80
<i>Talaromyces cellulolyticus</i>	425 334	130.74	0.84
<i>Rasamsonia eburnea</i>	349 258	130.61	2.07
<i>Bacillus acidicer</i>	433 342	130.52	5.04
<i>Talaromyces rugulosus</i>	364 273	130.52	1.00
<i>Bacillus thuringiensis</i>	396 305	130.46	3.89
<i>Brevibacillus sp.</i>	415 324	130.44	18.72
<i>Bacillus drementensis</i>	383 292	130.41	6.75
<i>Talaromyces verruculosus</i>	424 333	129.78	3.18
<i>Penicillium piscarium</i>	366 275	129.74	4.37
<i>Penicillium sclerotiorum</i>	347 256	129.72	9.06
<i>Hamigera terricola</i>	365 274	129.68	10.32
<i>Penicillium arenicola</i>	431 340	129.65	1.34
<i>Aspergillus tamarii</i>	377 286	128.86	1.03
<i>Penicillium sp.</i>	408 317	128.04	1.92
<i>Bacillus sp.</i>	382 291	127.84	3.64
<i>Aspergillus tubingensis</i>	386 295	127.66	3.89
<i>Penicillium emersonii</i>	355 264	127.65	3.75
<i>Talaromyces subinflatus</i>	385 294	127.26	6.60
<i>Bacillus thuringiensis</i>	405 314	127.08	3.00
<i>Penicillium vasconiae</i>	361 270	126.96	12.44
<i>Talaromyces columbinus</i>	362 271	126.69	4.05
<i>Bacillus bingmayongensis</i>	413 322	126.67	6.86
<i>Bacillus luciferensis</i>	379 288	126.51	5.46
<i>Hamigera striata</i>	394 303	126.36	2.04
<i>Bacillus thuringiensis</i>	406 315	126.30	9.00
<i>Bacillus toyonensis</i>	389 298	126.30	11.82
<i>Bacillus mycoides</i>	412 321	126.15	13.95
<i>Bacillus acidicer</i>	387 296	125.85	5.37
<i>Bacillus wiedmannii</i>	390 299	125.67	7.25
<i>Talaromyces variabilis</i>	363 272	125.33	5.78
<i>Bacillus thuringiensis</i>	374 283	125.32	4.08
<i>Penicillium simplicissimum</i>	360 269	125.04	10.32
<i>Aspergillus stramenius</i>	371 280	124.92	3.52
<i>Penicillium brefeldianum</i>	350 259	124.86	17.10
<i>Penicillium bialowiezense</i>	346 255	124.72	3.00
<i>Penicillium scabrosum</i>	353 262	124.56	14.67
<i>Penicillium megasporum</i>	369 278	124.44	2.10
<i>Penicillium donkii</i>	400 309	123.96	0.84
<i>Penicillium jensenii</i>	370 279	123.75	5.58
<i>Galactomyces candidus</i>	368 277	123.36	14.37
<i>Bacillus sp.</i>	395 304	123.18	8.82
<i>Aspergillus niger</i>	378 287	123.00	8.28
<i>Bacillus pseudomycoides</i>	421 330	122.22	4.86
<i>Penicillium hispanicum</i>	359 268	118.14	19.95
<i>Penicillium manginii</i>	354 263	117.96	13.15
<i>Penicillium meridianum</i>	345 254	117.72	14.92
<i>Paenibacillus sp.</i>	403 312	110.07	38.28
<i>Paenibacillus alginolyticus</i>	411 320	106.52	43.96

Example 7: Enhancement of Ethanol Yield with Phospholipase-Expressing Yeast Strains

(592) Commercial Amp corn mash was obtained from Trenton Agri at 35.9% (w/w) of dry solids content and was diluted with tape water to 32.0% (w/w). After dilution, the pH value of the mash was adjusted to 5.1 with 39% (w/v) NaOH solution. Urea and lactrol were added into the pH adjusted Amp mash to final concentration at 150 ppm and 3 ppm, respectively. The prepared corn mash was aliquoted into 250 mL flasks (100 g/flask).

(593) To propagate yeast, 50 mL of 6% YPD and 100 μ L of yeast-glycerol stock solution were mixed in a 125-mL flask and then incubated at 32° C. for overnight. After incubation, 45 mL of propagation was transferred to a 50 mL centrifuge tube and centrifuged at 3500 rpm for 10 minutes. The liquid fraction was decanted and deionized water was used to twice wash the cells. The cells were resuspended in 10 mL of deionized water and the total and dead cell accounts were measured using a NucleoCounter® YC-100.

(594) The exogenous α -glucoamylase (Spirizyme Achieve-T™; Novozymes A/S) was added into the flasks containing corn mash per DOE and mixed well. Then the pre-determined amount of yeast suspension was added and mixed well. The fermentation was performed at 32° C. for 54 hours. After the fermentation, 5 g of slurry was taken out and the fermentation was stopped by adding 50 μ L of 40% H.sub.2SO.sub.4. The liquid was separated from solid by centrifuging whole slurry at 900 g for 10 min. The ethanol concentration was measured using HPLC.

(595) Compared with the control strain MHCT-484, both phospholipase-expressing yeast strains showed significantly increased final ethanol yield (13.30% for control strain MHCT-484, to 13.43% and to 13.42% for strains HP21-F04 and HP21-F05, respectively).

Example 8: Enhancement of Oil Extraction with Phospholipase-Expressing Yeast Strains

(596) Commercial Amp corn mash was obtained from Trenton Agri at 35.9% (w/w) of dry solids content and was diluted water to 32.0% (w/w). After dilution, the pH value of the mash was adjusted to 5.1 with 39% (w/v) NaOH solution. Urea and lactrol were added into the pH adjusted Amp mash to final concentration at 150 ppm and 3 ppm, respectively. The prepared corn mash was aliquoted into 250-mL flasks (100 g/flask).

(597) To propagate yeast, 50 mL of 6% YPD and 100 μ L of yeast-glycerol stock solution were mixed in a 125-mL flask and then incubated at 32° C. for overnight. After incubation, 45 mL of propagation was transferred to a 50-mL centrifuge tube and centrifuged at 3500 rpm for 10 minutes. The liquid fraction was decanted, and deionized water was used to twice wash the cells. The cells were resuspended in 10 mL of deionized water and the total and dead cell accounts were measured using a NucleoCounter® YC-100.

(598) Exogenous α -glucoamylase (Spirizyme Achieve-T™; Novozymes A/S), was added into the flasks containing corn mash per DOE and mixed well. Then the pre-determined amount of yeast suspension was added and mixed well. The fermentation was performed at 32° C. for 54 hours. After fermentation, 10 mL of 95% hexane was added to 95 grams of whole slurry. The slurry-hexane mixture was mixed well and then centrifuged at 3000 $\times$ g for 10 minutes. After centrifuge, the top layer was transferred into a 15-mL tube using positive displacement pipettes. The oil was extracted from the slurry again with the same method. The total weight of transferred liquor was measured. The oil content and density were measured on densiometer. The final extracted oil was calculated as:

Extracted oil (g/flask)=first extracted oil (g/flask)+second extracted oil (g/flask)

(599) Compared with the control strain MHCT-484, both lipase-expressing yeast strains HP21-F04 and HP21-F05 improved the final oil yield (0.518 g/flask and 0.966 g/flask for strains HP21-F04 and HP21-F05, respectively compared to 0.483 g/flask for control strain MHCT-484).

Example 9: Defoaming Capabilities with Phospholipase-Expressing Yeast Strains

(600) Commercial Amp corn mash was obtained from Trenton Agri at 35.9% (w/w) of dry solids content and was diluted with tape water to 32.0% (w/w). After dilution, the pH value of the mash was adjusted to 5.1 with 39% (w/v) NaOH solution. Urea and lactrol were added into the pH

adjusted Amp mash to final concentration at 150 ppm and 3 ppm, respectively. The prepared corn mash was aliquoted into 250-mL flasks (100 g/flask).

(601) To propagate yeast, 50 mL of 6% YPD and 100 μ L of yeast-glycerol stock solution were mixed in a 125-mL flask and then incubated at 32° C. for overnight. After incubation, 45 mL of propagation was transferred to a 50-mL centrifuge tube and centrifuged at 3500 rpm for 10 minutes. The liquid fraction was decanted, and deionized water was used to twice wash the cells. The cells were resuspended in 10 mL of deionized water and the total and dead cell accounts were measured using a NucleoCounter® YC-100.

(602) The exogenous α -glucoamylase, Spirizyme Achieve-T™ (Novozyems A/S), was added into the flasks containing corn mash per DOE and mixed well. Then the pre-determined amount of yeast suspension was added and mixed well. Also, the pre-determined exogenous lipase was added and used as control. The fermentation was performed at 32° C. for 54 hours.

(603) The defoaming capability of control MHCT-484 and lipase-expressing yeast HP21-F04 was monitored using video camera after 12-hour fermentation. Compared with control yMHCT48, lipase-expressing yeast HP21-F04 showed significant defoaming capability (FIG. 4).

Claims

1. A recombinant *Saccharomyces* yeast cell comprising a heterologous polynucleotide encoding a phospholipase and a heterologous polynucleotide encoding a glucoamylase, wherein the phospholipase has a mature polypeptide sequence at least 90% sequence identity to the amino acid sequence of SEQ ID NO: 236, and wherein the cell is capable of higher ethanol yield and/or reduced foam accumulation when compared to fermentation using the same process and an identical cell without the heterologous polynucleotide encoding the phospholipase under the same conditions.
2. The recombinant yeast cell of claim 1, wherein the fermenting organism comprises a heterologous polynucleotide encoding an alpha-amylase.
3. The recombinant yeast cell of claim 1, wherein the fermenting organism comprises a heterologous polynucleotide encoding a protease.
4. The recombinant yeast of claim 1, wherein the cell is a *Saccharomyces cerevisiae* cell.
5. The recombinant yeast of claim 1, wherein the phospholipase has a mature polypeptide sequence having at least 95% sequence identity to the amino acid sequence of SEQ ID NO: 236.
6. The recombinant yeast of claim 1, wherein the phospholipase has a mature polypeptide sequence having at least 97% sequence identity to the amino acid sequence of SEQ ID NO: 236.
7. The recombinant yeast of claim 1, wherein the phospholipase has a mature polypeptide sequence having at least 98% sequence identity to the amino acid sequence of SEQ ID NO: 236.
8. The recombinant yeast of claim 1, wherein the phospholipase has a mature polypeptide sequence having at least 99% sequence identity to the amino acid sequence of SEQ ID NO: 236.
9. The recombinant yeast of claim 1, wherein the phospholipase has a mature polypeptide sequence that comprises or consists of the amino acid sequence of SEQ ID NO: 236.
10. A method of producing a fermentation product from a starch-containing or cellulosic-containing material comprising: (a) saccharifying the starch-containing or cellulosic-containing material; and (b) fermenting the saccharified material of step (a) with the recombinant yeast of claim 1.
11. The method of claim 10, wherein saccharification of step (a) occurs on a starch-containing material, and wherein the starch-containing material is gelatinized.
12. The method of claim 10, wherein saccharification of step (a) occurs on a starch-containing material, and wherein the starch-containing material is ungelatinized.
13. The method of claim 10, wherein saccharification of step (a) occurs on a starch-containing material, and wherein the method comprises liquefying the starch-containing material by contacting the material with an alpha-amylase prior to saccharification.

14. The method of claim 10, wherein liquefying the starch-containing material and/or saccharifying the starch-containing material is conducted in presence of exogenously added protease.
 15. The method of claim 10, wherein the recombinant yeast comprises a heterologous polynucleotide encoding an alpha-amylase.
 16. The method of claim 10, wherein the recombinant yeast is a *Saccharomyces cerevisiae* cell.
 17. The method of claim 10, wherein the method results in higher yield of fermentation product and/or reduced foam accumulation when compared to the same process using an identical cell without the heterologous polynucleotide encoding the phospholipase under the same conditions.
 18. The method of claim 10, wherein the phospholipase of the recombinant yeast has a mature polypeptide sequence having at least 95% sequence identity to the amino acid sequence of SEQ ID NO: 236.
 19. The method of claim 10, wherein the phospholipase of the recombinant yeast has a mature polypeptide sequence having at least 97% sequence identity to the amino acid sequence of SEQ ID NO: 236.
 20. The method of claim 10, wherein the phospholipase of the recombinant yeast has a mature polypeptide sequence that comprises or consists of the amino acid sequence of SEQ ID NO: 236.
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